

A Self-Consistent Spatially-Resolved Cluster Dynamics Model for Irradiated Zirconium

Nasr M. Ghoniem^{a,*}, Giacomo Po^b, Matthew Maron^b, Ruben Escobar^b, Benjamin Ramirez Flores^c,
Kristopher Baker^c, Thomas Black^d, James Hollenbeck^d

^aDepartment of Mechanical and Aerospace Engineering, University of California Los Angeles, Los Angeles, CA 90095, USA

^bDepartment of Mechanical and Aerospace Engineering, University of Miami, Coral Gables, FL 33146, USA

^cKnolls Atomic Power Laboratory, Fluor Marine Propulsion, LLC, Niskayuna, NY 12309, USA

^dBettis Atomic Power Laboratory, Fluor Marine Propulsion, LLC, West Mifflin, PA 15122, USA

Abstract

We present a self-consistent, spatially resolved cluster-dynamics model for the irradiated microstructure of α -zirconium, with the construction that converts its continuum fields into discrete dislocations and the two-time-scale method that integrates them. The model is self-consistent in two senses. First, the bias factors of *all* variants follow from anisotropic diffusion theory: every capture efficiency in a loop growth law, and every sink strength those loops impose on the mobile species, comes from the *same* diffusion tensors that transport them, through one anisotropy parameter per species rather than from independently fitted bias factors. Second, the transfer between the continuum fields and the discrete population preserves what defines that population — stored defects exactly, loop number to the integer draw, the dispersion of the size distribution to the truncation of the sampling — so one calculation may run as continuum, discrete or hybrid. The immobile population is resolved into nine crystallographic families — three prismatic $\langle a \rangle$ variants of both characters and a three-state basal vacancy chain — each carrying the first three moments of its size distribution, closed by a log-normal ansatz. The mobile species obey a steady reaction–diffusion problem with Dirichlet conditions at grain boundaries, the immobile families pointwise ordinary differential equations; freezing the mobile field removes the spatial coupling by construction, so that 2.1×10^8 stiff equations become 73 518 independent 36-dimensional systems and the cost of one factorization falls by 10^{10} . Grain boundaries are sinks for both populations by different mechanisms: a Dirichlet condition for the mobile species, and for the loops a size-gated removal of the fraction large enough to reach the boundary. Applied to a 500 nm hexagonal single crystal at 573 K to 40 dpa, it resolves mobile boundary layers of 8–46 nm ordered by species mobility and narrowing twofold as the sink density grows; closes both atom balances on production, the boundary absorbing 96% of interstitial production as mobile species and 27% of vacancy production as loops; and predicts a loop-denuded zone ~ 5 nm wide across which the mean loop diameter recovers from 6.8 to 144 nm. The parameters are calibrated against 78 transmission-electron-microscope measurements as a maximum *a posteriori* estimate under stated priors. Because every bias follows from the diffusion tensors, the anisotropy cannot absorb a shortfall by refitting a bias; it is fixed from transport, not from loop counts, with $p^v = 1.02$ and $p^i = 0.70$ as inputs, implying an interstitial migration anisotropy of 0.106 eV at 573 K, open to atomistic test. On that fixed anisotropy the calibration reproduces the basal loop diameter without adjusting it — 100.7 nm against 95 nm measured — with the prismatic density to $0.88\times$ and its diameter to $1.21\times$. What it does not reach is the basal density, which remains $2.1\times$ the measurement, and a prismatic temperature trend a factor of six low near 600 K; with the anisotropy fixed, the shortfall rests on the basal capture prefactor, hence in absorption rather than transport.

Keywords: zirconium, cluster dynamics, spatially resolved, dislocation loops, diffusional anisotropy, grain boundary, irradiation damage

Contents

1	Introduction	1
2	Defect Classes and Reactions	3
3	Cluster Dynamics Formulation	8
3.1	Master equation and the reduced system	8
3.2	The mobile block	9
3.3	The immobile block	10
3.4	Conditions for loop growth	14
3.5	The size distribution and its closure	17
3.6	Summary of the equations	17
4	Numerical Solution by Operator Splitting	18
5	Continuum-to-Discrete Self-Consistency	20
6	Calibration Against Experiment	23
7	Results: a 500 nm Hexagonal Single Crystal to 40 dpa	27
7.1	Geometry, discretization and system size	27
7.2	Mobile defect fields	28
7.3	Immobile fields and their moments	30
7.4	Global conservation	33
7.5	Grain-boundary effects and the denuded zone	34
7.6	Volume averages and comparison with experiment	35
7.7	Continuum-to-discrete conversion	37
7.8	Evolution of the loop size distributions	40
8	Discussion and Conclusions	45
Appendix A	Derivation of the Two-Block Equations	50
Appendix B	Iterative FEM Solution Algorithm for Mobile Species	58
Appendix C	Continuum-to-Discrete Conversion Algorithm	59
Appendix D	Capture efficiencies from the diffusion tensor	62
Appendix E	Thermal Emission of Vacancies	62
Appendix F	State-vector layout and the conservation accumulators	64

*Corresponding author.

Email address: ghoniem@ucla.edu (Nasr M. Ghoniem)

1. Introduction

Zirconium and its alloys — Zircaloy-2 and Zircaloy-4 in light-water reactor fuel cladding, Zr-2.5Nb in CANDU pressure tubes, and the low-tin variants M5 and ZIRLO in pressurized-water reactor service — operate under prolonged neutron exposure at displacement rates of 10^{-8} to 10^{-7} dpa s $^{-1}$ and accumulate 5–30 dpa over a fuel-cycle lifetime [1, 2]. The microstructural changes that follow — nucleation and growth of $\langle a \rangle$ prismatic loops, the comparatively delayed appearance of $\langle c \rangle$ -component basal vacancy loops, dislocation channeling under load, anisotropic irradiation growth and creep, and stress-driven hydride reorientation — are the principal source of property degradation in Zr-based components, and they manifest as hardening, loss of uniform elongation, axial growth of fuel assemblies and creep-down of pressure tubes [3–6]. Predicting those changes across the joint envelope of dose, temperature, texture and applied stress is the central quantitative problem of fuel-element and pressure-tube qualification.

Cluster dynamics, in the form it took in the late 1970s and early 1980s [7–11], remains the workhorse mesoscopic theory for that prediction. It treats each defect species and size as a population whose concentration evolves under coupled master rate equations balancing in-cascade production, diffusion-controlled reactions, thermal emission and absorption at fixed sinks [12–14]. Its advantages are access to time scales far beyond molecular dynamics [15, 16], explicit resolution of the cluster-size distribution and therefore of every quantity entering a macroscopic property model [17, 18], and a cost low enough relative to kinetic Monte Carlo that it can serve as the meeting point between atomistic parameterization [19, 20] and engineering-scale prediction [5, 21].

In its classical form, however, cluster dynamics is a spatially averaged, mean-field theory. Every concentration is a single number for the whole material, every sink is represented by a sink strength k^2 already averaged over an assumed random and statistically uniform distribution of absorbers, and the transport connecting a mobile defect to a sink appears only through that average. Three distinct assumptions are folded into this, and each fails under conditions that are not exotic. The first is spatial uniformity: a sink strength is meaningful only if the diffusion field around each absorber relaxes to a common far-field value before it meets the next absorber, and near a free surface, a grain boundary or a denuded band it does not, the mobile concentration varying by orders of magnitude over a distance comparable with the boundary-layer width. The failure is structural rather than quantitative, because a single average concentration then describes no point in the material; in the calculations reported below, the outermost shell of a 500 nm grain carries essentially the entire prismatic loop population, so a domain average of the loop density is a statement about the boundary layer and not about the material. The second is diluteness. The Avrami-type kernels that represent loop–loop and loop–network interaction are derived by assuming absorbers are placed independently and at random; once the volume fraction swept by growing loops becomes appreciable the placement is no longer independent, since loops that have coalesced are correlated by construction, and the mean-field estimate of the overlap probability ceases to estimate anything. That is the point at which the population must be represented as discrete objects rather than as a density, and locating it is a prediction the model can make about itself. The third is an isotropy inherited from cubic metals. Self-interstitial migration in α -Zr is strongly anisotropic, prism-plane mobility exceeding basal mobility by roughly an order of magnitude, and vacancy migration is anisotropic more weakly [19, 20]; most cluster-dynamics treatments of zirconium nevertheless retain isotropic diffusion constants taken from the body-centered cubic literature and recover the missing directionality through independently fitted bias factors. That conflation permits a model to hold simultaneously that diffusion is isotropic and that its sinks are biased by the anisotropy of that same diffusion, and it severs the relation between atomic-scale parameters and predicted dimensional change.

Spatially resolved cluster dynamics replaces the averaged concentrations by fields $c(\mathbf{x}, t)$ obeying reaction–diffusion equations on a meshed domain, so that the transport a mean-field sink strength summarizes is instead solved for [22–24]. Three consequences follow that a mean-field model cannot supply. The boundary layer becomes resolved rather than parameterized: a grain boundary is no longer a number $k_{gb}^2 \sim 6\sqrt{k^2}/d$ but a Dirichlet condition, and the depleted zone in front of it — whose width is set by the competition between the local diffusivity and the local sink density, and which therefore differs for each mobile species and evolves through the transient — becomes an output. The denuded zones seen in

irradiated microstructures are a direct consequence, and they are among the few observations that speak to the transport parameters rather than to the reaction parameters. Quantities degenerate in a mean-field model also become distinguishable: [Section 3.4](#) shows that diffusional anisotropy opens a window of the mobile flux ratio over which prismatic interstitial and basal vacancy loops grow together, but that the window is a condition on one number, which a mean-field model carries once and a spatially resolved model carries as a field, so that the observed coexistence of both loop characters can be a statement about different points of one grain rather than about a single balance holding everywhere. Finally, the fields are the natural input to a discrete model, since dislocation dynamics, micromechanical models of channeling and fracture models all require defect positions and sizes rather than densities; a spatially resolved continuum field carries the information a discrete population must reproduce, and the conversion between the two becomes a well-posed problem with conservation laws attached rather than a rescaling of a single average [\[25–27\]](#).

The model presented here is self-consistent in two distinct and independent senses, and the term is used in this paper only for their conjunction. In the first, internal to the continuum model, every bias factor follows from anisotropic diffusion theory: each capture efficiency appearing in a loop growth law, and each sink strength the same loops impose on the mobile species, is generated from the same anisotropic diffusion tensors that transport those species, through the tensorial form of the diffusional-anisotropy-difference correction [\[28–31\]](#). This holds for all variants, the three prismatic habits of each character and the three basal states alike, so one anisotropy parameter per mobile species fixes the bias of every one of the nine families, and no independently fitted bias factor appears anywhere in the model. What the constraint rules out is the possibility described above, of diffusing the mobile species with one anisotropy while biasing its sinks with another, fitted independently of it. In the second sense, which holds between the continuum and the discrete descriptions, the transfer between them preserves the quantities that make them descriptions of one microstructure. [Section 5](#) states the requirements and their ranking: stored defects exactly, loop number to the integer draw, the dispersion of the size distribution to the truncation of the sampling, and the sink strength reported rather than forced. That second sense is what makes a hybrid calculation meaningful, continuum where the population is dilute and discrete where it is not, with the choice made on grounds of dose and computational cost rather than of physics. The two senses meet at the loop’s capture cross-section, which the first makes a function of the transport tensors alone and which the second requires the discrete population to reproduce; a failure of either therefore appears in the transfer ledger of [Section 5](#) as a discontinuity in sink strength, which is why that ledger is reported and not merely checked.

The immobile population is correspondingly expanded to nine crystallographically resolved families: the three prismatic variants a_1, a_2, a_3 of both interstitial and vacancy character, and a three-state basal vacancy chain running from the stacking-fault pyramid through the faulted loop to the perfect loop. The expansion follows the experimental loop character [\[32–36\]](#), which reports prismatic $\langle a \rangle$ loops of both signs and basal $\langle c \rangle$ -component loops that are predominantly faulted and that form by collapse of a pyramidal precursor rather than by accretion. Each family carries the first three moments of its size distribution, so that the width of the distribution and not only its mean is transported, and the conversion to a discrete population can draw loop sizes from a distribution rather than assigning every loop at a point the same size.

This paper reports the formulation, its numerical solution, and its first application to a spatially resolved single crystal. It presents an SRCD formulation for α -Zr that is self-consistent in the first sense above, with the capture efficiencies and sink strengths of all nine families generated from the diffusion tensors and no independent bias ([Sections 2 and 3](#)); a two-time-scale operator split that makes of order 10^6 stiff ordinary differential equations per substep tractable, together with the cost argument justifying it and the splitting error that pays for it ([Section 4](#)); a continuum-to-discrete conversion that is self-consistent in the second sense, with its conservation guarantees stated and ranked and its residual discrepancies measured, and a grain-boundary treatment that makes the boundary a sink for immobile loops as well as for mobile defects ([Section 5](#)); a calibration against 78 experimental measurements of loop density and diameter, presented as a maximum *a posteriori* estimate under stated priors and with an explicit account of which parameters the data determine and which the bounds determine ([Section 6](#)); and a 500 nm hexagonal single-crystal calculation to 10 dpa reporting the mobile and immobile fields, the closed atom balance including the boundary channel, the predicted denuded zone, the discrete loop population the fields imply, and the reconstructed size distribution of every variant ([Section 7](#)).

The remainder is organized as follows. [Section 2](#) fixes the defect classes and the reactions among them. [Section 3](#) gives the formulation, beginning from the general master equation and reducing it to the equations actually integrated. [Section 4](#) is the solution method, [Section 5](#) the continuum-to-discrete map and the grain-boundary models, and [Section 6](#) the calibration. [Section 7](#) is the single-crystal study and [Section 8](#) draws the conclusions and states what remains open. Four appendices collect the capture-efficiency derivation, the moment closure, the peripheral-emission model that replaces fitted annealing lifetimes, and the state-vector layout.

2. Defect Classes and Reactions

The primary irradiation defects in hcp zirconium are vacancies, self-interstitial atoms and their clusters. Because the lattice is elastically and crystallographically anisotropic, a cluster is not specified by its size alone: its Burgers-vector character and habit plane determine its contribution to irradiation growth, to hardening, and to the sink field seen by every other species. The observed loop population in irradiated Zr and its alloys is conventionally grouped into $\langle a \rangle$ loops and $\langle c \rangle$ -component loops [[35](#), [37](#), [38](#)], with

$$\mathbf{b}_a = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad \mathbf{b}_c \parallel [0001], \quad (1)$$

the latter carrying a Burgers-vector component along the hexagonal axis.

Five principal cluster families span the immobile part of that population, together with the four species that remain mobile and feed it [[3](#), [5](#), [39–41](#)]. The immobile families are

$$I_n^a, \quad V_n^a, \quad S_n, \quad V_n^{cf}, \quad V_n^{cp}, \quad (2)$$

where I_n^a is an interstitial prismatic $\langle a \rangle$ loop of n excess self-interstitials on a $\{10\bar{1}0\}$ habit, V_n^a a vacancy prismatic loop of the same habit and Burgers vector but opposite character, S_n a basal stacking-fault pyramid of n aggregated vacancies bounded by basal $\frac{1}{6}\langle 20\bar{2}3 \rangle$ and prismatic facets and formed directly from collapsed cascade vacancy debris, V_n^{cf} a faulted basal vacancy $\langle c \rangle$ -component loop and V_n^{cp} its perfect counterpart. The last two are one population at two stages of a single transformation, separated by the fault they do or do not carry. The faulted loop has Burgers vector $\frac{1}{6}\langle 20\bar{2}3 \rangle$ with edge component $c/2$, removes one basal layer, and is the $\langle c \rangle$ loop that neutron-irradiation micrographs predominantly report [[32–34](#)]; the perfect loop has $\mathbf{b} = [0001]$, edge component c , removes two layers, and forms when a supercritical faulted loop unfaults and sheds its stacking fault [[42–44](#)]. Both are carried because the vacancy content stored per unit area is then a property of the state rather than a fitted average of the two: the perfect loop stores twice the vacancies per unit area that the faulted one does, so the unfaulting transfer of [Eq. \(25\)](#) conserves vacancies exactly while halving the area they occupy, and the radius, the line density and the capture cross-section of the population all change although its stored content does not. A model holding a single basal state must absorb that change into a fitted mean instead. The basal population as a whole appears above a temperature- and dose-rate-dependent threshold, by collapse of supercritical pyramids, and drives breakaway irradiation growth at high fluence [[40](#), [45](#), [46](#)]. Neither helium nor a three-dimensional void population is carried, because Zr alloys generate a negligible (n, α) helium yield and multi-vacancy cascade debris in α -Zr collapses onto basal or prismatic habits rather than forming compact cavities [[3](#), [5](#)].

The families of [Eq. \(2\)](#) are fed, and drained, by the four species that migrate,

$$\mathcal{M} = \{C_v, C_i, C_{2i}, C_{3i}\}, \quad (3)$$

the monovacancy, the mono-interstitial and the di- and tri-interstitial, with signed defect contents $-1, +1, +2$ and $+3$. The two small interstitial clusters are state variables in their own right rather than a correction folded into C_i , and three properties make the distinction load-bearing rather than bookkeeping. They are produced directly, since a fraction of the interstitial debris of a displacement cascade emerges already clustered rather than as isolated self-interstitials [[47–49](#)]. They remain glissile and migrate with their own diffusion tensors [[50–52](#)], so each carries its own anisotropy parameter p^m in [Eq. \(29\)](#) and its own capture efficiency $Z_{sk,m}$ at every family in [Eq. \(31\)](#); an anisotropy assigned to C_i alone would misstate the

directionality of the flux the clusters carry, which is the quantity the coexistence criterion of [Section 3.4](#) is written in. And they are the homogeneous nucleation channel for the interstitial prismatic loops, through the $2i + 2i$, $i + 3i$ and $2i + 3i$ reactions that produce a cluster past the mobile cut-off, so the mobile ladder and the loop populations are not separable even in principle. Each also recombines with vacancies, at rates $R_{v,2i}$ and $R_{v,3i}$ distinct from $R_{i,v}$, and dissociates thermally against its own binding energy. The ladder is closed at three interstitials: anything larger is immobile and is carried as a loop, a truncation stated with its consequences in [Section 3](#).

This classification separates character, interstitial or vacancy, from crystallography, prismatic or basal, and both distinctions carry weight. Two families sharing a habit plane and a Burgers vector but differing in character, I_n^a and V_n^a , respond to the same mobile fluxes with opposite sign, while two families sharing a character but differing in habit, V_n^a and V_n^{cf} , present differently oriented capture cross-sections to an anisotropic diffusion field and therefore compete for the same vacancies on unequal terms. It is that second competition, taken up in [Section 3.4](#), which the self-consistent formulation exists to describe without a free parameter.

[Equation \(2\)](#) is size-resolved, each family being an infinite ladder in n . The model solved here replaces the ladder by a small number of moments of the size distribution per family ([Section 3.5](#)) and resolves instead the crystallographic variants that a size-resolved but variant-lumped model cannot distinguish. Writing the character as $s \in \{i, v\}$ and the habit as $k \in \{a_1, a_2, a_3, c_0, c_f, c_p\}$, the families carried are

$$\mathcal{F} = \underbrace{\{(i, a_1), (i, a_2), (i, a_3)\}}_{\text{prismatic interstitial}} \cup \underbrace{\{(v, a_1), (v, a_2), (v, a_3)\}}_{\text{prismatic vacancy}} \cup \underbrace{\{(v, c_0), (v, c_f), (v, c_p)\}}_{\text{basal vacancy chain}}, \quad (4)$$

nine in all, of which eight are dislocation loops and one is not: c_0 is the stacking-fault pyramid, the precursor from which the basal loops form. We write $\mathcal{F}_L = \mathcal{F} \setminus \{(v, c_0)\}$ for the eight loop families, and every statement below invoking a perimeter, a line density or a coalescence channel is a statement about $\mathcal{F}_L$.

Three features distinguish [Eq. \(4\)](#) from the lumped, stress-aligned pair of the earlier mean-field model, and each is a statement about crystallography rather than about the solver. The prismatic loops are resolved into their three variants of both characters, so that each carries its own habit normal $\hat{\mathbf{a}}_k$, the resolved stress on each is computed rather than parameterized, and the response to an arbitrary stress state is a redistribution among the three rather than a shift of one scalar; at zero deviatoric stress the three are equivalent by symmetry, receive equal shares, and sum to the lumped population exactly, which is the regression test on the change. There is no basal interstitial family, because basal interstitial $\langle c \rangle$ loops are not a reported population in neutron-irradiated Zr; a formal c_i population is retained in the growth inequalities of [Section 3.4](#) for symmetry but is not a state variable. And the basal vacancy population is a three-state chain $c_0 \rightarrow c_f \rightarrow c_p$, in which the pyramid is not planar and stores its vacancies with a nearly isotropic relaxation volume while the faulted and perfect states are planar and store different vacancy content per unit area; both transitions are thermally activated and both are one-way.

The two habits are mutually orthogonal: the normals $\hat{\mathbf{a}}_k$ of the three prismatic variants lie in (0001) while the basal normal is $[0001]$ itself. That orthogonality is why no single electron-beam direction images both populations at true size, a zone axis showing one family face-on viewing the other edge-on, so that any comparison of [Eq. \(4\)](#) against transmission-electron-microscope densities must combine at least two zone axes and a single-axis micrograph constrains at most one habit. It is also why the calibration of [Section 6](#) treats the $\langle a \rangle$ and $\langle c \rangle$ data as separate target sets rather than as one.

Each family carries a character sign η_s , $+1$ for interstitial and -1 for vacancy, and three spatially resolved fields: the cluster density $n_{sL}^k(\mathbf{x})$ in clusters per host atom, the defect content $c_{sL}^k(\mathbf{x})$ in point defects stored per host atom, and the second content $q_{sL}^k(\mathbf{x})$ of [Section 3.5](#), which carries the width of the size distribution. All three are dimensionless, so a number density per unit volume is n/Ω and a stored-defect density is c/Ω , with Ω the atomic volume. The quantity converting loop area into stored defects is not $|\mathbf{b}|$ but the edge component $\mathbf{b} \cdot \mathbf{n}$, since a loop of radius R holds $\pi R^2(\mathbf{b} \cdot \mathbf{n})/\Omega$ defects, so that the mean radius, the mean size and the line density of family (s, k) follow from its content and density as

$$R_{sL}^k = \lambda_k \sqrt{\frac{c_{sL}^k}{n_{sL}^k}}, \quad \lambda_k = \sqrt{\frac{\Omega}{\pi(\mathbf{b} \cdot \mathbf{n})_k}}, \quad m_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k, \quad (5)$$

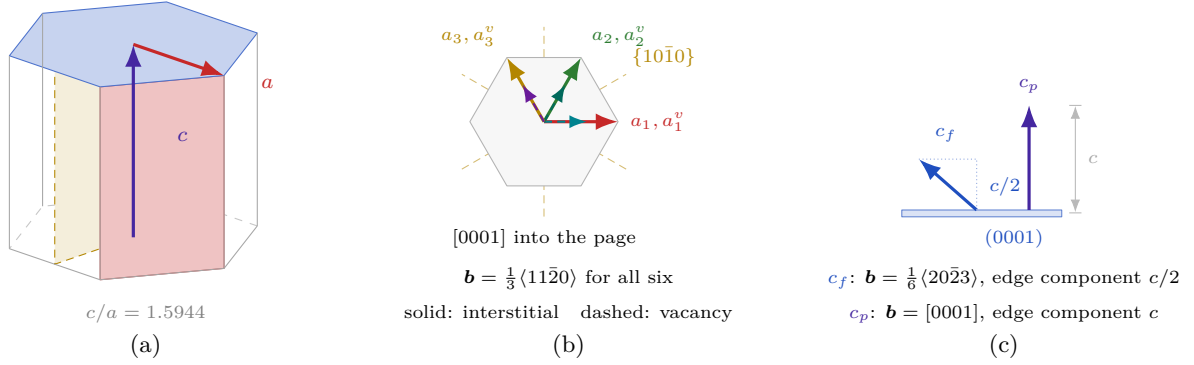


Figure 1: The hcp cell of α -Zr and every Burgers vector the model carries, drawn at the physical axial ratio $c/a = 1.5944$ rather than the ideal $\sqrt{8/3} = 1.6330$. **(a)** The cell, with the three planes that matter: the basal (0001) plane (blue), the habit of c_0 , c_f and c_p ; a first-order prismatic $\{10\bar{1}0\}$ face (red), a side face of the prism; and a second-order $\{11\bar{2}0\}$ plane (yellow, dashed), a diagonal *through* the cell rather than a face of it. The arrows are the two shortest lattice translations, $a = \frac{1}{3}\langle 11\bar{2}0 \rangle$ in the basal plane and $c = [0001]$ along the cell axis. **(b)** Plan view down $[0001]$: the three prismatic variants at 60° to one another, each drawn twice — long and solid for the interstitial family, short and dashed for its vacancy partner, which shares the habit plane and the Burgers vector and differs only in what it stores. The dashed yellow lines are the $\{10\bar{1}0\}$ traces, which bisect the Burgers directions. **(c)** The basal pair, seen edge-on. The perfect state c_p has $\mathbf{b} = [0001]$ normal to its habit, so its edge component is the full c ; the faulted state c_f has the inclined $\mathbf{b} = \frac{1}{6}\langle 20\bar{2}3 \rangle$, whose edge component is $c/2$ — one removed basal layer against two. That factor of two is why $\lambda_{c_p} = \lambda_{c_f}/\sqrt{2}$ in Table 1, and it is why the axial ratio has to be drawn correctly. The pyramid c_0 has no Burgers vector and so appears in neither panel. The inclined pyramidal planes containing $\langle c+a \rangle = \frac{1}{3}\langle 1123 \rangle$ carry *no* family: that vector is the glide vector accommodating c -axis strain in deformation, and its planes are not observed habit planes for irradiation-induced loops, so the absence is an experimental statement and not a truncation.

with one geometric length per habit: λ_a shared by all six prismatic families, and λ_{c_f} , λ_{c_p} for the two basal states. Equation (5) describes a planar object and does not apply to the pyramid, whose stored vacancies fill a volume rather than bound an area; its size follows instead from the volumetric relation

$$R_{vL}^{c_0} = \left(\frac{m_{vL}^{c_0} \Omega}{\sqrt{8}} \right)^{1/3}, \quad (6)$$

so that the pyramid carries no λ_k , no Burgers vector and no line density. What it does carry is a capture cross-section, and Section 3 shows that a cross-section is the only thing either the growth law or the sink strengths ever ask of a family. Table 1 lists the nine families with the crystallography that fixes their geometry.

Table 1: The nine families of Eq. (4), with the crystallography that fixes each one's geometry. $\lambda_k = \sqrt{\Omega/\pi(\mathbf{b} \cdot \mathbf{n})_k}$ is the length converting $\sqrt{m}$ into a radius, Eq. (5). The lattice constants are $a = 0.323$ nm, $c = 0.515$ nm, hence $c/a = 1.5944$ — the physical ratio and not the ideal $\sqrt{8/3} = 1.6330$ — and $\Omega = (\sqrt{3}/4)a^2c = 2.3266 \times 10^{-29}$ m³.

Family	Habit	$\mathbf{b}$	Character	λ_k (nm)	Comment
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3}\langle 11\bar{2}0 \rangle$	i	0.151	The dominant early population; $\mathbf{b} \cdot \mathbf{n} = a$.
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3}\langle 11\bar{2}0 \rangle$	v	0.151	Same habit and $ \mathbf{b} $ as the interstitial variants; separated from them only by capture character.
c_f	(0001)	$\frac{1}{6}\langle 20\bar{2}3 \rangle$	v	0.170	Faulted, $\mathbf{b} \cdot \mathbf{n} = c/2$, one removed basal layer. The dominant observed $\langle c \rangle$ loop.
c_p	(0001)	$[0001]$	v	0.120	Perfect, $\mathbf{b} \cdot \mathbf{n} = c$, two removed layers, hence $\lambda_{c_p} = \lambda_{c_f}/\sqrt{2}$.
c_0	(0001) base	partials	v	—	The stacking-fault pyramid: not planar, so Eq. (5) is replaced by Eq. (6) and it carries no λ_k . Precursor to c_f .

The coupled kinetics are generated from a directed multigraph, the reaction admissibility graph [53], whose vertices are the admissible cluster states and whose edges are the elementary reactions permitted by the polarity, habit-plane and mobility constraints above. The master equations follow by a deterministic graph walk, the time derivative at each vertex being the signed sum of the rate kernels of all incoming and outgoing edges incident on it. Vertices partition by polarity,

$$\mathcal{V}^+ = \{C_i, C_{2i}, C_{3i}\} \cup \{(i, a_k)\}, \quad (7)$$

$$\mathcal{V}^- = \{C_v\} \cup \{(v, a_k)\} \cup \{(v, c_0), (v, c_f), (v, c_p)\}, \quad (8)$$

so that the interstitial prismatic families lie on the positive branch and the six vacancy families on the negative one. Table 2 catalogs the fifteen edge classes and Fig. 2 draws the graph, and both are drawn against the equations summarized in Section 3.6 rather than against a description of them. Every class in the table appears as a named term of the master equation Eq. (9), and the correspondence is one to one in both directions: nothing in the table is missing from that equation, and no term of that equation lacks an edge here.

Three of those classes deserve comment, because they are absent from the mean-field treatment this model replaces and are easily overlooked. Cascade production is not one edge but two partitions of the same displacement rate: a part G_m deposited into the mobile species, and a part J_k that nucleates loops directly, with its own share for the prismatic families of *both* characters as well as for the basal family. Alongside that cascade route the interstitial prismatic family has a second, homogeneous nucleation channel fed by the $2i + 2i$, $i + 3i$ and $2i + 3i$ reactions among the mobile clusters, so that the mobile ladder and the loop populations are not separable even in principle. And the grain boundary appears twice, as a Dirichlet condition on the mobile species and as a size-gated removal acting on the loops themselves; the second has no counterpart in a mean-field treatment, where a loop has no size relative to a boundary that is itself only a sink strength. That removal has a companion gated the opposite way, the dissolution of loops that have shrunk below the minimum stable size, and the two are one process class read in the two senses: the first takes the loops above a threshold and sends their defects out of the crystal, the second takes those below one and returns them to the mobile pool.

Table 2: Admissible process classes of the reaction admissibility graph. Each elementary reaction maps to one edge class and one rate kernel, given at the equation indicated, and each is one named term of Eq. (9).

Process	Kernel	Given in
P1 vacancy–interstitial recombination, and vacancy capture by a mobile cluster	$R_{i,v}, R_{v,2i}, R_{v,3i}$	Eq. (13)
P2 clustering among mobile species	$\beta^{m,n}$	Eq. (13)
P3 thermal dissociation of a mobile cluster	$\alpha^{m,n}$	Eq. (13)
P4 cascade production into the mobile species	G_m	Eq. (12)
P5 cascade nucleation of loops, both characters	J_k	Eq. (26)
P6 homogeneous loop nucleation from mobile clusters	$R_{i,3i} + R_{2i,2i} + R_{2i,3i}$	Eq. (26)
P7 capture at a loop or the pyramid, of either polarity	ϕ_{km}	Eq. (18)
P8 peripheral emission from a vacancy loop	S_{vL}	Eq. (E.4)
P9 loop–loop coalescence	φ_{LL}	Eq. (22)
P10 loop–network coalescence	φ_{LN}	Eq. (22)
P11 absorption of a mobile species at the network	$R_{m,s}$	Eq. (15)
P12 basal chain transfer, $c_0 \rightarrow c_f \rightarrow c_p$	ν_{col}, ν_{uf}	Eq. (25)
P13 absorption of a mobile species at a grain boundary	Dirichlet	Eq. (17)
P14 size-gated absorption of <i>loops</i> at a grain boundary	ν_{gb}	Eq. (47)
P15 dissolution of loops below the minimum stable size	ν_{van}	Eq. (24)

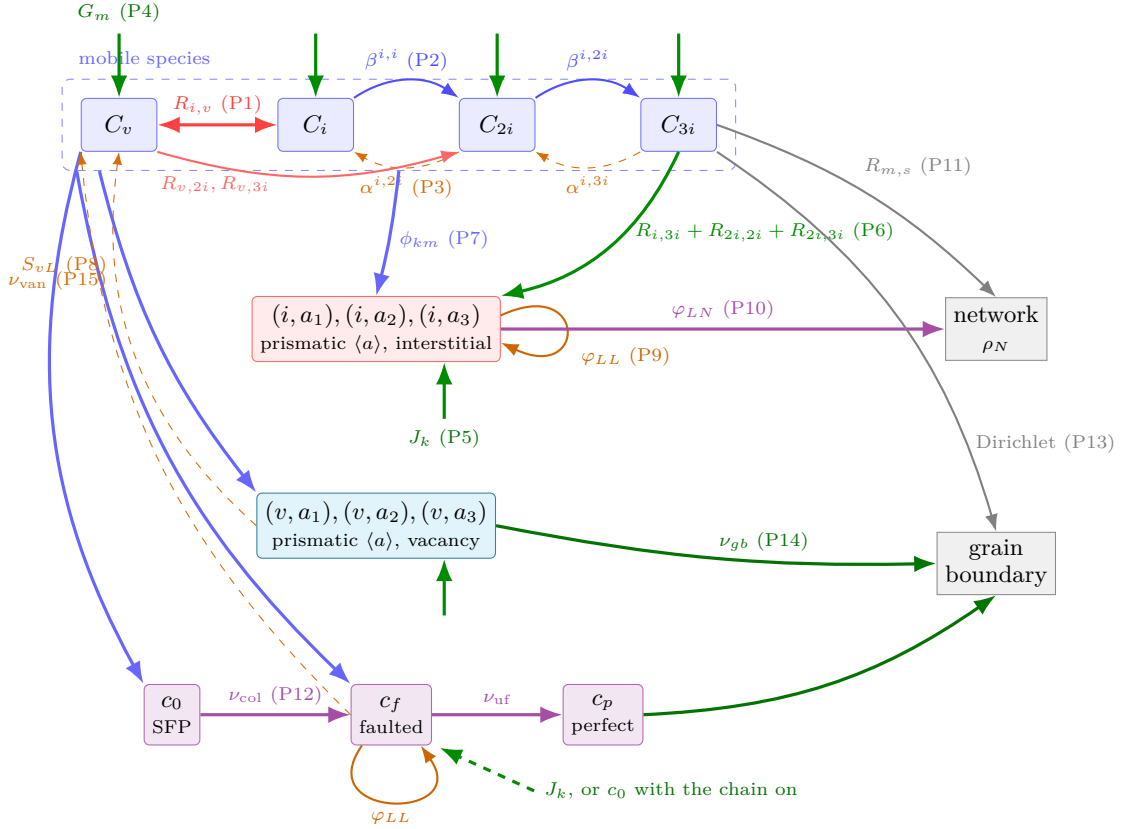


Figure 2: Reaction admissibility graph for the nine families of Eq. (4), carrying every edge class of Table 2. Cascade production is drawn as a short stub with no origin vertex at each population it feeds, being a partition of one displacement rate rather than a transfer between two populations: G_m (P4) into the mobile species, J_k (P5) into the loops directly. The four mobile species are boxed because capture onto a loop sums over all of them with a sign set by polarity, so one bundled arrow per family is faithful where four would be unreadable. Dashed orange edges run backwards to the mobile pool (P3, P8, P15). The three prismatic variants of each character are drawn as one vertex: they exchange nothing with one another and differ only in habit normal.

3. Cluster Dynamics Formulation

3.1. Master equation and the reduced system

The general cluster-dynamics master equation for a population indexed by a species label α and a defect content n is a balance of gains and losses over the edges of the graph of Fig. 2. It is written here with one term per edge class of Table 2, because everything the rest of this paper does to it is a restriction or a moment of *this* equation and nothing is added to it later:

$$\begin{aligned}
\frac{\partial C_n^\alpha}{\partial t} = & \underbrace{-\nabla \cdot \mathbf{J}_n^\alpha}_{\text{transport}} + \underbrace{G_n^\alpha}_{\text{P4, P5}} + \underbrace{\frac{1}{2} \sum_{\beta\gamma} \sum_{n'n''} \nu_{\beta n', \gamma n''}^\alpha \mathcal{K}_{n', n''}^{\beta\gamma} C_{n'}^\beta C_{n''}^\gamma}_{\text{P1, P2, P6, P7}} \\
& + \underbrace{\frac{1}{2} \sum_{n'} C_{n', n-n'}^\alpha C_{n'}^\alpha C_{n-n'}^\alpha - C_n^\alpha \sum_{n'} C_{n, n'}^\alpha C_{n'}^\alpha}_{\text{P9, coalescence}} + \underbrace{\varepsilon_{n+1}^\alpha C_{n+1}^\alpha - \varepsilon_n^\alpha C_n^\alpha}_{\text{P3, P8}} \\
& + \underbrace{\sum_{\beta} [\theta_n^{\beta \rightarrow \alpha} C_n^\beta - \theta_n^{\alpha \rightarrow \beta} C_n^\alpha]}_{\text{P12}} - \underbrace{[\mathcal{D}_N^\alpha(n) + \mathcal{D}_{gb}^\alpha(n) + \mathcal{D}_{\text{van}}^\alpha(n)] C_n^\alpha}_{\text{P10, P11, P14, P15}},
\end{aligned} \tag{9}$$

closed on the mobile species by the grain-boundary condition

$$C_n^\alpha|_{\partial\Omega_D} = C_n^{\alpha, \text{eq}}, \quad \alpha \in \mathcal{M} \quad (\text{P13}), \tag{10}$$

so that the boundary enters the formulation twice and at the same level as every other channel: as Eq. (10) on the species that reach it by diffusion, and as $\mathcal{D}_{gb}^\alpha$ on the loops that reach it by growing.

The symbols are the following. $\mathbf{J}_n^\alpha = -\mathbf{D}_n^\alpha \nabla C_n^\alpha$ is the diffusive flux, which vanishes identically on every immobile index. G_n^α is the cascade source, one displacement rate partitioned over the mobile species (P4) and over the loop families at their nucleation sizes (P5). $\mathcal{K}_{n', n''}^{\beta\gamma}$ is the rate constant of the encounter between (β, n') and (γ, n'') and $\nu_{\beta n', \gamma n''}^\alpha$ the net number of (α, n) that one such event produces — negative when (α, n) is a reactant, -2 when a single event consumes two of it — so that one term carries recombination, mobile clustering, homogeneous loop nucleation and capture at a loop, and carries them with their stoichiometry rather than with a sign convention imposed by hand.

$C_{n', n''}^\alpha$ is the coalescence kernel, the rate at which two members of the *same* family holding n' and n'' defects merge into one holding $n' + n''$. It is separated from $\mathcal{K}$ because both of its partners are immobile, which is exactly the property that keeps it out of the size ladder of Section 3.1 and makes it a Smoluchowski coagulation rather than a transition on that ladder; its gain and loss terms are the standard pair, and the $\frac{1}{2}$ in the gain removes the double count over the ordered pairs $(n', n - n')$.

ε_n^α is the thermal emission rate, covering both a mobile cluster shedding a point defect (P3) and a vacancy loop emitting across its own perimeter (P8). $\theta_n^{\beta \rightarrow \alpha}$ is a first-order transfer that moves a cluster from one family to another at fixed defect content, which is the basal chain (P12) and nothing else in this model. And the three $\mathcal{D}$ are first-order removals at fixed sinks: the dislocation network, absorbing mobile species (P11) and whole loops (P10); the grain boundary, absorbing loops large enough to reach it (P14); and the dissolution of loops that have shrunk below the minimum stable size (P15). All three are written as functions of n because two of them are size-gated, and that gate is the only reason the last two are separate channels rather than one.

Equation (9) is one equation per (α, n) and per mesh node, and is not solvable as written for the size ranges of interest.

Two reductions bring it to the system integrated here, and they cut in different directions.

The first truncates the mobile ladder. Only the monomers and the smallest interstitial clusters migrate. The mobile set is the $\mathcal{M}$ of Eq. (3), closed at $\mathbf{m}_M = [-1, +1, +2, +3]^T$ — monovacancy, mono-, di- and tri-interstitial — with the signed entries m^k carrying the polarity $\varsigma^k = m^k/|m^k| \in \{\pm 1\}$, positive for an

interstitial species and negative for a vacancy. That ς is the only polarity symbol in this paper: each immobile family carries the same sign as its character, $\varsigma^s = +1$ for $s = i$ and -1 for $s = v$, and the $\varsigma_s(m)$ that appears in the growth law Eq. (18) and in the flux sums Eq. (A.9) is nothing but the product $\varsigma^s \varsigma^m$ of the two. It is not to be confused with the diffusional anisotropy p^m of Eq. (29), which is a positive real and shares no meaning with it beyond a species index. Everything larger is immobile and is carried by the second reduction. The mobile block retains its spatial derivative, because transport is what it is for.

The second truncates the immobile ladder in a different variable, keeping the variants and giving up the size resolution. For each family the ladder in n is replaced by the first three moments of the size distribution of each family in Eq. (4),

$$n_{sL}^k = \int f_k dm, \quad c_{sL}^k = \int m f_k dm, \quad q_{sL}^k = \int m^2 f_k dm, \quad (11)$$

closed by the log-normal ansatz of Section 3.5. The immobile block loses its spatial derivative entirely: a loop does not diffuse, and the only thing that couples one node's loops to another's is the mobile field. That is not an approximation — it is the property that makes the numerical method of Section 4 possible.

3.2. The mobile block

Because the mobile species reach quasi-steady state on a time scale $\tau_M \sim (k^2 D)^{-1}$ of order microseconds, while the immobile families evolve on the dose scale ($1 \text{ dpa} = 10^7 \text{ s}$ at $G = 10^{-7} \text{ dpa s}^{-1}$), the mobile block is solved as a *steady* reaction-diffusion problem for the immobile state currently held. Writing $\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$ in per-atom units,

$$\mathbf{0} = -\nabla \cdot \mathbf{J} + \mathbf{P}_M + \frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathcal{D}) \mathbf{c}_M + \mathbf{S}_{vL}, \quad (12)$$

where upright sans-serif bold denotes arrays in *species space* and bold italic ordinary Cartesian vectors and tensors, so that the entries $\mathbf{J}^x = -\mathbf{D}^x \nabla c^x$ of the stacked flux $\mathbf{J}$ are themselves Cartesian vectors and $\nabla \cdot$ acts on each. One species-space array is set in bold calligraphic rather than bold sans: the absorption matrix $\mathcal{D}$, which would otherwise be a third object named D alongside the diffusion tensors $\mathbf{D}_{\text{ten}}$ and their orientational averages $\mathbf{D}_d$.

The terms are, in order: the cascade production $\mathbf{P}_M = G\varepsilon \mathbf{f}$, with G the displacement rate, ε the surviving efficiency and $\mathbf{f}$ the partition of production among the mobile species; the second-order reactions $\mathbf{R}_2$, whose entries are the pairwise rate constants

$$\beta^{m,n} = \frac{A^{m,n}}{\Omega} (\bar{D}^m + \bar{D}^n)(r^m + r^n), \quad r^m = \begin{cases} \left(\frac{3\Omega}{4\pi}\right)^{1/3}, & |m^m| = 1, \\ \left(\frac{|m^m|\Omega}{\pi b}\right)^{1/2}, & |m^m| > 1, \end{cases} \quad (13)$$

with the recombination-volume factor β_r carried on the interstitial-vacancy channel and the combinatorial factors retained on the like-species channels; the thermal emission matrix $\mathbf{E}$, whose entries are $\alpha^{m,n} = \beta^{m,n} \exp(-E_{m,n}^b/k_B T)$; the linear absorption matrix $\mathcal{D}$; and $\mathbf{S}_{vL}$, the vacancies returned to the pool by peripheral emission from vacancy loops (Appendix E).

The absorption matrix is where the immobile population acts back on the mobile one, and it is written so that no defect can be counted twice:

$$\mathcal{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_L, \quad \mathbf{D}_d = \text{diag}\{\bar{D}^v, \bar{D}^i, \bar{D}^{2i}, \bar{D}^{3i}\}, \quad (14)$$

whose first term carries the *network alone*,

$$C_s^v = \frac{a^2}{z_c} \rho_N, \quad C_s^i = \frac{a^2}{z_c} Z_N \rho_N, \quad (15)$$

in which ω is not an independent parameter but the jump attempt frequency $\omega^m = z_c \nu^m \exp(-E_{\text{mig}}^m/k_B T)$, tied to the orientationally averaged diffusivity of Eq. (29) by $\bar{D}^m = (a^2/z_c) \omega^m$. Written out, the product is

$$\omega \odot \mathbf{C}_s = (\bar{D}^v \rho_N, Z_N \bar{D}^i \rho_N, Z_N \bar{D}^{2i} \rho_N, Z_N \bar{D}^{3i} \rho_N),$$

the lattice constant and the coordination number having cancelled between the two factors. The network term is therefore the classical $\bar{D}^m k_N^2$ with $k_N^2 = Z_{N,m} \rho_N$, and splitting it into a frequency and a sink concentration is a bookkeeping convention inherited from the 0-D model, not a second physical quantity. The loop term below is built the same way, which is what allows the two to be added.

The second term carries the loops through the diagonal sink-strength block, in which $Z_{sk,m}$ is the capture efficiency of family (s, k) for species m , built in Section 3.4 from the diffusion tensors themselves,

$$\mathbf{K}_{\mathcal{L}} = \text{diag}\{\mathbf{k}^2\}, \quad k_m^2 = \sum_{(s,k) \in \mathcal{F}_L} Z_{sk,m} \rho_{sL}^k + Z_{vc0,m} \alpha_{\text{sfp}} \pi R_{vL}^{c0} \frac{n_{vL}^{c0}}{\Omega} + Z_{N,m} \rho_N. \quad (16)$$

The loops must not also appear in Eq. (15), or their absorption would be counted once as a network sink and once as loop growth. The middle term of Eq. (16) is the stacking-fault pyramid, which has no line density and enters through a compact capture cross-section instead; Eq. (20) below shows that a cross-section is the only thing either side ever asks of a family.

The block is closed by its boundary conditions. On a free surface or a grain boundary the mobile species are pinned at thermal equilibrium,

$$c^v = c^{v,\text{eq}} = \exp(-E_v^f/k_B T), \quad c^i = c^{2i} = c^{3i} = 0 \quad \text{on } \partial\Omega_D, \quad (17)$$

which is process class P13 of Table 2. This replaces the mean-field grain-boundary sink strength $k_{gb}^2 \simeq 6\sqrt{k^2}/d$ by a boundary condition, and the depleted layer in front of it becomes an output rather than a parameter. Faces declared periodic instead carry periodic conditions on all four species.

That block is assembled in Galerkin weak form, discretized on the second-order tetrahedral mesh and solved by Newton iteration; Appendix B sets the algorithm out in full. Three of its features are worth carrying into the reading of any output. The absorption matrix $\mathbf{D}$ of Eq. (14) is evaluated at every quadrature point from the immobile state held fixed there, not interpolated from nodal values, which is what makes the immobile field enter as a coefficient rather than as an unknown. What is solved for is an increment, so the Dirichlet data enter once, through the initial guess, and never again. And the mobile field is represented by superposition, $\mathbf{c}_M = \mathbf{c}_M^{\text{FEM}} + \mathbf{c}_M^{\text{DD}}$, with $\mathbf{c}_M^{\text{DD}}$ the analytic field of any discrete segments present; only the corrective part $\mathbf{c}_M^{\text{FEM}}$ is solved for, and the discrete field enters through the Dirichlet data, which are reduced to $\mathbf{c}_M^{\text{eq}} - \mathbf{c}_M^{\text{DD}}$. It is that last point that makes the mobile block of an output file the corrective part and not the concentration, whenever discrete loops are present.

3.3. The immobile block

The block has five channels, and they are given in turn. All nine families obey one growth law, in which the stored content changes under the net biased flux of the correct sign:

$$c_{sL}^k|_{\text{growth}} = \Pi_{sL}^k \eta_s \sum_{m \in \{v,i,2i,3i\}} |m| \bar{D}^m Z_{sk,m} c^m \varsigma_s(m), \quad (18)$$

with $Z_{sk,m}$ the capture efficiency of Eq. (31) and $\varsigma_s(m) = \varsigma^s \varsigma^m$ the polarity product of Section 3.1: +1 when the polarity of species m matches the character s , so that absorbing it grows the family, and -1 otherwise, so that absorbing it shrinks the family. The geometric prefactor is

$$\Pi_{sL}^k = \frac{\lambda_k}{\ell} \xi_k \sqrt{n_{sL}^k c_{sL}^k}, \quad \ell \equiv \frac{z_c \Omega}{2\pi a^2}, \quad (19)$$

in which λ_k is the family's own radius scale of Eq. (5), ℓ a length built from the capture number and the atomic volume, and ξ_k the one fitted sink scale per habit (Table 9). And because $\rho_{sL}^k = 2\pi\lambda_k \sqrt{n_{sL}^k c_{sL}^k}$ by Eq. (5),

$$\Pi_{sL}^k = \xi_k \frac{\rho_{sL}^k}{2\pi\ell} \quad \text{identically, for every } (s, k) \in \mathcal{F}_L. \quad (20)$$

The prefactor in the growth law is the family's own sink density divided by $2\pi\ell$. That identity is what makes Eq. (18) and Eq. (16) two readings of one quantity, and it is what carries the growth law onto the pyramid unchanged: a family needs no geometry beyond the capture cross-section it presents, and the pyramid supplies that cross-section as $\Pi_{vL}^{c_0} = \xi_{c_0} \alpha_{\text{sfp}} R_{vL}^{c_0} n_{vL}^{c_0} / 2\ell$.

Nucleation is the second channel. Cascade production deposits directly into the immobile families at a rate $G_k = G \epsilon_k$ per family, each new cluster carrying m_k^{nuc} defects, so that $\dot{n}_{sL}^k|_{\text{nuc}} = J_{sL}^k = G_k / m_k^{\text{nuc}}$ and $\dot{c}_{sL}^k|_{\text{nuc}} = G_k$. The partition ϵ_k and the nucleation sizes are calibrated quantities.

Alongside that cascade route the interstitial prismatic families carry a homogeneous one, from the three mobile-mobile reactions that put a cluster past the cut-off of Eq. (3):

$$J_{iL}^{a_k}|_{\text{hom}} = w_{a_k} (R_{i,3i} + \frac{1}{2}R_{2i,2i} + R_{2i,3i}), \quad \dot{c}_{iL}^{a_k}|_{\text{hom}} = w_{a_k} (4R_{i,3i} + 2R_{2i,2i} + 5R_{2i,3i}), \quad (21)$$

with w_{a_k} the share of the three prismatic variants, $\sum_k w_{a_k} = 1$. Neither set of weights is free. Each content weight is the defect content that same reaction removes from $\mathbf{c}_M$ — $1 + 3$ for $i+3i$, 2 for every di-interstitial lost to $2i+2i$, and $2 + 3$ for $2i+3i$ — so what is deposited is exactly what is withdrawn, and the three clustering entries of Eq. (A.3) are the audit of it. Each number weight is one loop per *event*, which is where the $\frac{1}{2}$ comes from: $R_{i,3i}$ and $R_{2i,3i}$ are unlike-pair rates and count events already, while $R_{2i,2i}$, like $R_{i,i}$ in Eq. (A.2), is a di-interstitial *loss* rate and two are lost per event. Content over number is then 4, 4 and 5 defects on the three channels — the clusters $i+3i$, $2i+2i$ and $2i+3i$ actually make.

The third channel removes loops rather than feeding them. Loops are lost by overlap with other loops and with the network. With $N = n/\Omega$, $R = \lambda \sqrt{c/n}$, and the radii capped at the relevant spacing ($R_{LL} = \min(R, (\Omega/n)^{1/3})$, $R_{LN} = \min(R, \rho_N^{-1/2})$), the Avrami overlap probabilities and rate scales are

$$\begin{aligned} \varphi_{LL} &= 1 - \exp(-\kappa_{LL} \frac{4}{3} \pi R_{LL}^3 N), & \varphi_{LN} &= 1 - \exp(-\kappa_{LN} \pi R_{LN}^2 \rho_N), \\ \nu_{LL} &= c_{LL} v_{\text{abs}} N^{1/3}, & \nu_{LN} &= c_{LN} v_{\text{abs}} \sqrt{\rho_N}, \end{aligned} \quad (22)$$

giving the number and content losses

$$\Delta N_{LN} = \nu_{LN} \varphi_{LN} n, \quad \Delta N = \nu_{LL} \varphi_{LL} n + \Delta N_{LN}, \quad \Delta C = \nu_{LN} \varphi_{LN} c, \quad \Delta Q = \Delta_k (\bar{m}^2 \Delta N - 2\bar{m} \Delta C). \quad (23)$$

Here ΔQ is the second-moment counterpart of ΔN and ΔC — the q that leaves with the loops these two channels remove — and $\Delta_k \equiv q_{sL}^k n_{sL}^k / (c_{sL}^k)^2$ is the shape factor defined with the closure at Eq. (43), which is ≥ 1 for any non-negative size distribution and equals $e^{s_k^2}$ for the log-normal used there. The bracket is the shape-preserving combination: a channel carrying no size distribution of its own removes $\bar{m}^2 \Delta N - 2\bar{m} \Delta C$ of the second moment, which is the statement that it leaves the shape of the family untouched and moves only its mean and its number. The like-loop channel merges two loops into a larger one and therefore *conserves content*, removing density only; the loop-network channel removes both, and the removed content is booked as network sink absorption so the point-defect balance stays closed. The second moment moves in opposite directions on the two, which is why ΔQ carries one sign and not two: substituting $\Delta C = \bar{m} \Delta N_{LN}$ collapses it to $\Delta Q = \Delta_k \bar{m}^2 (\Delta N_{LL} - \Delta N_{LN})$, a gain from the merge and a loss from the removal. Both halves are derived in Appendix A, the second exactly and the first up to the one closure this model needs for coalescence. The rate scale is set by the *absorbed*-flux climb speed v_{abs} rather than the net growth velocity, so coarsening persists at saturation. The capping is what makes Eq. (22) saturate rather than diverge as the number density collapses, and $\varphi_{LL} \rightarrow 1$ is the model's own statement that its mean-field treatment of coalescence has expired — the detector used in Section 5.

The fourth channel runs the other way. Vacancy loops exchange vacancies across their own perimeter at a rate set by a size-dependent binding energy computed from the anisotropic loop self-energy (Appendix E),

one expression serving both characters, so that the same mechanism shrinks a vacancy loop and grows an interstitial one and the classical climb threshold emerges rather than being imposed. This channel replaces the two fitted annealing lifetimes of the mean-field model. It returns vacancies to the mobile pool through $\mathbf{S}_{vL}$ in Eq. (12).

Emission removes content and never a loop, so on its own it leaves coalescence as the only sink on vacancy loop *number*; the disappearance channel supplies the missing one. It is the physical content of the lifetime the emission channel deleted, written as a size condition rather than as a fitted rate: loops that have shrunk below a minimum stable size m_{van} dissolve at ν_{van} , and the fraction that qualifies is read off the same log-normal. Writing

$$\Phi_k^{(j)}(m_\theta) = \frac{1}{2} \operatorname{erfc} \left(\frac{\ln m_\theta - \mu_k - j s_k^2}{s_k \sqrt{2}} \right)$$

for the fraction of the family's j -th moment carried by loops *larger* than m_θ — the same gate Eq. (46) uses at a grain boundary, with (μ_k, s_k) the log-normal parameters of Eq. (43) — and taking it in the opposite sense,

$$\Xi_0^k = [1 - \Phi_k^{(0)}(m_{\text{van}})] n_{sL}^k, \quad \Xi_1^k = [1 - \Phi_k^{(1)}(m_{\text{van}})] c_{sL}^k, \quad \Xi_2^k = [1 - \Phi_k^{(2)}(m_{\text{van}})] q_{sL}^k, \quad (24)$$

the defects returning to the mobile pool as a dissolving loop's must. Because $\Phi^{(2)} > \Phi^{(1)} > \Phi^{(0)}$ the complements reverse that ordering, so what Eq. (24) removes is *smaller* than the family mean: the density falls while the surviving mean size rises. That is the one trade coalescence cannot make — it removes the largest loops and shortens the survivors' lives — and it is why the two channels are not interchangeable.

The fifth channel exists only on the basal branch. Its three states are linked by two one-way activated transfers,

$$c_0 \xrightarrow{\nu_{\text{col}}} c_f \xrightarrow{\nu_{\text{uf}}} c_p, \quad (25)$$

both size-gated, and both conserving vacancies exactly: the content leaving one state enters the next, and only the *number* changes, because the states store different vacancy content per unit area (Table 1) [44, 54, 55]. The chain is basal and has no prismatic counterpart: *ab initio*-parameterized formation energies place the prismatic vacancy $\langle a \rangle$ loop *below* both the compact cavity and the faulted basal loop, so the planar prismatic loop is already the ground-state geometry rather than something a compact cluster must convert into [44]. None of the basal-chain rates is calibrated; the formulation gives their Arrhenius form and no measurement in this work supplies the barriers, so the chain is switched off in the calculation of Section 7 and its consequences are discussed in Section 8.

The immobile block is what Eq. (9) becomes after the second reduction of Section 3.1 and three successive approximations, and naming them is the quickest way to say what the equations below are. Restricting Eq. (9) to indices that do not diffuse deletes the transport term outright, which is what leaves the block pointwise; within one family the surviving pairwise terms move a loop along the size axis m and do nothing else, so the family's size density $f_k(m, t)$ obeys a birth–death equation on the integer m . Treating m as continuous and truncating the Kramers–Moyal expansion at second order turns that into a Fokker–Planck equation, the first approximation, and it is second order in the jump size measured against the width of the distribution rather than in the jump size itself. Multiplying by m^j and integrating gives an *exact* hierarchy in the moments; retaining only $j = 0, 1, 2$, the three fields of Eq. (11), is the second. That hierarchy does not close, because a perimeter-limited rate carries $\sqrt{m}$ and so calls for half-integer moments, and supplying them from the log-normal Eq. (43) is the third — the only place the closure reaches the rest of the model, as the factor $\Delta_k^{-1/8}$ carried on Π_{sL}^k .

The channels that are not size jumps are not appended to that result; they are the remaining terms of Eq. (9), and the same operation $\int m^j(\cdot) dm$ produces them. Which terms those are is fixed by the partition of the index set and not by a modelling choice: the pairwise kernel $\mathcal{K}$ with one mobile partner is what becomes the ladder, and what is left over is the coagulation $\mathcal{C}$, whose partners are both immobile and which therefore moves no loop along m ; the transfer θ , which moves a cluster between families at fixed content; and the three removals $\mathcal{D}$. Each of the last four is first order in its own moments or quadratic in them, and its j -th moment integral is elementary. The coagulation gives $-\frac{1}{2} \iint \mathcal{C} f f$ at $j = 0$, *exactly zero* at $j = 1$ because a merge conserves content, and $+\iint m m' \mathcal{C} f f$ at $j = 2$; so the statement that coalescence *raises*

the second moment, which [Section 3.5](#) needs and which is the sign most easily inverted, is a property of [Eq. \(9\)](#) and not a rule imposed on the result. A removal with an n -independent rate gives $-\nu\{n, c, q\}$ and a size-gated one the same thing weighted by the gated fraction of each moment. [Appendix A](#) carries every step out, and [Table A.1](#) is the accounting that no term of [Eq. \(9\)](#) is used twice or dropped.

The result is 27 fields, pointwise, with no spatial coupling. For each family $(s, k) \in \mathcal{F}$ at each node independently,

$$\dot{n}_{sL}^k = J_{sL}^k - \Delta N - \Phi_{\min}^k \pm \Gamma_{\theta}^n - \nu_{gb} \Theta_0^k - \nu_{\text{van}} \Xi_0^k, \quad (26)$$

$$\dot{c}_{sL}^k = \dot{c}_{sL}^k|_{\text{growth}} + G_k - \Delta C - m_{\min} \Phi_{\min}^k \pm \Gamma_{\theta}^c - \nu_{gb} \Theta_1^k - \nu_{\text{van}} \Xi_1^k, \quad (27)$$

$$\dot{q}_{sL}^k = \dot{q}_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 + \Delta Q - m_{\min}^2 \Phi_{\min}^k \pm \Gamma_{\theta}^q - \nu_{gb} \Theta_2^k - \nu_{\text{van}} \Xi_2^k, \quad (28)$$

whose terms are collected in [Table 3](#). Two features of the set are worth reading off it directly. Each channel appears in all three equations with the weights 1, m and m^2 evaluated at the size where it acts — m_k^{nuc} for nucleation, $m_{\min}$ for the floor current — which is what makes the triple (n, c, q) the moments of one distribution rather than three independent fields; and the only term that is not a rate times a moment is the growth flux, which is where the mobile block enters.

Table 3: The terms of [Eqs. \(26\) to \(28\)](#), in the order they appear. Every one is derived in [Appendix A](#) as a moment of a term of [Eq. \(9\)](#). The first four come from the Fokker–Planck equation in m and its moment hierarchy; the last four are moments of the terms that are not steps along m — the coagulation, the transfer and the three removals.

Term	Given by	What it is
$J_{sL}^k, G_k, J_{sL}^k (m_k^{\text{nuc}})^2$	Eq. (21)	Nucleation, one source deposited at a single size m_k^{nuc} : cascade production G_k/m_k^{nuc} per family and, for the interstitial prismatic families, the homogeneous clustering route. $G_k = J_{sL}^k m_k^{\text{nuc}}$ is the same current weighted by m , which is why it needs no separate symbol in Eq. (28) .
$\dot{c} _{\text{growth}}$	Eq. (18)	The net biased absorption flux, Π_{sL}^k times the signed sum over the four mobile species. This is the only place the mobile block enters, and the only term that is not a rate times a moment.
$\dot{q} _{\text{FP}}$	Eq. (41)	The same flux resolved to second order: a drift that carries the mean and a spreading term that broadens the family even when the mean is stationary.
$\Phi_{\min}^k$	Eq. (A.11)	The Fokker–Planck flux out through the lower bound $m_{\min}$, booked at exactly $m_{\min}$ defects per loop, hence the weights 1, $m_{\min}$, $m_{\min}^2$. It is identically zero for a monodisperse family.
$\Delta N, \Delta C, \Delta Q$	Eq. (23)	Coalescence, both partners immobile. Loop–loop merging removes number at fixed content and <i>raises</i> the second moment; loop–network removes number and content and lowers it.
$\Gamma_{\theta}^n, \Gamma_{\theta}^c, \Gamma_{\theta}^q$	Eq. (25)	The basal-chain transfers, negative for the donor state and positive for the receiver, conserving vacancies exactly. Zero for every prismatic family.
$\nu_{gb} \Theta_j^k$	Eq. (47)	Size-gated absorption at a grain boundary: the fraction of each moment carried by loops large enough to touch the nearest face. The defects leave the crystal.
$\nu_{\text{van}} \Xi_j^k$	Eq. (24)	Dissolution of loops below the minimum stable size, the same gate in the opposite sense. The defects return to the mobile pool.

Together with the evolving network density ρ_N and the atom-balance accumulators of [Section 7.4](#), this is a system of ordinary differential equations at each node — $3 \times 9 + 1$ fields plus accumulators — in which no neighbor’s state appears. The mobile species are the only thing that couples one node to another, and freezing them, as [Section 4](#) does, removes the spatial coupling by construction rather than by approximation.

3.4. Conditions for loop growth

Each mobile species carries a diffusion tensor $D_{ij}^x = D_{0ij}^x \exp(-E_{ij}^{xm}/k_B T)$, diagonal in the crystal frame with the basal components equal and the c -axis component distinct. The whole of that anisotropy is expressed by one number per species,

$$p^m = \left(\frac{D_{\parallel}^m}{D_{\text{basal}}^m} \right)^{1/6}, \quad (29)$$

and the migration energies are generated from it at *fixed* effective diffusivity,

$$E_{\langle 11 \rangle}^m = E_{\langle 22 \rangle}^m = E_{\text{eff}}^m + 2k_B T \ln p^m, \quad E_{\langle 33 \rangle}^m = E_{\text{eff}}^m - 4k_B T \ln p^m, \quad (30)$$

so that the orientation-averaged diffusivity $\bar{D}^m = (\det \mathbf{D}^m)^{1/3}$ is exactly the scalar D_m of the mean-field kinetics. No mobility is redefined by the anisotropy — only its directional weighting. This is the technical device that makes the formulation self-consistent: Eqs. (29) and (30) are a change of variable in which p^m is the only anisotropy the model has, and Section 3.4 uses that same p^m and no other parameter.

That anisotropy is what the capture efficiencies are built from. Every capture efficiency is a product of factors, one per degeneracy to be broken:

$$Z_{sk,m}(\boldsymbol{\sigma}) = \underbrace{Z_m^0}_{\text{elastic bias}} \underbrace{A_{h(k)}(p^m)}_{\text{DAD, orientation}} \underbrace{X_{s,m}}_{\text{character}} \underbrace{\left[1 + \frac{\Delta Z_{h(k),m}^{\text{SIPA}}(\sigma_{kk})}{Z_m^0 A_{h(k)}(p^m)} \right]}_{\text{stress, SIPA}}, \quad (31)$$

with $h(k) \in \{a, c\}$ the habit of family k and the Woo diffusional-anisotropy factors [28, 29, 31]

$$A_a(p) = \frac{p + p^{-2}}{2}, \quad A_c(p) = p. \quad (32)$$

Each factor breaks exactly one degeneracy and is silent on the others. $A_{h(k)}$ depends on the habit alone — on the angle between the loop normal and the c axis — and therefore assigns one value to all six prismatic families and one to all three basal states, the pyramid included, whose base is (0001) so that $h(c_0) = c$. It separates prismatic from basal but is degenerate in character. $X_{s,m}$ is the character factor, which separates an interstitial from a vacancy loop of the same habit [55, 56]; the coexistence window it opens has log-width $\ln \chi$ with $\chi = X_{iI} X_{vV} / (X_{iV} X_{vI})$.

Equation (31) contains no free bias. $A_{h(k)}$ is built from the same p^m that Eq. (30) puts into the diffusion tensors, so a change in the transport anisotropy moves the growth laws and the sink strengths of Eq. (16) together. The two remaining constants, Z_m^0 and $X_{s,m}$, are calibrated (Section 6); the orientation dependence is not.

The forms in Eq. (32) are not monotone in the same way, and the consequence is quantitative and easily missed. $A_c(p) = p$ increases without bound, while $A_a(p) = (p + p^{-2})/2$ has its minimum at $p = 2^{1/3} = 1.2599$. A single change of p^v therefore pushes the basal and prismatic families in *opposite* directions, starving $\langle c \rangle$ growth while accelerating $\langle a \rangle$ shrinkage, or the reverse. Any statement about the effect of anisotropy that does not distinguish the two habits is incomplete.

With the efficiencies in hand, the question of which families can grow at once becomes a question about one number. Whether a family grows reduces to a single number, and that is the difficulty this subsection resolves. Each compares one flux ratio against a threshold of its own, so under identical local conditions an interstitial and a vacancy loop of one orientation cannot both grow — yet both are observed. What follows identifies which idealization must fail, shows that diffusional anisotropy alone cannot be it, and gives the two mechanisms that can. It is also what forces the formulation to be self-consistent in sense (a) of Section 1.

Define the mobile flux aggregates and their ratio,

$$\Phi_I = \sum_{n \in I_{\text{mob}}} n \bar{D}^{I_n} c^{I_n}, \quad \Phi_V = \bar{D}^v c^v, \quad x \equiv \frac{\Phi_V}{\Phi_I}, \quad (33)$$

reducing to $\Phi_I = \overline{D}^i c^i$, $\Phi_V = \overline{D}^v c^v$ in the monomer approximation. With $Z_{s,m}$ the capture efficiency of family s for species m , the driving force on a family of character sign $\eta_s = \pm 1$ is

$$F_s = \eta_s (Z_{s,I} \Phi_I - Z_{s,V} \Phi_V), \quad (34)$$

so each family grows or shrinks according to which side of its own threshold x falls on:

$$\text{interstitial family } s \text{ grows} \iff x < \frac{Z_{s,I}}{Z_{s,V}}, \quad \text{vacancy family } s \text{ grows} \iff x > \frac{Z_{s,I}}{Z_{s,V}}. \quad (35)$$

That is the whole of the growth problem, and it is uncomfortable. Every family reduces to *one* threshold, the ratio of its own two efficiencies, and x is a single number: whichever side of a shared threshold x lies on, one of the families sharing it is shrinking. Yet both prismatic characters are observed together in irradiated Zr [32–34, 55]. Coexistence therefore says some idealization has failed — homogeneity, identical capture coefficients, nucleation pathways, variants, or irradiation history.

Diffusional anisotropy removes the first. With Eqs. (29) and (32) the prismatic and basal thresholds separate,

$$x_a = \frac{Z_I^0 A_a(p^i)}{Z_V^0 A_a(p^v)}, \quad x_c = \frac{Z_I^0 A_c(p^i)}{Z_V^0 A_c(p^v)}, \quad (36)$$

and since $A_a(p) = (p + p^{-2})/2$ exceeds $A_c(p) = p$ for $p < 1$ and equals it at $p = 1$, $x_c < x_a$ whenever $p^i < p^v$. Prismatic interstitial and basal vacancy loops then grow *together* throughout

$$\boxed{x_c < x < x_a}, \quad \ln \frac{x_a}{x_c} = \ln \frac{g(p^v)}{g(p^i)}, \quad g(p) \equiv \frac{2}{1 + p^{-3}}, \quad (37)$$

with g strictly increasing, so the window opens exactly when $p^i < p^v$ and closes as $p^i \rightarrow p^v$; for $p^v \simeq 1$ its width is $(Z_I^0/Z_V^0)(1 - (p^i)^3)/2(p^i)^2$. At 200 °C it is open but narrow, $x_c \approx 1.0 < x < x_a \approx 1.2$, at $Z_I^0/Z_V^0 \approx 1.05\text{--}1.15$ and $p^i \approx 0.89\text{--}0.92$ [23, 24, 27, 28].

Three things are routinely conflated. The elastic bias opens nothing: Z_I^0/Z_V^0 multiplies *both* thresholds in Eq. (36) and cancels from the width, though it narrows the window by pushing x_c toward unity. The production bias positions x inside it — the excess of freely migrating vacancies left when more of the interstitial yield is locked into in-cascade clusters, $\eta = \varepsilon_V/\varepsilon_I \gtrsim 1$, of which 0–20% suffices. Neither substitutes for the other: at $p^i = p^v$ no η satisfies both conditions, and with η misplaced the model grows one family and dissolves the other. Third, anisotropy is degenerate in character. Equation (31) gives every family sharing a habit one factor $A_{h(k)}(p^m)$, so the interstitial and vacancy prismatic variants share a threshold,

$$\frac{Z_{a_k I, I}}{Z_{a_k I, V}} \xrightarrow{\text{DAD alone}} \frac{Z_{a_k V, I}}{Z_{a_k V, V}}, \quad (38)$$

the prismatic window closes identically, and the mechanism is silent on exactly the coexistence observed. Two mechanisms remove that degeneracy, and the formulation carries both.

The first, Resolution 1, promotes x to a field $x(\mathbf{x}, t) = \Phi_V(\mathbf{x}, t)/\Phi_I(\mathbf{x}, t)$. Interstitials migrate far faster than vacancies, so their channel responds first to any transient or gradient. In *time*, Zr’s nucleation transient of order 1 dpa is interstitial-rich, so $x \ll x_c$ and interstitial loops grow while vacancy loops nucleate and sit sub-critical; as the vacancy supersaturation builds, x rises into (x_c, x_a) and the vacancy families grow alongside those already established. In *space*, sinks — boundaries, surfaces, precipitates, the network — make the fields non-uniform, and the longer interstitial diffusion length drains interstitials to the far field while vacancies pile up locally, so x is depressed away from sinks and raised near them, and one specimen holds regions on both sides of the threshold; Section 7.2 measures that structure, boundary layers of 8–46 nm differing by species. It costs nothing — a spatially resolved formulation carries $x(\mathbf{x}, t)$ natively — but leaves $Z_{a_k I, \cdot} = Z_{a_k V, \cdot}$ intact and evaluates the two conditions at different points of space and time, so it predicts *segregated* coexistence, not local simultaneous growth.

The second, Resolution 2, removes the degeneracy itself. March-Rico, Blondel and Wirth [55] compute the defect-loop interaction energy atomistically and replace geometric intersection by a drift-diffusion capture,

$$\mathbf{J}_\alpha = -D_\alpha \left[\nabla c^\alpha + \frac{c^\alpha}{k_B T} \nabla E_{\text{int}}^{\alpha L} \right], \quad \alpha = i, v, \quad (39)$$

which reduces to effective capture radii entering additively in the reaction volume. *Spontaneous* radii are nearly equal for all loops and separate nothing; *thermal-drift* radii do, an interstitial loop's dilatational field attracting interstitials over a longer range than vacancies and conversely for a vacancy loop. That bias is the factor $X_{s,m}$ of Eq. (31), and with the character-splitting factor

$$\chi \equiv \frac{Z_{a_k I, I} / Z_{a_k I, V}}{Z_{a_k V, I} / Z_{a_k V, V}} = \frac{X_{iI} X_{vV}}{X_{iV} X_{vI}}, \quad (40)$$

the prismatic window $Z_{a_k V, I} / Z_{a_k V, V} < x < Z_{a_k I, I} / Z_{a_k I, V}$ is non-empty if and only if $\chi > 1$, of logarithmic width $\ln \chi$ — the character analogue of Eq. (37), which measures how far anisotropy separates the two habits where $\ln \chi$ measures how far thermal-drift capture separates the two characters *within* one. Because $\chi > 1$ opens a genuine interval, one uniform steady x satisfies both conditions at once — the stronger and more falsifiable claim.

Two further ingredients are not optional. In-cascade vacancy clustering supplies the embryos the capture bias acts on, homogeneous nucleation against the prevailing interstitial bias being prohibitively slow; and the *same* asymmetry generates η , so the two must not be fitted independently. Cascade overlap reduces the generation rate with dose, $\tilde{g}(\gamma) = \varepsilon(\gamma)g$, saturating over 0.01–0.1 dpa, which lets coarsening rather than unbounded nucleation dominate at high dose; as one scalar on both species it rescales dose without moving x , a rate effect and not a threshold effect. Table 4 summarizes the division of labor.

Table 4: What each ingredient does to the growth criterion Eq. (35). Only the last two separate loops of the same prismatic orientation; only character-resolved capture does so locally and instantaneously.

Ingredient	Acts on	What it does, and does not do
Elastic bias Z_I^0/Z_V^0	both thresholds together	Shifts x_a and x_c jointly; opens no window, and narrows the prismatic-interstitial/basal-vacancy one by pushing x_c toward unity.
Diffusional anisotropy, $p^i < p^v$	orientation	Opens the prismatic-interstitial/basal-vacancy window, width Eq. (37). Cannot separate loops of one orientation, Eq. (38).
Production bias η	position of x	Places x inside whatever window exists; creates none. At $x_a = x_c$ it merely switches which family grows.
Flux field $x(\mathbf{x}, t)$	position of x , resolved in space and time	Meets the two prismatic conditions at different doses or places, so coexistence is <i>segregated</i> . Native to a spatially resolved model.
Thermal-drift capture, $\chi > 1$	character	Opens the prismatic window, width $\ln \chi$, giving <i>local</i> simultaneous growth. Needs in-cascade embryos, and x positioned inside.

Two requirements follow, and together they are the content of Sections 2 and 3. The immobile families must be crystallographic: a scalar aligned/non-aligned split says *how much* of a population a stress favors but not *which* prismatic system, and means nothing under a multiaxial stress; resolving a_1, a_2, a_3 makes the resolved stress on each habit computed, and carrying both characters on each variant makes Eq. (40) representable. Splitting the basal population into faulted and perfect states is forced separately, by the different vacancy content per unit area the two store (Table 1).

And the mobile field must be spatially resolved, Resolution 1 being unavailable to a 0-D model: it needs $x(\mathbf{x}, t)$, hence Eq. (12) with real boundary conditions. That is one of two independent reasons for the spatial treatment, the other being that surfaces and grain boundaries are the dominant defect sink in any specimen small enough to examine (Table 10).

Two cautions remain, so neither is mistaken for a prediction. The thermal-drift radii behind Eq. (40) scale as $(T_m/T)^{m_d}$ with $T_m = 2128$ K, so for $m_d > 0$ the same-type bias is *strongest at low temperature* — while the observed vacancy $\langle a \rangle$ fraction *increases* with temperature. That is no contradiction — the fraction is controlled largely by nucleation, vacancy mobility and thermal emission from small vacancy clusters, none in χ — but reproducing it tests the combined model rather than coming for free. The related hazard is double counting: the same-type bias *is* an elastic-interaction effect, and so is the dislocation bias $Z_I^0/Z_V^0 > 1$, and only one should carry the interstitial–vacancy asymmetry. The calibration of Section 6 fits $\chi = 1$, so Z^0 carries it alone — which is to say the present parameter set exercises Resolution 1 and not Resolution 2.

3.5. The size distribution and its closure

Carrying only n and c assigns every loop at a point the same size, and the population that produces is monodisperse by construction. The second content q_{sL}^k removes that restriction. Its Fokker–Planck part is

$$\dot{q}_{sL}^k|_{\text{FP}} = 2\Pi_{sL}^k \left[\underbrace{\Sigma_k^{\text{net}} \bar{m}_{sL}^k \Delta_k^{3/8}}_{\text{drift}} + \underbrace{\frac{1}{2} \Sigma_k^{\text{gro}} \Delta_k^{-1/8}}_{\text{spreading}} \right], \quad (41)$$

to which every other channel of Eq. (9) contributes its own second moment, weighted by m^2 where Eqs. (26) and (27) weight it by m^0 and m^1 :

$$\dot{q}_{sL}^k = \dot{q}_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 + \Delta Q - m_{\min}^2 \Phi_{\min}^k \pm \Gamma_{\theta}^q. \quad (42)$$

where Γ_{θ}^q is the second moment of the basal-chain transfer of Eq. (25), negative for the donor state and positive for the receiver and zero for every prismatic family; it and its counterparts Γ_{θ}^n and Γ_{θ}^c are listed with every other term of the block in Table 3. The spreading term does not vanish when the family stops growing: at $\Sigma_k^{\text{net}} = 0$ the drift term is zero and $\dot{q}_{sL}^k = \Pi_{sL}^k \Sigma_k^{\text{gro}} \Delta_k^{-1/8} > 0$, so the distribution keeps broadening at fixed mean size and fixed number. There is no value of the two lower moments that expresses that, which is the argument for carrying the third.

Closing Eq. (42) with a log-normal $f_k(m)$ determines its two parameters from the three carried moments,

$$\bar{m}_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \Delta_k \equiv \frac{q_{sL}^k n_{sL}^k}{(c_{sL}^k)^2} = e^{s_k^2} \geq 1, \quad s_k^2 = \ln \Delta_k. \quad (43)$$

$\Delta_k \geq 1$ is Cauchy–Schwarz on a non-negative measure, so $\Delta_k < 1$ is not a loose tolerance but a defect: it says the three carried numbers are not the moments of any distribution. Every channel of Eq. (9) respects it automatically except the two that carry neither a distribution nor a gate, and Appendix A states what those two need. The term most easily got wrong is coalescence, and the reason is worth separating from the remedy: a coalescing loop does not leave the family, it *merges*, so fewer loops holding the same content are *larger* loops and coalescence *raises* q . That is not a rule but a consequence — the second moment of a coagulation operator has a non-negative integrand, Eq. (A.19) — and debiting q by the removed number times $\langle m^2 \rangle$, the natural-looking reading, inverts the sign of the largest single term in the equation.

The closure reaches the rest of the model in exactly one place, as $\Delta_k^{-1/8}$ on the sink prefactor, and it reaches the discrete conversion of Section 5 as the distribution loop sizes are drawn from.

3.6. Summary of the equations

Both blocks having now been assembled, this subsection is a reference rather than a development: it states what is integrated, in one place, so that the structure of the whole system can be read without following the derivation again.

Table 5: The two blocks of the integrated system. The difference in structure between them is the whole of the numerical method: the mobile block is one global nonlinear problem with no time derivative, and the immobile block is a set of independent ordinary differential equations, one per node.

	Mobile block	Immobile block
Fields	$\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$	$(n_{sL}^k, c_{sL}^k, q_{sL}^k)$ for each of the nine families
Given by	Eq. (12), closed by Eq. (17)	Eqs. (26) to (28), terms in Table 3
Structure	steady: no time derivative at all	transient: first order in dose
Coupling	every node to its neighbors, through $\nabla \cdot \mathbf{J}$ alone	none; no neighbor's state appears
Character	nonlinear, algebraic; solved by Newton on the finite-element space (Appendix B)	stiff, pointwise; integrated independently at each node
Each enters the other through	$\mathcal{D}$, Eq. (14), evaluated from the immobile state held fixed	the growth flux, Eq. (18), the only term that is not a rate times a moment

Two things follow from that contrast and are used throughout what remains. The mobile species are the only quantity that couples one node to another, so freezing them removes the spatial coupling by construction rather than by approximation — which is what Section 4 exploits, and why the immobile side of a march is a large number of small independent problems instead of one large coupled one. And the two blocks meet in exactly two places, the absorption matrix $\mathcal{D}$ and the growth flux, which are built from one set of capture efficiencies $Z_{sk,m}$ and debit the mobile species by precisely what the loop content gains. That single set of efficiencies, used on both sides of the exchange, is the operational content of the word *self-consistent* as this work uses it.

4. Numerical Solution by Operator Splitting

The system of Section 3.6 on the mesh of Section 7.1 carries 2 446 659 immobile and 362 468 mobile degrees of freedom, and as one coupled stiff system it would be hopeless. Three properties compound. It is stiff: recombination acts on microseconds and the loops evolve over 10^7 s, so an implicit integrator is forced and must factorize a Jacobian. It is large: that factorization is $O(N^3)$ at $N = 2\,809\,127$, and sparsity softens without rescuing it, a three-dimensional finite-element Jacobian still going as $\sim O(N^2)$ with fill as $O(N^{4/3})$. And it is long: a march to 10 dpa at 10^{-7} dpa s $^{-1}$ is 10^8 s, over which an adaptive integrator takes $\sim 10^2$ internal steps per unit of slow dynamics.

What makes the problem hard is also what resolves it. The mobile species reach quasi-steady state on $\tau_M \sim (k^2 D)^{-1}$, of order microseconds at 573 K, while the immobile families evolve on the dose scale — thirteen decades away. On the slow manifold they are therefore slaved: writing the two blocks as $\dot{\mathbf{c}}_M = \varepsilon^{-1} \mathcal{R}_M(\mathbf{c}_M; \mathbf{Q})$ and $\dot{\mathbf{Q}} = \mathbf{g}(\mathbf{Q}; \mathbf{c}_M)$ with $\varepsilon = \tau_M / \tau_Q \ll 1$, the singular limit $\varepsilon \rightarrow 0$ replaces the first by the algebraic constraint $\mathcal{R}_M(\mathbf{c}_M^*; \mathbf{Q}) = \mathbf{0}$. The spatial code implements that limit exactly: its mobile solver carries no time derivative, so Eq. (12) is the steady problem itself, not a transient taken to large times.

The two blocks are advanced by a Lie–Trotter split. Over one dose interval, with $\Delta t = (\gamma_{n+1} - \gamma_n)/G$, the scheme alternates four steps:

- S1. Solve the elastic problem with the current inelastic deformation for $\sigma_n(\mathbf{x})$ — constant here, retained for generality.

- S2.** Assemble the local coefficients from $\mathbf{Q}_n(\mathbf{x})$ and σ_n : the diffusion tensors, the efficiencies of Eq. (31), the sink operator of Eq. (14), $\mathbf{R}_2$, the variant weights and the boundary data.
- S3.** With $\mathbf{Q}_n$ fixed, solve $\mathcal{R}_M(\mathbf{c}_M^*; \mathbf{Q}_n) = \mathbf{0}$ for the quasi-steady mobile field — finite element, Newton, spatially coupled.
- S4.** At every node, integrate Eqs. (26) to (28) over Δt with the four mobile values frozen at $\mathbf{c}_M^*$ — pointwise and decoupled.

Two properties of that loop are easily lost, both found by their failure modes. The fast step must be inside it: replayed instead from a recorded calculation, $\mathbf{c}_M$ never responds to the immobile state the march is building, so a state away from quasi-steady state can never relax — and any spatially uniform seed is such a state, carrying no boundary layer where the true $\mathbf{c}_M^*(\mathbf{x})$ is pinned to equilibrium on every Dirichlet face. And solver settings must be shared across compared routes: the spatial code’s default makes the mobile solve a projected Newton iteration capping out at 53.6% error in c^i , so it is disabled throughout — left on one side only, a comparison measures that defect, not the physics.

With $\mathbf{c}_M^*(\mathbf{x})$ held fixed the immobile equations at different nodes are independent — not an approximation made for speed, but because the only thing coupling neighboring points is diffusion of the mobile species. The step reuses the same stiff integrator with a flag that zeroes the four mobile derivatives *after* every reaction term has been evaluated, so the immobile equations, the accumulators and ρ_N all read those values as frozen parameters. Each node is one case of an OpenMP batch with its own workspace, and identical states are deduplicated before dispatch — a measured 1.24 at the end of the march, higher early on when much of the domain shares one state. The integrator was chosen by measurement: CVODE with backward differentiation formulas and a dense solver at $\text{rtol} = 10^{-6}$, $\text{atol} = 10^{-20}$, whose error against a tight reference is 3.0×10^{-4} , where a Krylov solver gives 1.7×10^{-2} and the Runge–Kutta tables designed as IMEX pairs fail outright in pure implicit mode. The frozen system is 36-dimensional, so a dense factorization is $\sim 1.6 \times 10^4$ flops and sparsity offers nothing.

That decoupling makes the cost bearable. The slow step integrates 3 262 212 scalar ordinary differential equations per substep — but not as one system of three million. It is 72 928 independent systems of 36, and the distinction is worth ten orders of magnitude:

one implicit factorization	flops
split, $72\,928 \times 36^3$	3.4×10^9
coupled, $(90\,617 \times 31)^3$	2.2×10^{19}
ratio	6.5×10^9

Stiffness localizes the same way: each point’s system genuinely is stiff, of order 100 internal steps per substep, which is why BDF, but its cost scales with the dimension of the coupled block, not with how many blocks there are. The coupling did not vanish, then; it moved to where it is affordable. The tightly coupled problem is now the fast solve, 362 468 unknowns and 4 m 12 s on 24 cores against 1 m 12 s for an entire slow sweep over all 90 617 points, and what the split buys is frequency: with no time derivative in Eq. (12) it is solved eight times over the march rather than every timestep. At five Newton iterations per solve, hence ~ 50 s per linear solve, the counterfactual is $100 \times 24 \div 5 \approx 480$ factorizations, or 6.7 h for the mobile block *alone*, against the 29 min the slow side actually took — and it is linear in node count, as the structure predicts.

The bill arrives as splitting error. Freezing $\mathbf{c}_M^*$ over a substep ignores the drift of $\mathbf{Q}$, and hence of the sinks, within it — a first-order Lie–Trotter error, $O(\Delta\gamma)$ per step and one-sided: with strengthening sinks the loops are slightly over-grown. The accuracy dial is therefore the fast-solve cadence, not the dose-snapshot spacing — the march takes n_{sub} substeps per interval and re-solves the mobile field every n_{fem} of them. Two consequences follow, again from failure modes: the cadence counter must run over the whole march, since counted within an interval it goes silently inoperative whenever that interval holds fewer substeps than the cadence, and the substep count is fixed per dose interval rather than by a wall-clock cap on Δt , a 5 dpa

interval here being 5×10^7 s against which a 5×10^4 s cap would have spawned a thousand subprocesses. Three refinements go unused: a step cap on $\max_q \Delta \langle k^2 \rangle / \langle k^2 \rangle$; Strang splitting, $O(\Delta \gamma^2)$ for one extra fast solve; and an in-step Picard iteration between S3 and S4, which removes the error entirely.

Table 6 reports the measured cost on 24 cores, and two readings matter. The fast and slow sides cost nearly the same, 52% against 45%, the balance the cadence $n_{\text{fem}} = n_{\text{sub}} = 3$ was chosen to produce, so raising it buys accuracy at nearly linear cost in the fast side alone. And one point-integration costs $994 \mu\text{s}$ for 36 stiff equations over a dose interval, which is what makes a million of them per substep an hour rather than a week.

Table 6: Measured cost of the 500 nm hexagonal single-crystal march of Section 7, on 24 cores. The state carried is 40 wide — 4 mobile, 27 immobile, 6 conservation accumulators, the network density, and the two grain-boundary accumulators of Section 5 — of which 36 are integrated, the four mobile values being held fixed over each substep.

Quantity	Value	
CD nodes	90 617	
mobile field, 4 species	362 468 dof	
immobile field, 9 families $\times$ 3 moments	2 446 659 dof	
elastic displacement	271 851 dof	
state carried per point	40	
integrated per point (= Newton block)	36	
substeps	24	
points integrated per substep, after deduplication	72 928 of	90 617
deduplication factor	$\times 1.24$	
point-integrations over the march	1 750 272	
scalar ODEs integrated over the march	7.8×10^7	
fast solves ($8 \times$ finite-element Newton)	33 m 42 s	(52%)
slow substeps ($24 \times$ batch CVODE)	29 m 00 s	(45%)
checkpoint writes (32)	18.2 s	(0.5%)
march total	1 h 04 m 37 s	
one fast solve	4 m 12 s	
one slow substep	1 m 12 s	
one point-integration	994 μs	

5. Continuum-to-Discrete Self-Consistency

This section is sense (b) of Section 1: where sense (a) is a statement *inside* the continuum model, this one is *between* two descriptions of a microstructure, that the map from continuum fields to discrete loops preserves what makes them the same microstructure. The two are independent — a model may generate every efficiency from its diffusion tensors and still convert to a population storing the wrong number of defects.

The second is needed because every cluster-dynamics formulation is a mean-field theory of a population of discrete objects. Its kernels assume absorbers placed independently and at random, so that overlap depends on the volume fraction alone — a dilute-limit statement known to degrade as the swept fraction becomes appreciable [12, 14, 17]. Three mechanisms take it apart, at different doses. Coalescence destroys spatial independence, a merged loop occupying the union of two former positions, so the Avrami estimate stops estimating anything. Local elastic interaction breaks the far-field assumption behind Eq. (31), two loops within a few radii sharing a stress field whose drift term is not the sum of the two isolated ones [55, 56]. And gradients break the premise that the diffusion field has relaxed between absorbers — the failure Section 1 describes, which spatial resolution repairs for the mobile species but not for the loops.

The model detects the first itself, which is the useful property: φ_{LL} of Eq. (22) is its own estimate of the probability that a loop has met another, and approaching unity it is of order its own maximum and carries

no information. The coarsening dose

$$d_{\text{coarsen}} = \min\{\gamma : \varphi_{LL}(\gamma) \geq \varphi^* \text{ on a fraction } \geq f^* \text{ of interior nodes}\} \quad (44)$$

is therefore read off one of the model's own state variables rather than imposed from outside, at no cost. Which family reaches it first is a property of the formulation and not the material: between calculations differing only in whether the efficiencies come from the diffusion tensors or from fitted constants, the family that coarsens first swaps.

Repairing the dilute limit is not the only reason to want a discrete representation; it is also the interface to micromechanics. Dislocation dynamics computes the stress a population exerts and its resistance to a gliding line, a channeling model needs which obstacles a channel has swept, a fracture model the local obstacle spacing — and none takes a density as input, but positions, radii, Burgers vectors and habit planes. The objective is therefore stronger than a one-way export: that the two representations be equivalent within stated tolerances, so a calculation may run in either, or in a hybrid, with the choice made on dose and cost rather than physics. That is what attaches conservation laws to the conversion instead of making it a rescaling.

Those requirements cannot all be met exactly, and their ranking is a modeling decision:

1. Stored defects, exactly: $\sum_{\ell} m_{\ell}$ must equal the continuum content of the region the loops replace, a transfer that loses defects appearing downstream as a spurious sink.
2. Loop number, to the integer draw: a continuum density gives a real expected count and a discrete population an integer one, so the best available is $|N_{\text{draw}} - \Lambda| \leq \frac{1}{2}$.
3. The dispersion Δ_k , to the truncation of the size sampling — a requirement that did not exist before [Section 3.5](#), a monodisperse family having nothing to reproduce.
4. Sink strength, reported but not asserted on: the line length a discrete population presents differs from ρ_{sL}^k for physical rather than numerical reasons, ellipticity among them.

Only the eight loop families convert. The stacking-fault pyramid has no Burgers vector and no line, hence no discrete representation at all; folding its content into c_f would report basal vacancies as dislocation content the solver was then free to move.

The conversion proceeds in four steps — a nodal volume for every node, the number of loops each is owed, their sizes and shapes, and what the domain refuses to hold — and [Appendix C](#) sets them out in full, with the tessellation and the algorithm drawn in [Figs. C.1](#) and [C.2](#). Two of its consequences belong here, because they bear on how the transferred population may be read. The first is a limit on resolution rather than on accuracy: the continuum density is recovered exactly in the mean over the draw, but a family whose total is a few hundred loops cannot express structure finer than the spacing those few hundred points support, which is why [Section 7](#) reads the conversion as a check on the mean field rather than as a prediction of where individual loops lie. The second is a hard feasibility bound. A discrete-dislocation solver silently declines a loop whose nodes leave the grain, so the conversion predicts that refusal and keeps the refused share in the continuum, counted in defects rather than in loops — on one measured transfer, one refusal in six was 83% by count and 98% by stored defects. On the geometry of [Section 7](#) that makes the basal population effectively infeasible to transfer above ~ 0.1 dpa, a $\langle c \rangle$ loop of mean radius 25 nm overhanging any face it lies within 25 nm of.

The grain boundary requires two treatments, the objects differing in kind. For the mobile species it is a Dirichlet condition: [Eq. \(17\)](#) pins the concentrations at thermal equilibrium, absorption is the flux $-\mathbf{D}^x \nabla c^x \cdot \mathbf{n}$, and the depleted layer is solved for rather than parameterized, its width set by the local diffusivity and sink density and evolving through the transient. A loop cannot be given a boundary condition, being untransported with no flux to set. What it has is a size, and one of radius $R^k = \lambda_k \sqrt{m}$ centered a distance x from the nearest face intersects it when

$$m \geq m_{gb}(x) = \left(\frac{x}{\lambda_k} \right)^2, \quad (45)$$

so the absorbed fraction of each moment is the mass of the family's size distribution above that threshold. With $\Phi_k^{(j)}(m_\theta)$ the fraction of the j -th moment of Eq. (43) above m_θ ,

$$\Phi_k^{(j)}(m_\theta) = \frac{1}{2} \operatorname{erfc}\left(\frac{\ln m_\theta - \mu_k - j s_k^2}{s_k \sqrt{2}}\right), \quad (46)$$

the three removals of Eqs. (26) to (28) are

$$\Theta_0^k = \Phi_k^{(0)}(m_{gb}) n_{sL}^k, \quad \Theta_1^k = \Phi_k^{(1)}(m_{gb}) c_{sL}^k, \quad \Theta_2^k = \Phi_k^{(2)}(m_{gb}) q_{sL}^k, \quad (47)$$

at a rate ν_{gb} taken large, so the touching fraction is removed essentially instantaneously and the denuded-zone width is a prediction rather than a rate parameter.

Three properties of Eq. (47) are worth stating. It is automatically realizable: what remains after removing the mass above a threshold is a genuine sub-measure, so $\Delta_k \geq 1$ holds by construction. The defects do not return to the pool, the whole difference between it and the dissolution channel that credits freed vacancies back to $\mathbf{c}_M$: a loop swallowed by a free surface takes its defects out of the crystal, so Eq. (47) is booked to a grain-boundary ledger alone. Adding it to the dissolution term too would return the atoms twice, a conservation residual of the wrong sign the only symptom. And the distribution must be truncated at the crystal, the log-normal having unbounded support where the crystal has not: uncapped, the far tail of Eq. (43) holds loops larger than the grain, which Eq. (45) faithfully absorbs at every distance, reporting absorption at a grain center no loop can reach. Capping at $m_{\max} = (R_{\max}/\lambda_k)^2$ with $R_{\max}$ the inradius replaces Eq. (46) by the mass between threshold and cap over the mass below it,

$$\tilde{\Phi}_k^{(j)}(m_\theta) = \frac{\Phi_k^{(j)}(m_\theta) - \Phi_k^{(j)}(m_{\max})}{1 - \Phi_k^{(j)}(m_{\max})}, \quad (48)$$

which reduces to $\Phi_k^{(j)}$ exactly as $m_{\max} \rightarrow \infty$. The carried moments are still inverted for (μ_k, s_k) as if untruncated, so closure and support disagree at order $\Phi_k^{(j)}(m_{\max})$; reconciling them would mean inverting a truncated log-normal from truncated moments, which has no closed form. The inconsistency is bounded by the discarded tail, which is the tail with no business existing. Measured on Section 7, truncation leaves the near field unchanged to four digits, sets the absorption rate 1 μm from a boundary to zero where it had been finite, and raises it at 200 nm by 4%.

Whether this holds together is settled by auditing every transfer on three totals, each summed over the crystal whichever representation holds them:

(I) loop number	$N = \sum_j n_{sj}^k V_j + \#\{\text{discrete loops}\},$
(II) stored defects	$C = \sum_j \frac{c_{sj}^k}{\Omega} V_j + \sum_\ell \frac{\pi R_\ell^2 (\mathbf{b} \cdot \mathbf{n})_k}{\Omega},$
(III) sink strength	$S = \sum_j 2\pi R_{sj}^k n_{sj}^k V_j + \sum_\ell 2\pi R_\ell.$

(49)

(I) and (II) are conservation statements, holding to the sampling tolerance of one loop and to machine precision. (III) is a modeling statement, and the one that catches mistakes: a discontinuity in it means the two representations disagree about what a loop of a given content is, a crystallography error rather than a numerical one. The residual sink discontinuity fell from a factor of 4.85 to 2% by removing an inconsistency in the $\langle c \rangle$ Burgers vector — the continuum sizing a basal loop with the full $[0001]$ where the discrete side used $\frac{1}{2}[0001]$ — and not by tuning.

One constraint follows: a transfer is all-or-nothing per family. The continuum carries one mean size per family per node, so a size-selective transfer is not representable. More importantly, a family discrete *and* still present in the continuum is a sink twice over, once through its Green's function and once through Eq. (16), and a loop whose own sink remains in the continuum samples its own depletion field at its own line. Zeroing the transferred family removes both, and they are the same fault.

6. Calibration Against Experiment

The formulation of [Section 3](#) contains several classes of input, and confusing them is the commonest way to misreport a calibration. [Table 7](#) separates them by what a change to each *does*; only the last class is fitted here, the model controls selecting which physics is active, the crystallography being measured, and the transport generated from one anisotropy parameter per species.

Table 7: Classes of model input, by what a change to each affects.

Class	Example	What a change does
Model controls	families and moments carried; emission model; basal chain and grain-boundary loop channel on or off	Selects which terms of Eqs. (26) to (28) are integrated at all: changing one changes the model, not its parameters.
Crystallography	$a, c, \Omega, (\mathbf{b} \cdot \mathbf{n})_k, \lambda_k$	Fixed by measurement (Table 1). Not free.
Transport	$p^m, E_{\text{eff}}^m, \nu_m, z_c$	Sets the diffusion tensors (Eq. (30)) and through them the orientation factors $A_{h(k)}(p^m)$ in every capture efficiency.
Elastic and character bias	$Z_m^0, X_{s,m}$ (through χ), Z_N	The parts of Eq. (31) that the diffusion tensors do <i>not</i> generate.
Reaction and geometry constants	$\varepsilon_k, m_k^{\text{nuc}}, c_{LL}, c_{LN}, \kappa_{LL}, \kappa_{LN}, \xi_k, m_{\text{min}}, \nu_{\text{vanish}}, m_{\text{vanish}}$	Nucleation, coalescence, sink prefactors and the small-size floor — the calibrated set.

The data are transmission-electron-microscope measurements of loop number density and mean diameter in neutron-irradiated Zr and its alloys, across twelve temperatures and four decades of dose. They divide sharply, and the division limits what can be determined ([Table 8](#)).

Table 8: Experimental targets. Coverage is very uneven: 71 $\langle a \rangle$ densities across twelve temperatures and four decades against five $\langle c \rangle$ points at two temperatures, all above 14 dpa. Nothing constrains the basal population below that, where the model still places resolvable loops; the extrapolation is revisited in [Section 8](#). High-temperature $\langle a \rangle$ diameters ($T \geq 650$ K) are excluded as vacancy loops, which the measurements do not separate from the interstitial population.

Set	Points	T (K)	Dose (dpa)	N / d given	Sources
$\langle a \rangle$	73	473–723	0.0067–24.5	71 / 64 (44 used)	Jostsons 1977, Northwood 1974/1976/1977/1979, Gilbert 1979, Harte 2017, Kobylyansky 2008
$\langle c \rangle$	5	573–583	14–35	5 / 5	Tournadre 2012, Harte 2017

Let $\boldsymbol{\theta}$ be the calibrated parameters and $\mathbf{y}_{\text{exp}}$ the measurements. The estimator minimizes

$$J(\boldsymbol{\theta}) = \sum_{f \in \{A, C\}} w_f \left[\frac{1}{|\mathcal{N}_f|} \sum_{r \in \mathcal{N}_f} \left(\log_{10} \frac{N_r(\boldsymbol{\theta})}{s_f^N N_r^{\text{exp}}} \right)^2 + w_f^d \frac{1}{|\mathcal{D}_f|} \sum_{r \in \mathcal{D}_f} \left(\log_{10} \frac{d_r(\boldsymbol{\theta})}{s_f^d d_r^{\text{exp}}} \right)^2 \right] + J_{\text{os}}(\boldsymbol{\theta}), \quad (50)$$

with s_f^N, s_f^d the transfer factors of [Section 6](#) and J_{os} a penalty on non-monotonic dose response. This is not an arbitrary least-squares choice but the maximum *a posteriori* estimator under a stated model of the data. With log-normal measurement errors — the right family for a quantity spanning four decades and reported as an order of magnitude plus a factor — and $p(\boldsymbol{\theta})$ uniform on the box of admissible values, Bayes’ rule gives

$$-\log p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}}) = \text{const} + \sum_{f,r} \frac{1}{2\sigma_f^2} \left(\log_{10} \frac{y_r(\boldsymbol{\theta})}{y_r^{\text{exp}}} \right)^2 - \log p(\boldsymbol{\theta}), \quad (51)$$

so minimizing [Eq. \(50\)](#) over the box is exactly maximizing [Eq. \(51\)](#), with $w_f \propto \sigma_f^{-2}$. The reported set is a MAP point and the weights state the relative credibility of the two data sets rather than tuning anything.

The logarithmic form matters and was not the original choice: an absolute-ratio residual $(d - d^{\text{exp}})/d^{\text{exp}}$ penalizes a model a factor of two *high* four times as heavily as one a factor of two *low*, and against a diameter the model underpredicts that asymmetry rewards shrinking. Equation (50) is symmetric under inversion of the ratio, as a ratio of positive quantities requires. What has not been done is sample the posterior: the pipeline this framing points at — a space-filling design screened by variance-based sensitivity analysis, a Gaussian-process emulator, and Markov-chain sampling of Eq. (51) on it — is tractable at 0.2–0.5 s per evaluation but has not been run, so everything below is a point estimate with no credible interval.

What that estimator is aimed at needs one further step, because it is a 0-D estimator and the calculation it feeds is 3-D, on a grain whose whole surface absorbs. The 0-D has no boundary, so the two report different populations at the same parameters, and a set landing the 0-D on the measurements lands the 3-D interior elsewhere. The factors shift the targets by the measured transfer so that the calculation, not the fit, is what agrees.

Those transfers do not point the same way for the two families, which is why one global slack would be wrong. A 3-D interior carries 0.75 of the 0-D $\langle a \rangle$ density, 2.0 times its $\langle c \rangle$ density and 0.74 of its $\langle c \rangle$ diameter. Each rests on a single calculation at a dose below the targets, and for $\langle c \rangle$ the direction is contested: a boundary that removes loops should leave an interior below the 0-D, and the run measuring it gives one above. Half the logarithm of each is therefore taken, the point least wrong under either reading — 1.15 on the $\langle a \rangle$ density, 0.71 on the $\langle c \rangle$.

The $\langle c \rangle$ diameter is pinned to the measurement outright, for a reason unrelated to transport: a basal loop of 220 nm has $r = 110$ nm in a 500 nm prism of half-width 250 nm, and the conversion of Section 5 refuses it, so the transferred population and the hardening cell both come out empty. It is a feasibility constraint on the calculation the parameters exist for, and it outranks the transfer argument; a fit allowed the full 1.35 shift reached 203–219 nm and was discarded on that ground. The diameter weight $w_f^d = 3$ follows for the same reason. With the logarithmic form both terms have the same shape, so buying density with size is a trade the estimator prefers and will make in whichever family is left unweighted — weighting only $\langle c \rangle$ moved the $\langle a \rangle$ diameter from 0.74 to 2.15 of experiment in one stage. A weight beats a tighter box, which would report as a rail and conceal whether the data wanted the value.

The transport parameters are fixed rather than fitted. In every earlier calibration the four bias parameters came to rest on their bounds, p^v at a ceiling implying $D_{\parallel}^v/D_{\text{basal}}^v = (1.6)^6 = 16.8$ — a strong claim about vacancy transport in α -Zr that nothing here supports, and the model absorbing a missing channel into a diffusion tensor, which is what Section 3 exists to prevent. The anisotropies are therefore inputs,

$$p^v = 1.02, \quad p^i = p^{2i} = p^{3i} = 0.70, \quad (52)$$

vacancies slightly anisotropic and interstitials strongly so with the basal plane fast. It satisfies the coexistence criterion of Section 3.4 with a window $g(p^v)/g(p^i) = 2.02$ against the 1.44 the fitted bias gave; it makes $Z_{\text{prism}}(i)/Z_{\text{basal}}(i) = 1.96$ against 1.16, so prismatic loops out-capture basal ones through the transport anisotropy rather than a fitted efficiency; and by Eq. (30) it asks for an interstitial migration split of 0.106 eV at 573 K, a falsifiable prediction where a railed p^v was not. Two caveats. The cluster entries do nothing — set to 0.50 or 1.00 they move every reported quantity by less than 10^{-3} , the di-interstitial jump rate being $\sim 5 \times 10^{-6}$ of the monomer's — so they are pinned with p^i for consistency, not effect, and this is where the model departs from treatments placing the anisotropy on the clusters alone. And Eq. (30) fixes energies, so a p^m quoted without a temperature is incomplete.

With the targets shifted onto the calculation and p^m pinned outside the search, the remaining parameters were fitted in stages, each a bounded Nelder–Mead search over a subset with the full set polished jointly at the end: the capture prefactors at fixed p^m ; the $\langle a \rangle$ temperature trend; the $\langle a \rangle$ level; a $\langle c \rangle$ repair, needed because two coalescence exponents are shared between the families; then the polish. Staging answers a strongly anisotropic objective and is not a claim about identifiability. A screen over the 26 candidates found nine inert here, a change over their whole range moving J by less than the convergence tolerance, and those were held fixed.

Two choices in that sequence were forced by watching it fail. Fitting the $\langle c \rangle$ removal channels before the prefactors commits the search to a coalescence basin it does not leave, ending with those coefficients on

their ceilings and a basal diameter of 8.6 nm: coalescence removes loops by shrinking the survivors where the data want fewer *and* larger, and only the small-size leak of Eq. (24) moves both the right way. And the seed must already have both diameters near experiment — started from the uncalibrated set the search reaches the same basin, $\langle c \rangle$ density correct at 1.26 and diameter at 0.21, which scores well and is useless.

The set the staged search returns is the one everything after this section uses. Table 9 lists it, Figs. 3 and 4 show it against the measurements and Fig. 5 is the parity plot over all 78 points. Agreement is the mean and standard deviation of $\log_{10}(\text{model/experiment})$, zero being unbiased and ± 0.3 a factor of two high:

Quantity	mean	s.d.	points
$\langle a \rangle$ density	-0.056	0.493	71
$\langle a \rangle$ diameter	+0.084	0.176	44
$\langle c \rangle$ density	+0.321	0.225	5
$\langle c \rangle$ diameter	-0.084	0.100	5

The $\langle c \rangle$ diameter is the row to read first, because it was obtained without being steered: at the fixed anisotropy of Eq. (52) the basal family sits at 100.7 nm against 95 nm measured and stays between 85 and 131 nm at every stage, rather than being dragged there by the diameter weight. That is the case for treating p^m as an input — a bias free to move puts basal loops wherever the density term wants them. The $\langle c \rangle$ density is 2.1 times the measurement, and five points at two temperatures, all above 14 dpa, cannot distinguish that from a fit. The $\langle a \rangle$ row is the informative one, 71 points across twelve temperatures, its scatter of $10^{0.493} = 3.1$ comparable to the spread among the measurements themselves.

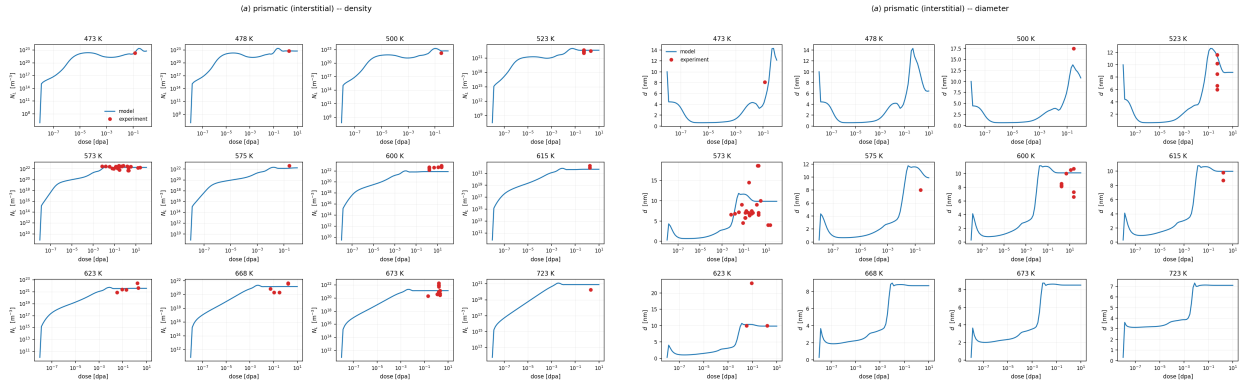


Figure 3: Prismatic $\langle a \rangle$ loops by temperature: number density (left) and mean diameter (right), 71 and 44 points.

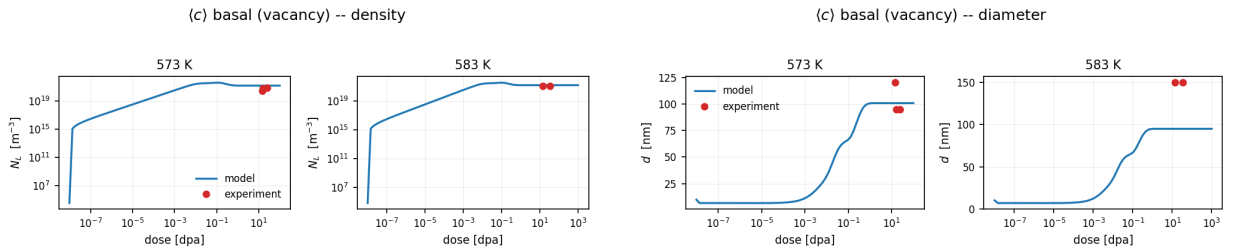


Figure 4: Basal $\langle c \rangle$ loops against the five available measurements, all above 14 dpa; below that the model extrapolates.

Two things the set does not achieve are worth separating from what it does, because both bear on how the results of Section 7 should be read. The first is that the $\langle a \rangle$ mean conceals a clear deficiency. Resolved by temperature the model runs high at both ends and low in the middle:

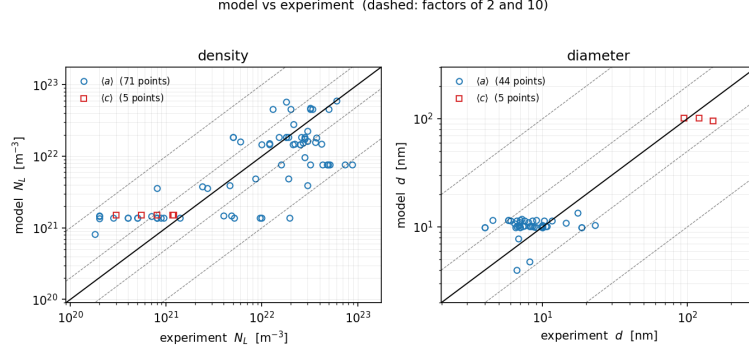


Figure 5: Parity plot over all 78 measurements. The diagonal is exact agreement and the dotted lines a factor of three.

Table 9: The calibrated parameter set. The transport anisotropies are *inputs*, Eq. (52), not fitted values. [†] rests on a bound and is a constraint rather than an optimum (Section 6); [‡] was not moved from the seed.

Parameter	Value	Parameter	Value	Parameter	Value
<i>transport, fixed by Eq. (52)</i>					
p^v	1.0200	p^i	0.7000	$p^{2i} = p^{3i}$	0.7000
<i>calibrated</i>					
Z_v^0	1.3000 [†]	c_{LL}^a	4.7720	ε_{2i}	0.0010 [†]
Z_i^0	0.8935	c_{LL}^c	1.2386	ε_{3i}	0.007376
ξ_c	2.0000 [†]	c_{LN}^a	94.615	ε_{iL}	0.000200 [†]
Z_N	1.0100 [‡]	c_{LN}^c	1.1198	ε_{vL}	0.005000 [†]
E_i^m (eV)	0.8970	κ_{LL}	20.000 [†]	m_a^{nuc}	53.79
E_{2i}^m (eV)	1.3629 [‡]	κ_{LN}	4.0000 [†]	m_c^{nuc}	400.00 [†]
E_{2i}^b (eV)	0.9570 [‡]	m_{min}	5.8286	c_{ρ_N}	0.0010 [‡]
χ	1.0000 [‡]	m_{vanish}	14.04	$k_{\rho_N}^{\text{rec}}$	$3.5 \times 10^{-5\ddagger}$
		ν_{vanish}	0.0010 [†]		

T (K)	473	523	573	575	600	623	673
N_a/N_a^{exp}	1.29	1.66	0.88	0.35	0.17	1.01	1.17
d_a/d_a^{exp}	0.59	1.37	1.34	1.38	1.17	0.77	—

Near 600 K the prismatic density is still a factor of six low, and 573–600 K is the band [Section 7](#) runs in. This is not a stage that failed to converge: four independent searches, seeded differently and carrying the levers a screen identifies as having the right temperature signature — E_i^m , the two coalescence exponents and the cascade partitions ε_{2i} , ε_{3i} , each of which lowers the low-temperature end while raising the middle — all reach the same place, the mean improving and the middle band not. The deficiency survives the change of anisotropy, the diameter weighting and its own dedicated stage. Two readings remain, and these data cannot separate them: the prismatic nucleation term may carry the wrong temperature dependence, being a cascade partition with none of its own, or the high-temperature points may not measure the same population, since above ~ 650 K the reported $\langle a \rangle$ loops include vacancy loops — already the reason their diameters are excluded from [Table 8](#). The second is testable by character-resolved measurements at 600–700 K, and is the more useful thing to ask of an experiment.

The second is that fixing p^m removed four rails and not the pressure that produced them. Four parameters in [Table 9](#) rest on the edge of the box: $Z_v^0 = 1.30$ and the basal sink scale $\xi_c = 2.00$ at their ceilings, and both coalescence exponents $\kappa_{LL} = 20$, $\kappa_{LN} = 4$ at theirs. Their pattern is more informative than any one of them. Z_v^0 and ξ_c both multiply the basal capture rate and are nearly degenerate, so they are one statement, not two: given a stated anisotropy the model cannot deliver the measured basal population without a prefactor no transport argument supplies. That locates the shortfall in the capture prefactor and not the diffusion tensor. The two κ on their ceilings say the same of coalescence — the fit wants more overlap removal than the admissible range allows. That is sharper than the earlier calibrations supported — with p^v free it absorbed the discrepancy and the diagnosis stopped at “the fit wants an implausible anisotropy”; pinned, it reappears where it cannot be mistaken for transport. Atomistic constraints on p^m would confirm or refute [Eq. \(52\)](#) directly, and [Eq. \(30\)](#) would accept them without refitting anything else.

Two caveats on carrying this set into [Section 7](#). The fit is the march’s slow step term for term — one templated function, the same binary and switches, differing only in whether the mobile species are frozen — so the parameters transfer exactly. What does not transfer is what the 0-D cannot see: there the vacancy pool is free and re-equilibrates where the march pins it to the fast solve, and the two need not move a quantity the same way. And the shifts above target a 3-D interior not yet re-measured with this set.

7. Results: a 500 nm Hexagonal Single Crystal to 40 dpa

Every result below comes from one calculation: the calibrated parameter set of [Table 9](#), all nine families, three moments each, peripheral emission active, the basal chain inactive, and the grain-boundary loop channel of [Section 5](#) active. Conditions are $T = 573$ K, $G = 10^{-7}$ dpa s $^{-1}$, zero applied stress, and thirteen dose intervals from 10^{-4} to 40 dpa. The upper limit is set by the measurements rather than by the model: the $\langle c \rangle$ data reach 35 dpa, so a calculation stopping at 10 would have to be extrapolated before it could be compared with them. Where a comparison is drawn against the same case *without* the loop channel, that companion differs in nothing else.

7.1. Geometry, discretization and system size

The domain is a hexagonal prism of 500 nm across flats and 800 nm along [0001], meshed with a refined boundary layer 100 nm deep in which the target element size falls from 60 to 20 nm ([Fig. 6](#)). The refinement is not cosmetic: [Section 7.2](#) measures interstitial boundary layers of 16–28 nm, so an unrefined mesh at 60 nm would resolve the steepest gradient in the problem with a single element.

The cluster-dynamics trial functions live on second-order elements, so the field nodes are more numerous than the mesh vertices: 90 617 of them, filling a crystal of 1.295×10^8 nm 3 , or 5.568×10^9 atoms. The resulting system sizes are those of [Table 6](#): 362 468 mobile degrees of freedom solved as one coupled algebraic system

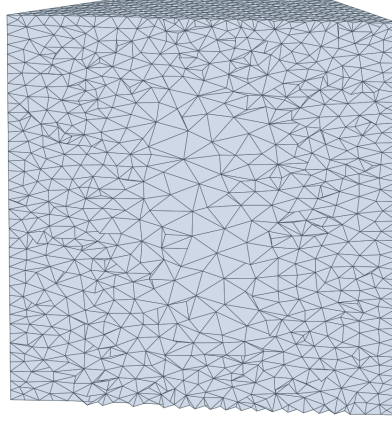


Figure 6: The finite-element mesh, cut in half so that the interior is visible: 60 922 tetrahedra on 12 581 mesh vertices. The refinement toward the six prismatic faces and the two basal faces is the boundary layer, 100 nm deep, within which the element size falls from 60 to 20 nm. Crystal axes as in Fig. 1: $[10\bar{1}0]$, $[\bar{2}\bar{1}\bar{1}0]$ and $[0001]$.

eight times, and 2 446 659 immobile ones marched as 72 928 independent 36-dimensional stiff systems at each of 24 substeps — 7.8×10^7 scalar ordinary differential equations over the march, in 1 h 04 m on 24 cores.

The whole surface is a boundary. No face is periodic, so every face carries Eq. (17), and the maximum distance from any node to the nearest face is 215 nm — the inradius, and the cap $R_{\max}$ used in Eq. (48).

7.2. Mobile defect fields

Figure 7 shows the four mobile species early in the transient (10^{-4} dpa) and at 40 dpa, on two orthogonal mid-cuts. Every panel has the same structure — an interior plateau and a depleted shell — and the differences between them are entirely in the width of that shell and in the level of the plateau.

The boundary layer is a diffusion length and it orders the species. Measuring the layer as the distance at which the binned median recovers $1 - e^{-1}$ of the interior plateau:

Dose (dpa)	C_v	C_i	C_{2i}	C_{3i}
10^{-4}	16.5 nm	27.5 nm	46.0 nm	46.0 nm
10^{-2}	14.9 nm	27.5 nm	46.0 nm	46.0 nm
10	8.0 nm	16.5 nm	18.2 nm	18.2 nm

Two readings, and both are consequences of $L \sim \sqrt{D/k^2}$ rather than of anything fitted. *Across species at fixed dose*, the layer widens with mobility: the vacancy, least mobile, is depleted over the shortest distance, while the interstitial clusters — which reach a sink from further away — carry the widest layer. *Across dose at fixed species*, every layer narrows, by a factor of two between 10^{-4} and 10 dpa, because k^2 has grown with the loop population that the same calculation is building. The boundary layer is therefore not a fixed geometric feature of the domain; it is a running measure of the sink density, and it tightens as the microstructure develops.

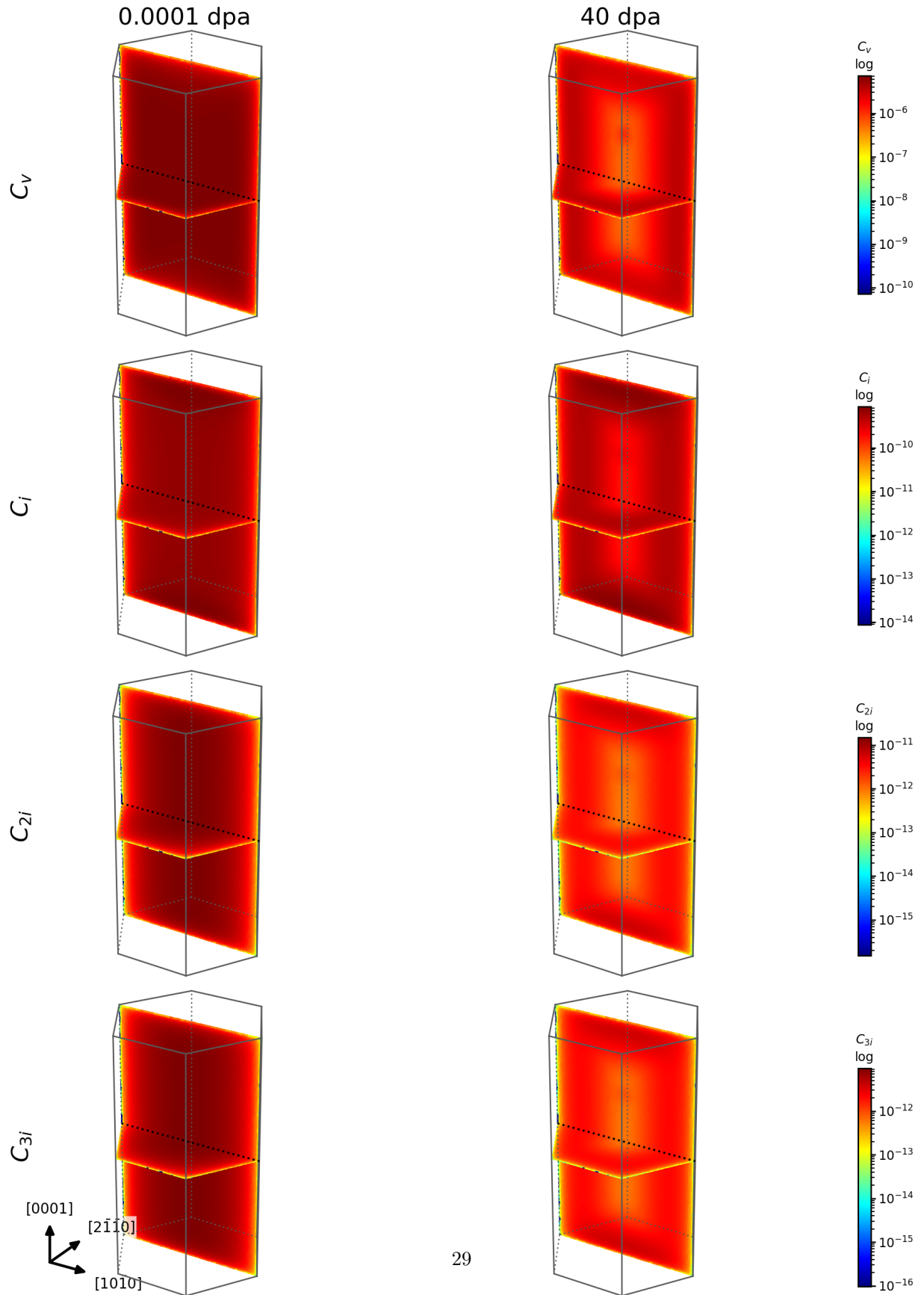


Figure 7: Mobile species at 10^{-4} dpa (left) and 40 dpa (right), on one vertical and one horizontal mid-cut. Rows, top to bottom: C_v , C_i , C_{2i} , C_{3i} ; each row has its own logarithmic color scale, so levels are comparable across dose within a row and not between rows. The interior plateau falls and the depleted shell narrows between the two doses, both because the sink density has risen.

Anisotropy acts on the shape of the layer, not only its width. Because the domain has two basal faces and six prismatic ones, the layer can be measured separately toward each set, using only the nodes for which that set is the nearer boundary:

Species	10 ⁻⁴ dpa		40 dpa	
	toward basal	toward prismatic	toward basal	toward prismatic
C_v	84.8 nm	11.7 nm	24.0 nm	6.8 nm
C_i	12.8 nm	28.8 nm	7.5 nm	18.3 nm
C_{2i}	22.0 nm	45.2 nm	6.8 nm	24.0 nm
C_{3i}	24.0 nm	45.2 nm	6.8 nm	24.0 nm

The vacancy layer is 7.2 times *wider* toward the basal faces than toward the prismatic ones, which is the sign the $p^v = 1.6$ this calculation was run at requires: vacancies are $(p^v)^6 = 16.8$ times more mobile along [0001], and a species that reaches a sink from further away is depleted over a longer distance. The scale expected from $L \sim \sqrt{D/k^2}$ is $\sqrt{16.8} = 4.1$, so the measured 7.2 is not a clean reading of the diffusivity ratio, and the interstitial rows show why. At $p^i = 1.0$ the interstitials diffuse isotropically, yet their layer is *narrower* toward the basal faces by a factor of 0.44 — an effect of domain shape rather than of transport, since a node near a prismatic face is close to several faces at once and is depleted more than its distance to the nearest one would suggest. Anisotropy and geometry therefore act in opposite directions here, and only their combination is directly observable.

Two consequences. First, an SRCD calculation on a shaped grain measures the convolution of the diffusion tensor with the grain geometry, so extracting one requires the other to be known — which is an argument for calibrating p^m atomistically rather than from a boundary-layer profile (Section 8). Second, the narrowest layer in the problem is the interstitial one toward a basal face, 7.5 nm at 10 dpa, against a 20 nm boundary element: the steepest gradient the calculation contains is resolved by well under one element, and it is at exactly such a location that an earlier calculation on this geometry drove a single node beyond the integrator’s reach.

Interior median concentrations fall over the march — C_v from 7.22×10^{-6} to 1.50×10^{-6} , C_i from 6.75×10^{-10} to 3.80×10^{-10} , and the clusters by nearly an order of magnitude — which is the signature of a rising sink density at constant production.

7.3. Immobile fields and their moments

Figures 8 and 9 show the three carried moments of the faulted basal family and of one prismatic interstitial variant, at the same two doses. The three rows are not three views of one thing; they answer different questions, and their spatial structures differ.

Number is boundary-peaked, content is interior-peaked, and that is the physics. Cascade nucleation is spatially uniform, but the only channel that removes a loop in the interior is coalescence, whose rate is driven by the absorbed mobile flux — and that flux vanishes where the Dirichlet condition pins the mobile species. Near a boundary, loops are therefore nucleated and not removed, so n rises steeply toward the surface. They are also not *fed*, because the same vanishing flux is what would grow them, so c does not rise with n and the mean size c/n collapses. Without the channel of Section 5 that shell dominates every domain average, which is the artifact Section 7.6 quantifies.

The second moment does work, and it is realizable everywhere. The dispersion $\Delta_k = qn/c^2$ of Eq. (43) must be ≥ 1 for the three carried numbers to be the moments of any distribution at all. Measured over the interior:

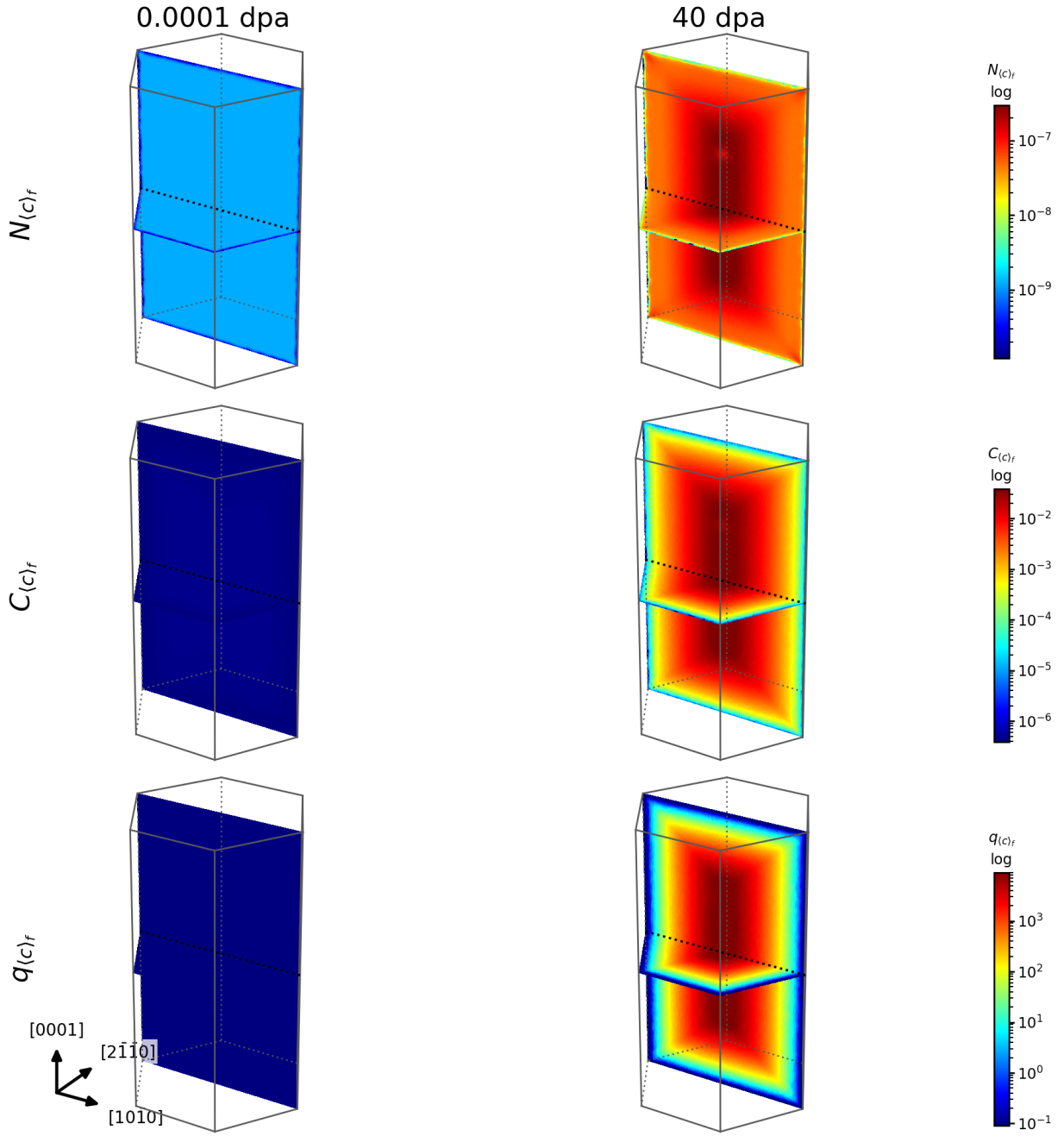


Figure 8: The basal faulted family c_f at 10^{-4} and 40 dpa : number density n (top), stored content c (middle), second content q (bottom).

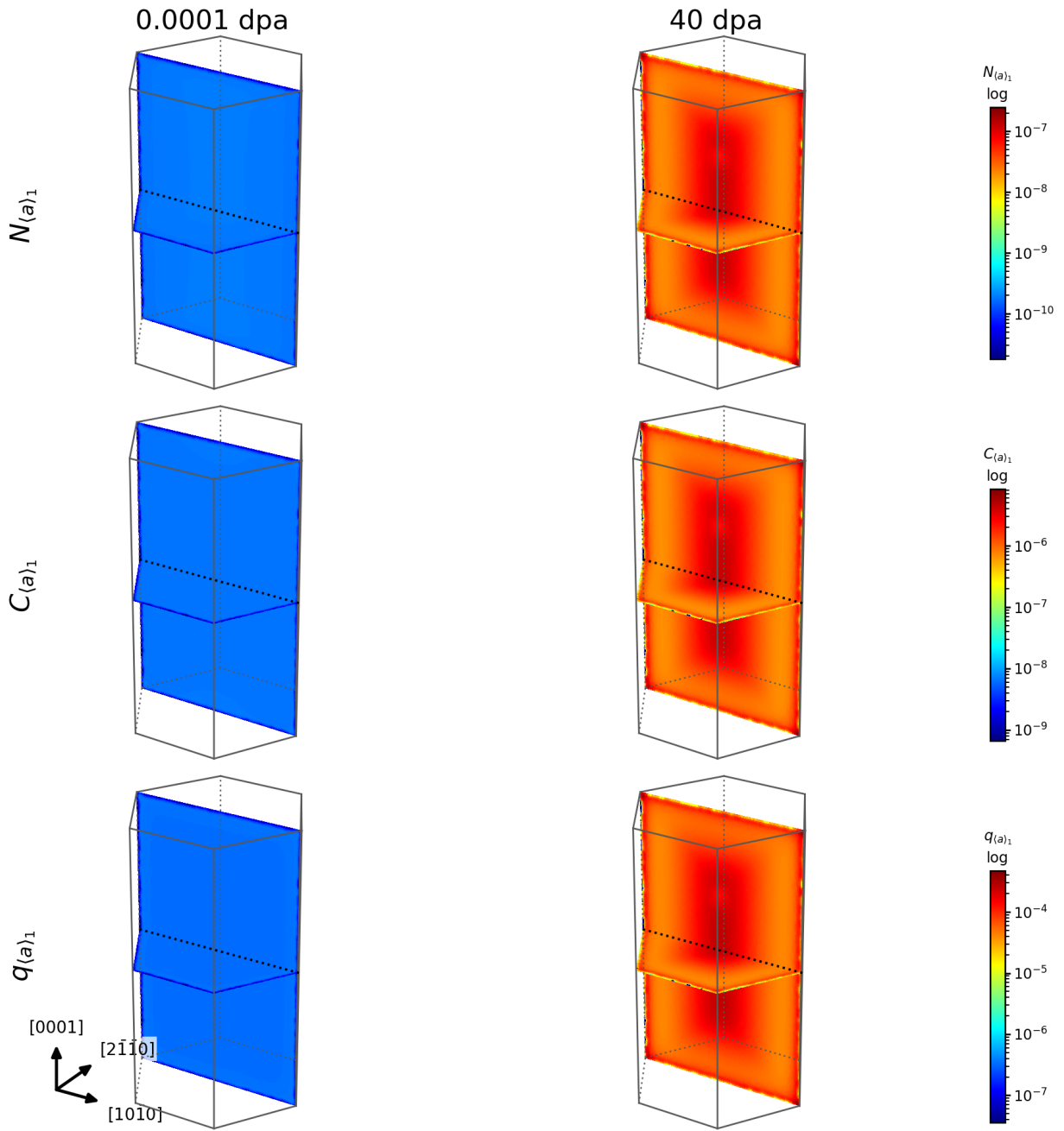


Figure 9: The prismatic interstitial variant a_1 at the same two doses, same three moments. The variants a_2 and a_3 are identical at zero applied stress and are not shown.

Dose (dpa)	c_f		a_1		a_1^v	
	median	min	median	min	median	min
10^{-4}	1.003	1.001	1.002	1.001	1.001	1.001
10^{-2}	1.473	1.098	1.883	1.090	1.509	1.076
1	1.559	1.250	2.724	2.442	2.215	2.112
10	1.585	1.266	2.705	2.439	2.175	2.104

Three things follow. At 10^{-4} dpa every family is monodisperse to three decimal places, as it must be — the loops present are all fresh nuclei deposited at m_k^{nuc} . By 1 dpa the prismatic distribution has broadened to $\Delta = 2.72$, i.e. $s = \sqrt{\ln \Delta} = 1.00$, a log-normal a full decade wide at $\pm 1\sigma$: a model carrying only n and c would represent that population by a single size. And $\Delta \geq 1$ holds at *every* interior node at every dose, minimum 1.001, which is the check that the channels debiting q are consistent with those debiting n and c — a check that no tolerance can substitute for, since $\Delta < 1$ is not an inaccuracy but a statement that the state is not a moment triple.

The basal family broadens less than the prismatic one (1.59 against 2.71) because its number loss is dominated by the boundary channel, which removes the large end and therefore *narrows* what remains, while the prismatic family broadens under growth with no comparable truncation.

7.4. Global conservation

The six atom-balance accumulators integrate the production, recombination and sink absorption of each character, and the two accumulators of [Section 5](#) integrate the loop content taken by the boundary. In a spatially resolved calculation on a Dirichlet domain the balance cannot close on those channels alone, and the residual is not solver error: it is the atoms that left through the surface as *mobile* species. The magnitude settles the matter — nothing in an integration at $\text{rtol} = 10^{-6}$ is tens of percent wrong.

[Table 10](#) is the balance over the whole march, per host atom.

Table 10: The atom balance over $0 \rightarrow 10$ dpa, volume-averaged, per host atom. Production and recombination are exactly equal between the characters, as Frenkel-pair production and recombination must be. The two channels that differ are where the physics is: the network absorbs more interstitials than vacancies (the bias), and the boundary absorbs vacancy *loops* 60 times more than interstitial ones.

Channel	interstitial	vacancy	% prod (i)	% prod (v)
produced	10.0000	10.0000	100.00	100.00
recombined	5.0832	5.0832	50.83	50.83
absorbed at the network	0.7885	0.5908	7.88	5.91
loops at the boundary	0.0085	0.5123	0.08	5.12
stored in the microstructure	0.0000	0.0020	0.00	0.02
mobile flux to the boundary (by difference)	4.1198	3.8117	41.20	38.12

Three readings.

Both columns close on production. $10.0000 = 5.0832 + 0.7885 + 0.0085 + 0.0000 + 4.1198$ and likewise for the vacancies, so once the loop channel is carried explicitly, what remains in the residual is the mobile flux alone and the two do not contaminate each other. Before that channel was booked separately, its 5.1% of vacancy production was silently absorbed into the residual, and because the residual is closure by difference, nothing looked wrong.

The boundary is the dominant sink, by a wide margin. It takes 41% and 38% of production as mobile species, against 7.9% and 5.9% at the network. That is a property of the geometry — a 500 nm grain with every face absorbing — and it is precisely what a mean-field model represents by a single k_{gb}^2 and cannot resolve.

The loop channel is strongly asymmetric. It removes 0.512 vacancies per atom against 0.0085 interstitials, a factor of 60. The asymmetry is geometric, not biased: $\langle c \rangle$ loops reach a mean radius of 25 nm where $\langle a \rangle$

loops reach 3 nm, so a basal loop intersects a face from far deeper inside the crystal. Note also that the boundary removes *250 times more* vacancy loop content than the microstructure retains (0.512 against 0.0020), which is the quantitative form of the statement that on this grain size the loop population is boundary-controlled.

Figure 10 shows the channels against dose and the fractional decomposition. The fractions sum to unity once the boundary channels are included; without them the sum reached only ~ 0.3 .

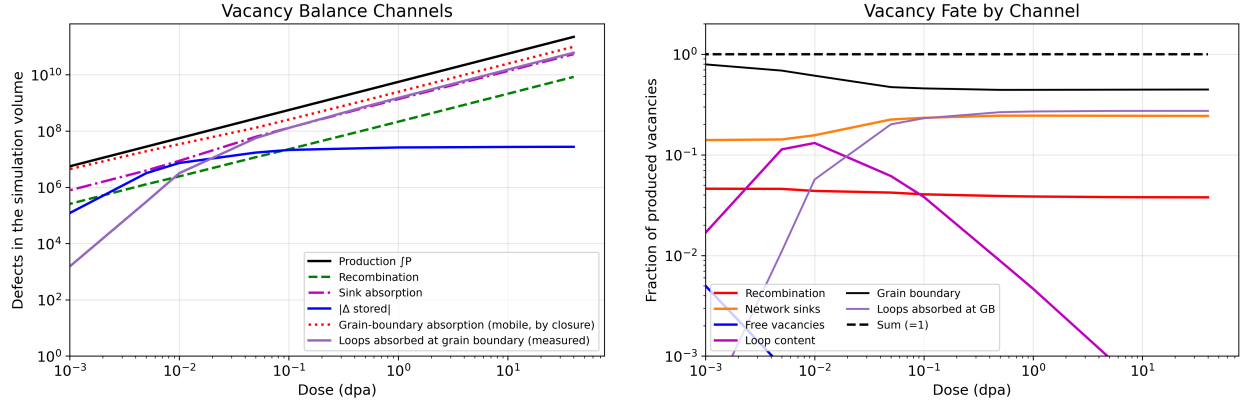


Figure 10: Vacancy balance against dose: cumulative channels (left) and their fractions of production (right). The interstitial counterparts are similar except that the loop-at-boundary channel is 60 times smaller.

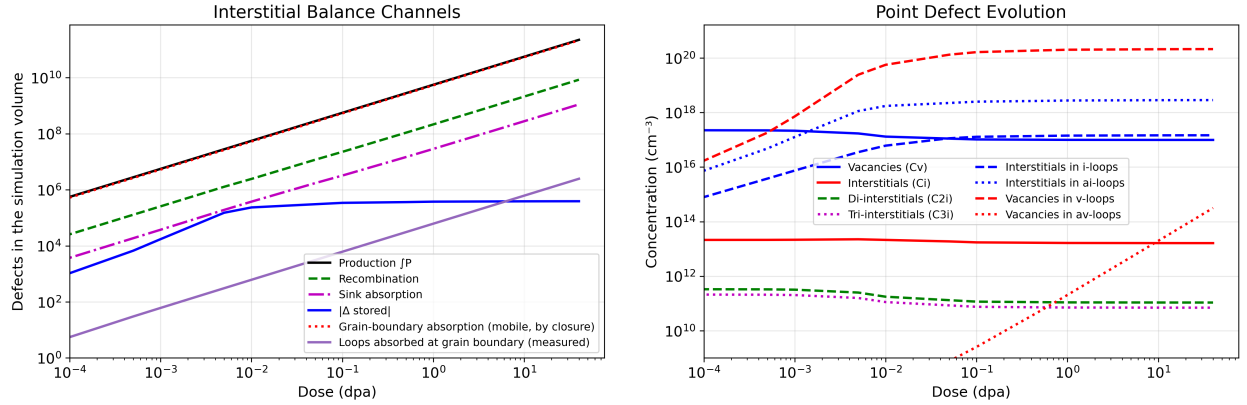


Figure 11: Left: the interstitial balance against dose, the counterpart of Fig. 10. Right: the volume-averaged mobile concentrations, whose fall through the march is the rising sink density of Section 7.2 seen as an average rather than as a field.

7.5. Grain-boundary effects and the denuded zone

Figure 12 is the central result of the loop channel: loop density and mean diameter against distance to the nearest face, with and without it.

x (nm)	N_a/twin	N_c/twin	d_a (nm)	d_c (nm)
0–1	10^{-6}	10^{-6}	5.0	5.4
3.5–4.9	0.646	0.608	4.6	3.5
6.7–9.1	0.703	1.36	7.1	10.1
23.6–32.3	2.95	5.45	5.7	32.5
83.3–114.3	0.415	4.21	9.2	50.2
156.8–215	0.670	12.6	8.9	57.9

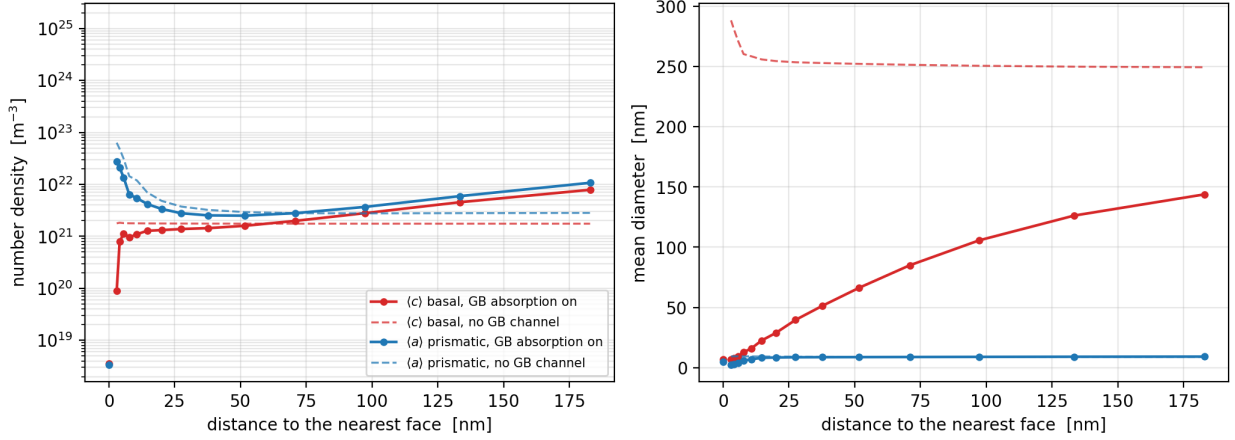


Figure 12: The loop-denuded zone at 40 dpa. Solid: grain-boundary loop absorption active. Dashed: the identical calculation without it. Left, number density; right, content-weighted mean diameter. The abscissa is the minimum distance to any face, and the ordinate on the left is logarithmic over six decades.

The zone is real, sharp, and about 5 nm wide. The face layer is swept to 10^{-6} of the companion calculation — which is the concentration floor, so as complete a removal as this model can express — and the depletion is still measurable at 5–9 nm. Beyond that the profile is no longer denuded. The mean diameter recovers monotonically from 3.5 nm at the boundary to 58 nm in the interior, which is the size gradient a denuded zone is identified by in a micrograph: not merely fewer loops near the boundary, but *smaller* ones, because the large end of the distribution is what Eq. (47) removes first.

The ledger and the density profile measure different things, and both are correct. Of the total absorption, only 1.7% occurs within 2 nm of a face and 93% between 5 and 100 nm. That is not a contradiction of the profile above: the face layer is swept within the first dose interval, so for 99% of the march there is nothing left there to absorb, while the region further in keeps *supplying* loops that grow into contact. The density says where loops are absent; the ledger says where the flux into the boundary came from over the whole history.

A local channel with non-local consequences. The interior $\langle c \rangle$ density is 4–5 times the companion calculation’s at *every* distance, including 200 nm from any face where Eq. (45) removes nothing. The cause is the operator split working as intended: removing the shell’s loops removes their sink strength from Eq. (16), the fast solve returns a different mobile field everywhere, and the interior responds to it. A model in which the boundary were only a mean-field sink strength could not produce this, because the boundary’s effect on the interior would be a single number rather than a changed field.

How this compares with experiment needs stating carefully. Loop-denuded and void-denuded zones at grain boundaries are a standard observation in irradiated metals [57–60], and the mechanism assumed here — the boundary as a strong sink whose influence extends a diffusion length into the grain — is the standard interpretation. We do not compare quantitatively against a measured zone width in zirconium: no such measurement is in the data set of Table 8, and the width predicted here (a few nm for $\langle a \rangle$, growing to tens of nm in the size profile) is offered as a prediction rather than as a validated result. It is also, at 2–5 nm for the prismatic family, comparable with the 20 nm element size of the boundary layer, so the $\langle a \rangle$ zone in particular is under-resolved and its width should be treated as an upper bound set by the mesh.

7.6. Volume averages and comparison with experiment

Figure 14 shows the volume-averaged trajectories.

Which average is quoted is not a presentational choice. On a Dirichlet domain the boundary shell is a sink for mobile defects and — without the channel of Section 5 — not a sink for loops, so it accumulates a

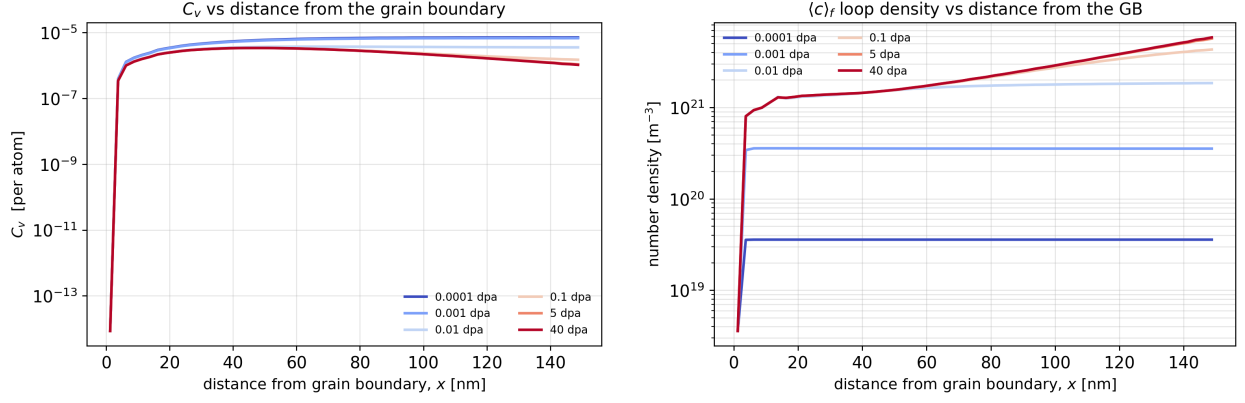


Figure 13: Profiles against distance from the boundary over the first 150 nm, with every dose overlaid. Left: the mobile vacancy concentration, whose recovery length is the boundary layer tabulated above and which narrows as the dose accumulates. Right: the basal $\langle c \rangle$ loop number density, whose profile is the denuded zone of Fig. 12 seen at every dose rather than only at the last.

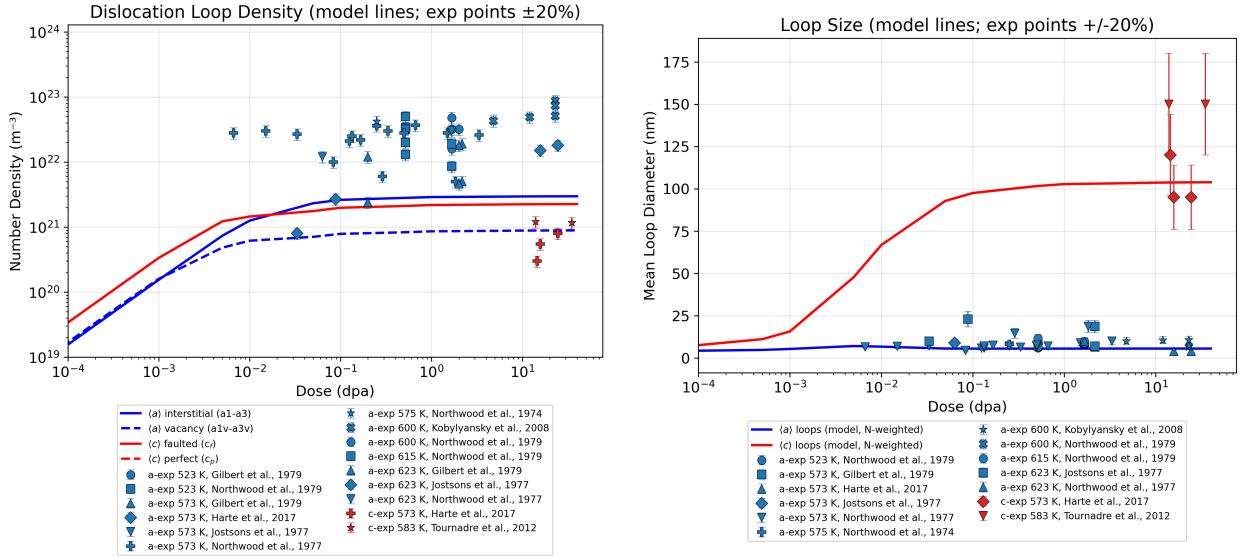


Figure 14: Volume-averaged loop density against dose (left) and mean size against the experimental measurements (right).

loop population that is numerous, tiny, and entirely unphysical as a description of the material. At 40 dpa:

	Region	N_a (m ⁻³)	d_a (nm)	N_c (m ⁻³)	d_c (nm)
loop channel off	domain	3.79×10^{23}	4.29	3.98×10^{23}	17.60
	interior	2.80×10^{21}	5.79	1.75×10^{21}	249.84
loop channel on	domain	3.86×10^{21}	5.58	2.26×10^{21}	103.79
	interior	7.06×10^{21}	5.98	5.31×10^{21}	133.01

Every entry is volume-weighted on the crystal itself rather than on its bounding box, and d is the number-weighted mean diameter of Fig. 14, so the table is that figure read at one dose. Without the channel the domain and interior means differ by a factor of 135 in N_a and 227 in N_c , and the domain-mean diameter *falls* to 18 nm while the interior mean grows to 250 nm — the domain average is measuring the boundary shell and nothing else. With the channel the disparity collapses to a factor of 1.8 in N_a and 2.4 in N_c , and it changes sign: the interior now carries *more* loops than the domain mean, which is the ordering a sink at the surface should produce. A domain average becomes a statement about the material. That is the strongest single argument for carrying the channel: it removes an artifact that otherwise makes the most natural summary statistic meaningless.

Against experiment, and the discrepancy. At 573 K and 24.5 dpa the measurements give $N_c = 8.0 \times 10^{20}$ m⁻³ and $d_c = 95$ nm. That dose is now inside the calculation rather than beyond it, and the comparison no longer needs an extrapolation: at 40 dpa the interior gives $N_c/N^{\text{exp}} = 6.6$ and $d_c/d^{\text{exp}} = 1.40$, and between 10 and 40 dpa neither quantity moves by more than a percent, so which snapshot is read makes no difference to either ratio. Too many loops, and now too *large*. The companion without the loop channel gives 2.2 and 2.63 — closer on the density and much worse on the size, which is expected and is not an argument against the channel: the parameters of Table 9 were fitted in a 0-D that has no boundary at all (Section 6), so the channel is an unmodeled loss acting on a population calibrated without it. A refit with the channel active is required before either number is quoted as a prediction.

The direction of the discrepancy is the informative part, and it has changed with the refit. Earlier parameter sets produced *more, smaller* $\langle c \rangle$ loops than are observed — the signature of a model removing loops faster than it grows them, so that the survivors are too young. This one produces *more and larger* loops: the density is 6.6× the measurement and the diameter 1.40×, so the stored basal content, which scales as $N_c d_c^3$, is 18× what is observed. The two errors no longer oppose each other, and that is a harder failure to excuse than the old one. It is not a size problem that a growth term could absorb; it is too much vacancy content retained in basal loops, and the channels that could remove it — coalescence, peripheral emission, the small-end leak — are the ones the refit has already tuned. Section 8 states what this implies.

7.7. Continuum-to-discrete conversion

The conversion of Section 5 was applied to the fields at four doses. The number of loops is not a parameter of the conversion: it is $\sum_j n_j V_j$, the number the field says is present.

Family	10 ⁻⁴ dpa	10 ⁻² dpa	1 dpa	40 dpa
c_f (basal, vacancy)	5	238	642	691
a_1 (prismatic, interstitial)	1	55	213	232
a_1^v (prismatic, vacancy)	1	23	69	75

Figure 15 shows the basal population at those four doses, drawn at true radius and true position, over the continuum content field it was drawn from. The two representations agree loop for loop: the figure is built from the same draw the conversion exports, so a reader comparing this figure with the exported loop table is comparing one object with itself.

Three observations.

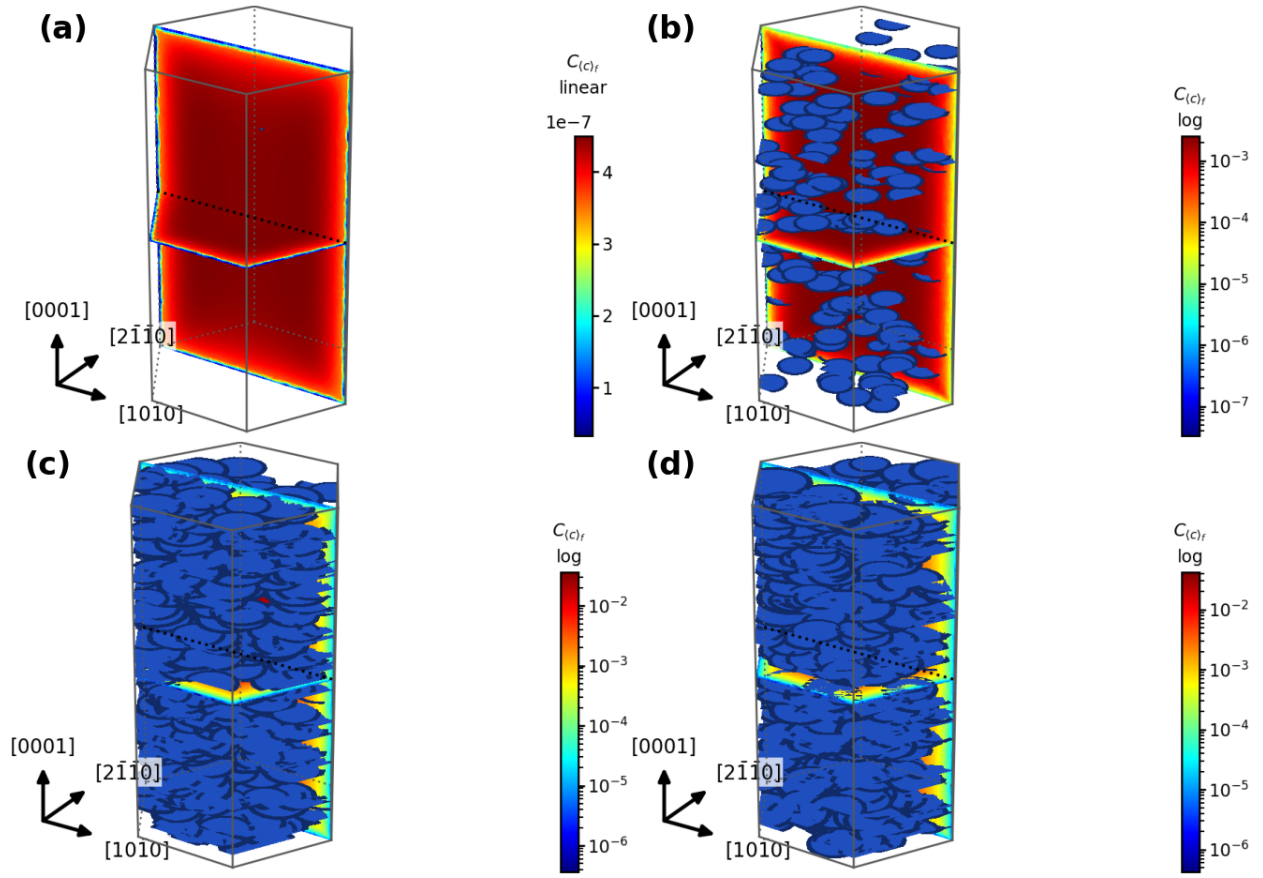


Figure 15: The discrete basal $\langle c \rangle$ population at (a) 10^{-4} , (b) 10^{-2} , (c) 1 and (d) 40 dpa, drawn at true radius over the continuum content field. Counts are 5, 238, 642 and 691. The loops are clipped at the crystal surface; see the text.

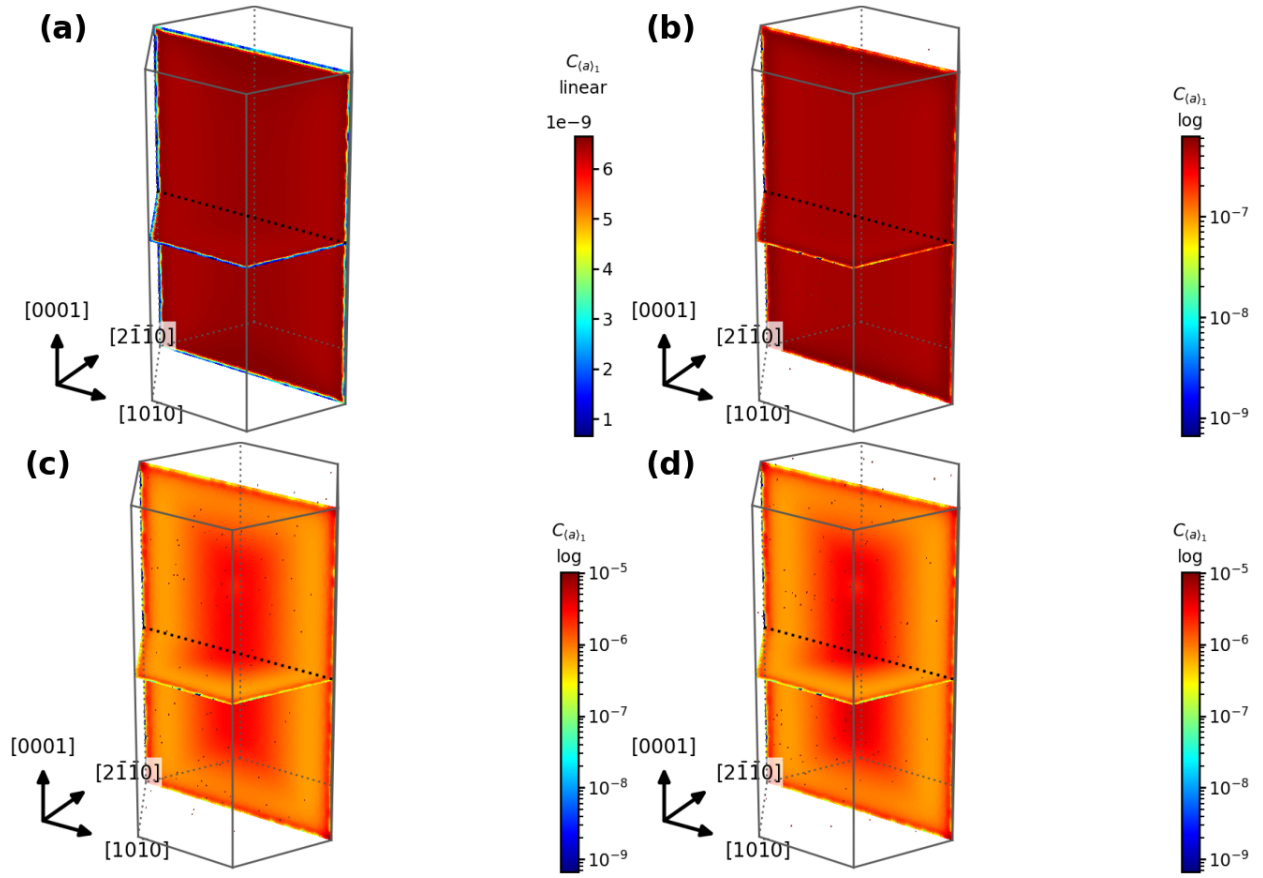


Figure 16: The discrete prismatic $\langle a \rangle_1$ population at the same four doses, (a) 10^{-4} , (b) 10^{-2} , (c) 1 and (d) 40 dpa, drawn at true radius. Counts are 1, 55, 213 and 232. At true scale these loops are 6–7 nm across in a 500 nm crystal and are correspondingly sparse; the same population drawn with exaggerated radii is legible but is not a population.

The count is not monotone, and it should not be. Between 10^{-2} and 1 dpa the prismatic count nearly quadruples, from 55 to 213, at a mean diameter that is unchanged at 2.14 nm: prismatic interstitial loops are being nucleated and are not growing. Under the previous parameter set this count *fell* over the same interval, which was coalescence merging loops into fewer larger ones; the refit has removed that behaviour from the prismatic family, and the conversion is where it becomes visible, because it reports number and size separately rather than the content that is their product.

Most of the basal loops do not fit. At 40 dpa, 419 of 691 $\langle c \rangle$ loops (60.6%) are centered within their own radius of a face and overhang it, by up to 88 nm; the corresponding prismatic figure is 3.5%. The drawn population is clipped at the surface, but the clip is a drawing convention and the exported loops are whole. This is the same geometric refusal that [Section 5](#) describes, and it means the basal handoff on a 500 nm grain is only partially feasible above ~ 0.1 dpa: a discrete-dislocation solver would decline those loops, and their defects would have to remain in the continuum. *The correction to the $\langle c \rangle$ Burgers vector that made the two representations agree made this worse*, not better, because it raised the basal radius by $\sqrt{2}$ and a larger loop is harder to fit.

The conversion is where the mean field is checked against itself. At 10^{-4} dpa the population is 5 basal and 6 prismatic loops in $1.3 \times 10^8 \text{ nm}^3$ — a population so dilute that a continuum density describing it is a statement about an ensemble and not about this crystal. At 40 dpa the basal loops are 131 nm across with a mean spacing of 57 nm — each loop is wider than the distance to its neighbour — which is the opposite regime, and it is the one in which the Avrami kernels of [Eq. \(22\)](#) are working hardest and are least defensible. The conversion spans both, and the ledger of [Eq. \(49\)](#) closes on stored defects to machine precision and on loop number to the integer draw at every one of the four doses.

7.8. Evolution of the loop size distributions

The three moments carried per family close on a log-normal ([Eq. \(43\)](#)), so the model holds a genuine size distribution at every node and not merely a mean. Two distributions are in play and they are not the same. At one node the closure gives a log-normal in the defect count m , with $\bar{m} = c/n$ and dispersion $\Delta = qn/c^2 = e^{s^2}$. Over the crystal the population is the *volume-weighted mixture* of those per-node log-normals, and a mixture of log-normals is not log-normal: it is broader than any of its components and may be multimodal where the field varies strongly. The mixture is the observable one — it is what a micrograph of a whole specimen samples — and it is what [Fig. 17](#) plots, as dN/dd against a linear diameter axis, which is the form a measured size distribution takes. Curves are drawn from 1 nm upward: below that a loop is not resolvable in a micrograph, and the $1/d$ factor relating dN/dd to $dN/d \ln d$ amplifies a tail that is in any case carried by the few anomalous nodes identified below.

Working in diameter is exact rather than numerical: if m is log-normal with parameters (μ, s) then $d = 2\lambda_k \sqrt{m}$ is log-normal with $(\ln 2\lambda_k + \mu/2, s/2)$, so each node contributes a closed-form curve and the mixture is accumulated directly on a logarithmic diameter grid. Every distribution below is taken from the final state *after* grain-boundary absorption, so the removals of [Eq. \(47\)](#) are already inside the moments being inverted.

The distributions are born monodisperse and broaden. At 10^{-4} dpa every family is a near-delta function: $\Delta = 1.001\text{--}1.003$, so $s = 0.03\text{--}0.06$ and the distribution is a spike a few percent wide at the nucleation size, 4.6–5.5 nm. This is not an artifact of the closure but its correct statement of the physics — at that dose every loop present is a fresh cascade nucleus deposited at m_k^{nuc} , and there has been no time for anything to broaden it.

By 10^{-2} dpa the spike has become a distribution spanning more than a decade, and by 1 dpa the shape has essentially stopped changing: the 1 dpa and 40 dpa curves are nearly coincident in every panel. The population saturates in shape as well as in the aggregates of [Section 7.6](#), which is the signature of a balance between nucleation, growth and coalescence rather than of a still-developing microstructure.

That width is part spatial and part local, and the split is measurable. [Figure 18](#) separates the two contributions, and the separation is unusually clean.

The boundary shell is exactly monodisperse, $\Delta = 1.000$ for every family at every dose, and it stays so. That is a physical result and not a numerical floor. Both terms that broaden a family in [Eq. \(41\)](#) are

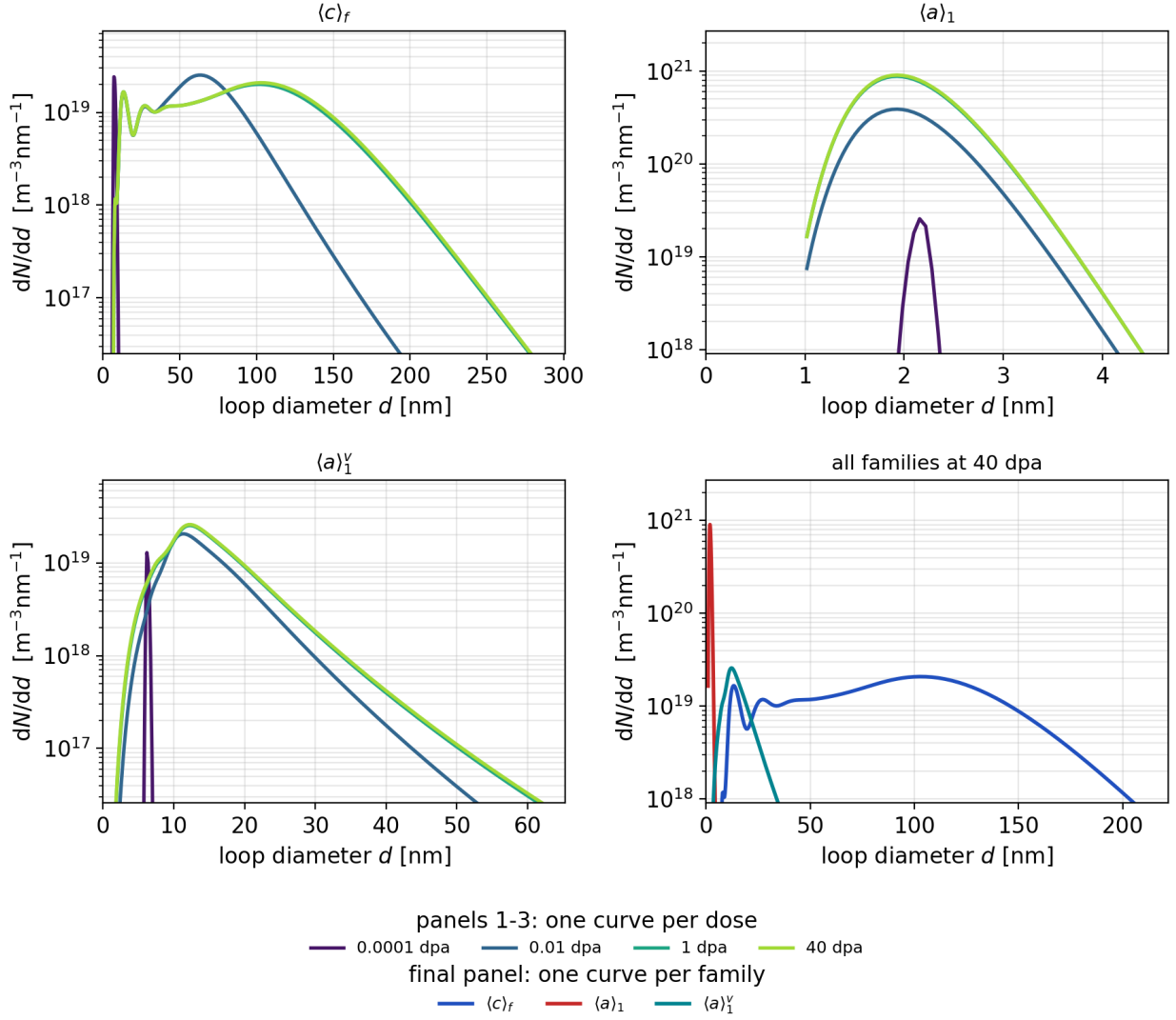


Figure 17: Reconstructed loop size distributions over the whole crystal at 10^{-4} , 10^{-2} , 1 and 40 dpa, for the basal faulted family and for one prismatic variant of each character; the fourth panel compares the three at 40 dpa. *Key, first three panels:* one curve per dose, dark violet 10^{-4} , blue 10^{-2} , teal 1 and green 40 dpa. *Key, fourth panel:* one curve per family, blue $\langle c \rangle_f$, red $\langle a \rangle_1$, teal $\langle a \rangle_1^v$. The ordinate is the population per nanometre of diameter per cubic metre, dN/dd , and each panel is shown over three decades below its own peak. The two remaining basal states carry no population in this calculation: the pyramid c_0 and the perfect state c_p are fed only by the basal chain, which is switched off because its rates are uncalibrated (Section 3.3).

Table 11: Aggregates of the reconstructed distributions. $\bar{d}$ is the content-weighted mean diameter and Δ the median per-node dispersion; $\Delta = 1$ is monodisperse and $\Delta = e$ is a distribution one e -fold wide in m . The interior quartile and the boundary shell are the innermost and outermost quarters of the crystal by distance to the nearest face.

	$\langle c \rangle_f$		$\langle a \rangle_1$		$\langle a \rangle_1^v$	
Dose (dpa)	$\bar{d}$ (nm)	Δ	$\bar{d}$ (nm)	Δ	$\bar{d}$ (nm)	Δ
<i>whole crystal</i>						
10^{-4}	5.51	1.002	4.64	1.001	4.78	1.001
10^{-2}	28.52	1.160	6.69	1.792	7.66	1.365
1	37.52	1.145	8.50	1.994	6.06	2.011
10	37.07	1.162	8.66	1.977	5.97	2.028
<i>interior quartile</i>						
10^{-4}	5.52	1.003	4.22	1.002	4.75	1.001
10^{-2}	28.42	1.473	6.53	1.883	5.91	1.509
1	51.30	1.559	12.04	2.724	2.83	2.215
10	50.62	1.585	12.11	2.705	2.84	2.175
<i>boundary shell</i>						
10^{-4}	5.41	1.000	5.21	1.000	4.79	1.000
10^{-2}	8.65	1.000	5.60	1.000	7.06	1.000
1	6.83	1.000	6.28	1.000	6.73	1.000
10	7.08	1.000	6.34	1.000	6.79	1.000

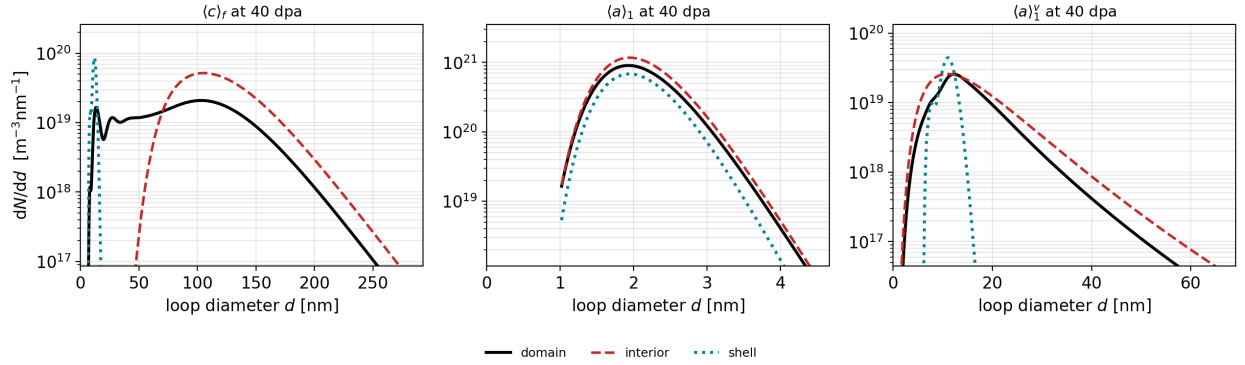


Figure 18: The 40 dpa distribution of each family over the whole crystal, its interior quartile and its boundary shell. *Key:* solid black, whole crystal; dashed red, interior quartile; dotted teal, boundary shell. The dispersions these curves carry, as Δ (whole crystal, interior, shell), are 1.070, 1.139, 1.000 for $\langle c \rangle_f$; 1.227, 1.233, 1.000 for $\langle a \rangle_1$; and 1.461, 2.513, 1.000 for $\langle a \rangle_1^v$. The shell is therefore monodisperse to three decimal places in every family, the interior is broad, and the whole-crystal spectrum is their mixture — its multimodality is spatial.

proportional to the absorbed mobile flux — the drift through Σ_k^{net} and the spreading through Σ_k^{gro} — and that flux vanishes where the Dirichlet condition pins the mobile species. In the shell, loops are nucleated at one size and are neither fed nor starved, so nothing acts on the width. They are also small, 6–7 nm, because they never grow.

The interior is broad, $\Delta = 1.59$ for the basal family and 2.71 for the prismatic interstitial one — the latter a log-normal with $s = \sqrt{\ln 2.705} = 1.00$, a full decade wide at $\pm 1\sigma$. A model carrying only n and c would represent that population by a single size.

The whole-crystal spectrum is therefore a mixture of a narrow shell peak near 7 nm and a broad interior distribution centered near 50 nm for the basal family, and *that* is the origin of the multimodality in Fig. 17, not any structure in a local distribution. The distinction matters for what happens next: a locally broad population coarsens, because its large loops grow at the expense of its small ones; a spatially sorted one does not, because its two peaks are not in contact.

Each variant then behaves as the growth conditions predict. The three families evolve differently, and Section 3.4 predicts exactly how. With the calibrated bias of Table 9 the thresholds of Eq. (36) are

$$x_c = \frac{Z_I^0 A_c(p^i)}{Z_V^0 A_c(p^v)} = 0.625, \quad x_a = \frac{Z_I^0 A_a(p^i)}{Z_V^0 A_a(p^v)} = 1.005, \quad (53)$$

so the co-growth window Eq. (37) is $0.625 < x < 1.005$. Evaluating the flux ratio Eq. (33) on the interior of this calculation, from the diffusion tensors the march actually used and its own mobile fields:

Dose (dpa)	x (interior)	Predicted
10^{-4}	0.935	$\langle c \rangle_f$ grows, $\langle a \rangle^i$ grows, $\langle a \rangle^v$ shrinks
10^{-2}	0.913	as above
1	0.695	as above
10	0.711	as above

The operating point sits inside the predicted window at every dose, and the three predictions are exactly what Table 11 shows in the interior:

- $\langle c \rangle_f$ grows strongly: its interior mean diameter rises $5.5 \rightarrow 28.4 \rightarrow 51.3$ nm and then holds. It is the family the window most favors, since x sits well above x_c .
- $\langle a \rangle_1$ grows and coarsens: its interior mean rises $4.2 \rightarrow 6.5 \rightarrow 12.1$ nm while its interior *count falls* from 43.3 at 10^{-2} dpa to 25.8 at 1 dpa. Fewer, larger loops holding more content is coalescence, and it is visible in the spectrum as the small end draining into a peak that moves right.
- $\langle a \rangle_1^v$ shrinks: its interior mean *falls* $5.9 \rightarrow 2.8$ nm between 10^{-2} and 1 dpa. It is the one family for which $x < x_a$ places it on the shrinking side, and it is shrinking. Its loops survive at all only because cascade nucleation keeps replacing them.

Figure 19 shows those three directions as the population itself, all seven loop variants drawn together in the crystal they occupy. The reconstruction of Fig. 17 is a distribution; this is the object the distribution describes, at true radii and sampled positions, and the two are the same $\sum_j n_j V_j$ loops seen as a spectrum and as a microstructure.

That three families move in three different directions, in the directions a criterion derived independently in Section 3.4 specifies, is the strongest internal consistency check in this paper. It also demonstrates Resolution 1 of Section 3.4 concretely: x is a field, it differs between the shell and the interior, and the families sort themselves accordingly.

Not everything in the reconstruction is well behaved, and the exception is worth reporting because it is structural rather than accidental. At 1 and 40 dpa a single node of the 90 663 — node 51 609, at $x = 214.2$ nm, the point of the prism *furthest* from any face — carries $\Delta = 4.7 \times 10^3$ and 6.2×10^3 respectively, against a median of 1.070. It is the only node above $\Delta = 3$ at any dose, and it holds under 0.002% of the basal

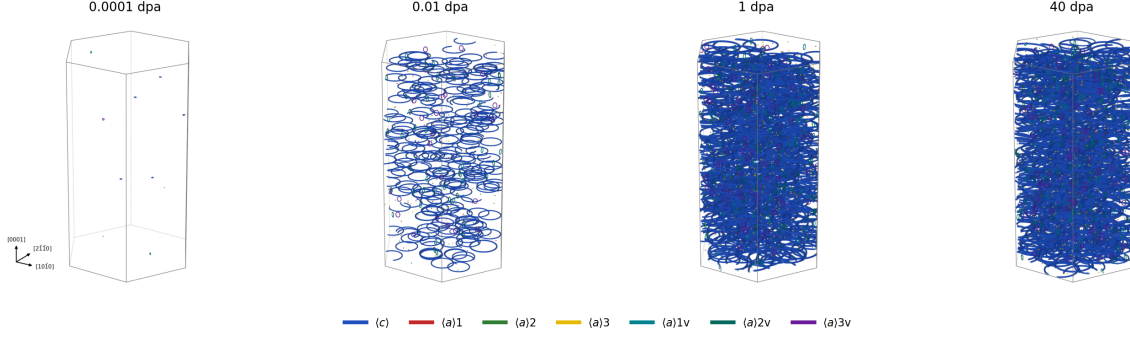


Figure 19: The discrete loop population of the interior at 10^{-4} , 10^{-2} , 1 and 40 dpa, every variant superimposed and drawn at its true radius and sampled position: basal $\langle c \rangle$ in blue, the three interstitial prismatic variants in red, green and gold, and the three vacancy prismatic variants in teal and purple. Each panel is the same crystal and the same viewpoint, so the panels are comparable across dose. The interior population runs $55 \rightarrow 333 \rightarrow 882 \rightarrow 1414$ loops over the four panels, and the basal share of it $13\% \rightarrow 50\% \rightarrow 45\% \rightarrow 43\%$ — so what the sequence shows is not the basal family becoming more numerous but its loops becoming very much larger, until at 40 dpa they dominate the drawing at a diameter an order of magnitude above the prismatic variants. The three interstitial prismatic variants stay exactly equal in number (232 each at 40 dpa), which is the isotropic variant weighting of Eq. (21) under zero applied stress; the three vacancy variants likewise (75 each). That domination by size is why Figs. 15 and 16 separate the families; here it is the subject.

population; it is what produces the weak shoulder above 100 nm in Fig. 17. Every other node stays below $\Delta = 1.5$, so the gap between the outlier and the rest of the crystal is three orders of magnitude and there is nothing in between.

The mechanism is in Eq. (41). The drift term carries $\Delta_k^{3/8}$ and the spreading term $\Delta_k^{-1/8}$, so a family that has already broadened broadens *faster* while its spreading is damped: the dispersion is self-amplifying through the drift. Everywhere in this calculation that feedback is held in check by coalescence and by the finite mobile flux; at the one point where the flux is largest and no boundary channel acts, it is not. The remedy is not a tolerance — Δ never violates $\Delta \geq 1$, so no realizability test catches it — but a bound on the drift exponent’s effect, and identifying one is left as work the closure needs.

The natural next experiment on this model is a post-irradiation anneal: switch off the source, hold the temperature, and follow the distributions. It is worth setting out what the formulation predicts, because the model has no fitted annealing lifetime — the two lifetimes of the mean-field treatment were deleted and replaced by peripheral emission against each family’s own size-dependent binding energy (Appendix E) — so an anneal is a genuine prediction rather than a fit.

Four expectations follow, and each is falsifiable against standard annealing experiments.

The flux ratio diverges, and the family ordering inverts. With the source off the interstitial supply collapses first, since interstitials are four orders of magnitude more mobile and drain to the sinks within $\tau_M \sim (k^2 D)^{-1}$. So $x = \Phi_V/\Phi_I \rightarrow \infty$, which by Eq. (35) places *every* vacancy family on the growing side and every interstitial family on the shrinking side. The prismatic interstitial loops — the dominant population at the end of irradiation — should therefore dissolve, while the prismatic vacancy family, which is shrinking under irradiation, should reverse. An anneal is the one experiment that reverses the sign of the interstitial–vacancy competition without changing a parameter.

The small end evaporates and the mean rises. Once production stops the driver is thermal emission, and the binding energy of Eq. (E.1) falls with loop size, so small loops emit fastest. The distribution should lose its small end while its large end persists — a rise in $\bar{d}$ accompanied by a fall in N , which is Ostwald ripening and is what annealing experiments on irradiated Zr report. In the moment language, $\dot{n} < 0$ and $\dot{c} < 0$ with $|\dot{n}|/n > |\dot{c}|/c$, and Δ falls: *annealing narrows the distribution*, where irradiation broadened it.

The anneal should be spatially structured, and a 0-D model cannot produce that. The interior enters the anneal broad ($\Delta = 1.6\text{--}2.7$) and the shell monodisperse ($\Delta = 1.000$). A distribution with a small end has one to evaporate and will coarsen; a monodisperse one has none and can only shrink uniformly. The prediction is therefore that the interior coarsens while the shell dissolves *en bloc*, and that the two-peak structure of

Fig. 18 collapses toward the large peak. Separately, the denuded zone should *widen* with annealing time, because the grain-boundary channel of Eq. (47) keeps removing whatever grows into contact while nothing replaces it.

It is the sharpest available test of the closure. Under irradiation the distribution is held near a driven balance and a wrong shape can be masked by a right mean. In an anneal the source term is gone, the shape evolves under emission alone, and the log-normal ansatz is exposed: if the true distribution develops the asymmetric large-size tail that ripening theory gives, a log-normal closure will misreport the lifetime of the population even while reproducing its initial mean. Comparing an annealed population against the isolated-loop law — which the same binding energy generates for a single loop, and which can be integrated independently — would separate an error in the emission physics from an error in the closure. That comparison is the natural companion to the present work and requires no new physics, only the source switched off.

8. Discussion and Conclusions

Self-consistency in sense (a) is achievable and is a constraint worth accepting. Every capture efficiency in Eq. (31) and every sink strength in Eq. (16), for all nine families, is generated from the same diffusion tensors that transport the mobile species, through one anisotropy parameter p^m per species. The formulation therefore cannot simultaneously hold that diffusion is isotropic and that its sinks are biased by the anisotropy of that diffusion, which is a state most cluster-dynamics treatments of zirconium are free to occupy. The price is that anisotropy can no longer absorb a shortfall by refitting a bias — and that price turns out to be worth paying, because it lets p^m be set from transport rather than from loop counts. With $p^v = 1.02$ and $p^i = 0.70$ fixed as inputs, the basal diameter comes out at 100.7 nm against 95 nm measured without being steered there, and it stays within 85–131 nm at every stage of the search. The prismatic families follow on the same fixed anisotropy: density $0.88\times$ the measurement over 71 points and diameter $1.21\times$ over 44, across twelve temperatures and four decades of dose, with a scatter of $10^{0.49} = 3.1$ that is comparable to the spread among the measurements themselves. A model that cannot hide a deficiency in a bias factor is one whose deficiencies stay visible, and one whose agreements mean something when they occur — the basal diameter was a factor of two low before the anisotropy was fixed, and no lever was applied to it.

The second sense holds where it can be made exact and is measured where it cannot. Stored defects transfer to machine precision and loop number to the integer draw, at every dose tested. The sink strength — which the transfer does not force — came within 2% of continuity only after an inconsistency in the $\langle c \rangle$ Burgers vector between the two representations was removed, and locating it is what the ledger of Eq. (49) is for: a discontinuity there is a crystallography error announcing itself, not a numerical tolerance to be widened. What sense (b) does not yet deliver on this geometry is feasibility. A quarter of the basal loops on a 500 nm grain do not fit inside it (Section 7.7), so a hybrid calculation there would have to carry the refused population in the continuum and account for it separately — and that same constraint is why the basal diameter was calibrated against the measurement rather than against a transfer factor.

Beyond what the two consistency constraints buy, spatial resolution changes the answer and not merely its presentation. The boundary layers of Section 7.2 are 8–46 nm wide, they order themselves by species mobility, and they narrow by a factor of two as the sink density grows, none of which a mean-field k_{gb}^2 can express. More consequentially, the loop population in the outermost shell is numerous and tiny and it dominates every domain average: without the boundary loop channel the domain and interior means of the $\langle c \rangle$ density differ by a factor of 114. A spatially averaged model does not have this problem because it cannot see it, and it reports the wrong number without any indication that it has.

That resolution is affordable, and the reason is worth separating from the physics. 2.1×10^8 stiff ordinary differential equations were integrated in about an hour on a workstation, not because a million-equation stiff system was solved but because the frozen-mobile construction turns it into 72 928 independent 36-dimensional ones — 6.5×10^9 fewer flops per factorization — while the genuinely coupled part, having no time derivative, is solved eight times rather than thousands. The bill arrives as a first-order splitting error whose dial is the fast-solve cadence, and the two sides of the calculation cost within 7% of each other, which is the balance that cadence was chosen to produce.

With the split in place, conservation can be audited channel by channel, and it closes once the boundary is carried explicitly. Both atom balances close on production to the digit, with the loop-at-boundary channel taking 5.1% of vacancy production against 0.08% of interstitial. That the channel had to be booked separately rather than left in the residual generalizes: closure by difference is not a check, because a channel omitted from it reappears silently in the remainder and nothing looks wrong. The channel that had to be added to close that balance is also the one that makes a prediction. The size-gated removal of Eq. (47) produces a denuded zone ~ 5 nm wide in which the loop density falls to the concentration floor and the mean diameter recovers monotonically from 6.8 to 144 nm, and it demonstrates something a mean-field treatment structurally cannot: a channel acting only within a diffusion length of a surface raises the $\langle c \rangle$ density in the deep interior by a factor of 4–5, because removing the shell’s sinks changes the mobile field everywhere.

What the calculation does not settle is the basal density — $2.1\times$ the measurement — and the shape of that residual is more useful than its size. The model carries more $\langle c \rangle$ loops than are measured while their diameter now agrees, which is a weaker and more specific statement than the earlier calibrations supported, where number and size were wrong together: number and size are linked, the number balance pinning total content and the growth balance pinning size, so a model removing loops faster than it grows them has survivors that are too young, and every lever exercised here moves along that constraint rather than across it. Raising coalescence brings the density down and collapses the diameter with it; the small-size floor moves the prismatic families and not the basal one; gating coalescence to the large end is worse still, the size bias being already inside the Avrami kernel so that applying it twice shrinks the family, weakens the rate and *raises* the density. Fixing p^m from transport sharpens the diagnosis rather than removing it: what now rests on its bound is the basal capture prefactor, Z_v^0 and ξ_c together, which is a statement about absorption and cannot be mistaken for one about diffusion. The mechanism the data ask for removes vacancy loop *number* without shortening the survivors’ lives — unfaulting to a glissile configuration that escapes to the network, a dissolution channel for the perfect basal state, or an absorption cross-section growing faster with size than R — and identifying it is the single most valuable next piece of physics. The prismatic families carry a smaller version of the same question: their density is reproduced in the mean, $0.88\times$, and low by a factor of six near 600 K, a temperature trend no lever in the calibrated set corrects. Section 6 sets out the two readings — a nucleation term with no temperature dependence of its own, or high-temperature data that do not measure the interstitial population — and the character-resolved measurement at 600–700 K that would separate them.

Two further reservations attach to the calibration. The basal chain is uncalibrated and switched off, because nothing in the data supplies the collapse and unfaulting barriers of Eq. (25); switched on with placeholder rates it inverts the family ordering, which is a property of the placeholders and not a prediction. Its terminal state c_p also has no number-loss channel beyond coalescence — it cannot unfault further and peripheral emission acts on content only — so under a sustained vacancy supply it is bounded only by the transfer feeding it, and whether it should have a sink of its own must be settled before ν_{uf} is fitted, the two not being separately identifiable from a steady c_p density. And the $\langle c \rangle$ data are thin: five points at two temperatures, all above 14 dpa, while the model places resolvable basal loops from 10^{-2} dpa with nothing constraining that extrapolation. No posterior was sampled either; Section 6 establishes that the reported set is a maximum *a posteriori* point under a stated likelihood and bounded uniform priors, without a credible interval to accompany it.

Four steps follow, in the order their results feed one another.

1. Refit with the boundary channel active, on the spatially resolved model rather than in 0-D, so the calibration and the calculation describe the same physics. The transfer factors of Section 6 are a stopgap for exactly this, and measuring them on the present march would replace them.
2. Constrain p^m atomistically. Because it is the only anisotropy the formulation has, a computed value enters Eq. (30) directly and either confirms Eq. (52) or replaces it, and the interstitial migration split of 0.106 eV it implies at 573 K is the specific quantity to compute.
3. Run the Bayesian pipeline. At 0.2–0.5 s per objective evaluation, a space-filling design screened by Sobol’ indices, a Gaussian-process emulator and Markov-chain sampling of Eq. (51) are all affordable,

and would replace the point estimate of Table 9 with a posterior, identify which parameters the data determine, and quantify how far the $\langle c \rangle$ predictions can be trusted on five measurements.

4. Close the loop to dislocation dynamics. The conversion of Section 5 is one-way: it exports a population and does not feed it back into a solve. Transferring a family when Eq. (44) fires, removing it from the continuum and continuing with the loops as discrete objects whose stress fields and climb are solved is what would make the model hybrid rather than continuum with an export. The obstacle is geometric and is measured in Section 7.7.

Acknowledgments

This work was performed under a Statement of Work with Fluor Marine Propulsion, LLC. The authors thank colleagues at Knolls Atomic Power Laboratory and Bettis Atomic Power Laboratory for discussions on the experimental microstructure of irradiated zirconium alloys.

References

- [1] B. Adamson, R. E. Coleman, C. M. Griffiths, Irradiation effects in zirconium alloys, Tech. Rep. IAEA-TECDOC-2020, International Atomic Energy Agency, Vienna, comprehensive IAEA review: irradiation growth, loop data, SPP evolution, void data, dose/temperature dependences for BWR, PWR, and CANDU Zircalloys. (2019).
- [2] C. Lemaignan, A. T. Motta, Zirconium alloys in nuclear applications, in: R. W. Cahn, P. Haasen, E. J. Kramer (Eds.), *Materials Science and Technology: A Comprehensive Treatment*, Volume 10B, VCH, 1994, pp. 1–51.
- [3] D. O. Northwood, Irradiation damage in zirconium and its alloys, *Atomic Energy Review* 15 (4) (1977) 547–610.
- [4] M. Griffiths, A review of microstructure evolution in zirconium alloys during irradiation, *Journal of Nuclear Materials* 159 (1988) 190–218, review: $\langle c \rangle$ -component loops and voids in Zr; formation mechanism; temperature window 625–775 K for voids; He role in cavity stabilization; irradiation growth mechanisms; SPP influence. doi:10.1016/0022-3115(88)90029-5.
- [5] F. Onimus, J.-L. Béchade, Radiation effects in zirconium alloys, in: R. J. M. Konings (Ed.), *Comprehensive Nuclear Materials*, Elsevier, 2012, Ch. 4.01, pp. 1–31. doi:10.1016/B978-0-08-056033-5.00064-4.
- [6] M. N. Cinbiz, D. A. Koss, A. T. Motta, The influence of stress state on the reorientation of hydrides in a zirconium alloy, *Journal of Nuclear Materials* 477 (2016) 157–164. doi:10.1016/j.jnucmat.2016.05.013.
- [7] M. Kiritani, Analysis of the clustering process of supersaturated lattice vacancies, *Journal of the Physical Society of Japan* 35 (1973) 95–107. doi:10.1143/JPSJ.35.95.
- [8] M. R. Hayns, The transient stages of damage accumulation in irradiated materials with point defect clusters, *Journal of Nuclear Materials* 59 (1976) 175–187. doi:10.1016/0022-3115(76)90163-5.
- [9] N. M. Ghoniem, A dynamic rate theory for the response of metals during steady state and pulsed irradiation, Ph.d. thesis (nuclear engineering), University of Wisconsin–Madison (1977).
- [10] N. M. Ghoniem, D. D. Cho, The simultaneous clustering of point defects during irradiation, *Physica Status Solidi (a)* 54 (1979) 171–178. doi:10.1002/pssa.2210540121.
- [11] N. M. Ghoniem, S. Sharafat, A numerical solution to the Fokker–Planck equation describing the evolution of the interstitial loop microstructure during irradiation, *Journal of Nuclear Materials* 92 (1980) 121–135. doi:10.1016/0022-3115(80)90148-8.
- [12] A. D. Brailsford, R. Bullough, The rate theory of swelling due to void growth in irradiated metals, *Journal of Nuclear Materials* 44 (2) (1972) 121–135. doi:10.1016/0022-3115(72)90091-8.
- [13] H. Wiedersich, On the theory of void formation during irradiation, *Radiation Effects* 12 (1972) 111–125. doi:10.1080/00337577208231128.
- [14] L. K. Mansur, Void swelling in metals and alloys under irradiation: an assessment of the theory, *Nuclear Technology* 40 (1) (1978) 5–34. doi:10.13182/NT78-2.
- [15] D. J. Bacon, F. Gao, Y. N. Osetsky, The primary damage state in fcc, bcc and hcp metals as seen in molecular dynamics simulations, *Journal of Nuclear Materials* 276 (2000) 1–12. doi:10.1016/S0022-3115(99)00165-8.
- [16] J. Marian, C. S. Becquart, C. Domain, S. L. Dudarev, M. R. Gilbert, R. J. Kurtz, D. R. Mason, K. Nordlund, A. E. Sand, L. L. Snead, T. Suzudo, B. D. Wirth, Recent advances in modeling and simulation of the exposure and response of tungsten to fusion energy conditions, *Nuclear Fusion* 57 (2017) 092008. doi:10.1088/1741-4326/aa5e8d.
- [17] S. I. Golubov, A. M. Ovcharenko, A. V. Barashev, B. N. Singh, Grouping method for the approximate solution of a kinetic equation describing the evolution of point-defect clusters, *Philosophical Magazine A* 81 (2001) 643–658. doi:10.1080/01418610108212164.
- [18] A. Holt, R. Microstructure of Zr-2.5Nb pressure tubes, *Journal of Nuclear Materials* 159 (1988) 310–338, zirconium CANDU pressure tube microstructure under neutron irradiation; He bubbles at grain boundaries ($\sim 5 \times 10^{20} \text{ m}^{-3}$, $\sim 1.5 \text{ nm}$); β -Zr decomposition; irradiation growth and creep. doi:10.1016/0022-3115(88)90030-1.
- [19] M. Christensen, W. Wolf, C. Freeman, E. Wimmer, R. B. Adamson, L. Hallstadius, P. E. Cantonwine, E. V. Mader, Diffusion of point defects, nucleation of dislocation loops, and effect of hydrogen in hcp-Zr: Ab initio and classical simulations, *Journal of Nuclear Materials* 460 (2015) 82–96. doi:10.1016/j.jnucmat.2015.02.013.

- [20] A. Patra, D. R. Trinkle, First-principles vacancy diffusion and Onsager transport coefficients in Zr, *Computational Materials Science* 170 (2019) 109176. doi:10.1016/j.commatsci.2019.109176.
- [21] A. A. Kohnert, B. D. Wirth, L. Capolungo, Modeling microstructural evolution in irradiated materials with cluster dynamics methods: a review, *Computational Materials Science* 149 (2018) 442–459. doi:10.1016/j.commatsci.2018.02.049.
- [22] M. Maron, Y. Li, I. Lalani, K. Baker, B. Ramirez Flores, T. Black, J. Hollenbeck, N. Ghoniem, G. Po, Spatially-resolved cluster dynamics modeling of irradiation growth, *International Journal of Plasticity* 181 (2024) 103989. doi:10.1016/j.ijplas.2024.103989.
- [23] F. Christien, A. Barbu, Cluster dynamics modelling of irradiation growth of zirconium single crystals, *Journal of Nuclear Materials* 393 (1) (2009) 153–161.
- [24] A. V. Barashev, S. I. Golubov, R. E. Stoller, Theoretical investigation of microstructure evolution and deformation of zirconium under neutron irradiation, *Journal of Nuclear Materials* 461 (2015) 85–94.
- [25] Y. Li, M. Maron, K. Baker, B. Ramirez Flores, T. Black, J. Hollenbeck, I. Lalani, N. Ghoniem, G. Po, Coupled cluster and dislocation dynamics modeling of microstructure evolution in irradiated materials, *Journal of the Mechanics and Physics of Solids* 206 (2025) 106366. doi:10.1016/j.jmps.2025.106366.
- [26] C. Ji, Y. Cui, Y. Li, N. Ghoniem, A concurrent irradiation-mechanics multiscale coupling model, *Journal of the Mechanics and Physics of Solids* 167 (2022) 105005. doi:10.1016/j.jmps.2022.105005.
- [27] Y. Li, G. Po, N. Ghoniem, Coupled cluster-dislocation dynamics of irradiation-induced defects, *Materialia* 14 (2020) 100891.
- [28] C. Woo, Theory of irradiation deformation in non-cubic metals: effects of anisotropic diffusion, *Journal of Nuclear Materials* 159 (1988) 237–256.
- [29] C. H. Woo, The sink strength of a dislocation loop in the effective medium approximation, *Journal of Nuclear Materials* 98 (3) (1981) 279–294. doi:10.1016/0022-3115(81)90154-9.
- [30] C. H. Woo, Intrinsic bias differential between vacancy loops and interstitial loops, *Journal of Nuclear Materials* 107 (1) (1982) 20–30. doi:10.1016/0022-3115(82)90554-3.
- [31] G. Po, A tensorial derivation of the DAD effect, Tech. rep., Department of Mechanical and Aerospace Engineering, University of Miami, internal technical note (2026).
- [32] A. Jostsons, P. Kelly, R. Blake, The nature of dislocation loops in neutron irradiated zirconium, *Journal of Nuclear Materials* 66 (3) (1977) 236–256.
- [33] M. Griffiths, A review of microstructure evolution in zirconium alloys during irradiation, *Journal of Nuclear Materials* 159 (1988) 190–218.
- [34] G. Carpenter, J. Watters, A study of electron irradiation damage in zirconium using a high voltage electron microscope, *Journal of Nuclear Materials* 96 (3) (1981) 213–226.
- [35] L. Tournadre, F. Onimus, J.-L. Béchade, D. Gilbon, J.-M. Cloué, J.-P. Mardon, X. Feaugas, O. Toader, C. Bachelet, Experimental study of the nucleation and growth of c-component loops under charged particle irradiations of recrystallized zircaloy-4, *Journal of nuclear materials* 425 (1-3) (2012) 76–82.
- [36] A. Harte, D. Jädnäs, M. Topping, P. Frankel, C. Race, J. Romero, L. Hallstadius, E. C. Darby, M. Preuss, The effect of matrix chemistry on dislocation evolution in an irradiated zr alloy, *Acta Materialia* 130 (2017) 69–82.
- [37] P. Kelly, R. Blake, The characterization of dislocation loops in neutron irradiated zirconium, *Philosophical Magazine* 28 (2) (1973) 415–426.
- [38] S. I. Choi, J. H. Kim, Radiation-induced dislocation and growth behavior of zirconium and zirconium alloys – a review, *Nuclear Engineering and Technology* 45 (3) (2013) 385–392. doi:10.5516/NET.07.2013.035.
- [39] G. J. C. Carpenter, J. F. Watters, R. W. Gilbert, Dislocations generated by neutron irradiation in zirconium, *Journal of Nuclear Materials* 159 (1988) 86–100.
- [40] X. Yi, M. L. Jenkins, M. Briceno, S. G. Roberts, Z. Zhou, M. A. Kirk, In situ study of self-ion irradiation damage in W and Zr at room temperature, *Philosophical Magazine* 93 (14) (2013) 1715–1738. doi:10.1080/14786435.2012.754110.
- [41] S. Doriot, B. Verhaeghe, J.-L. Béchade, D. Menut, D. Gilbon, J.-P. Mardon, B. Cazalis, J.-L. Bernier, Microstructural evolution of M5 alloy under PWR conditions, *Journal of ASTM International* 12 (2015).
- [42] B. Christiaen, C. Domain, L. Thuinet, A. Ambard, A. Legris, A new scenario for c vacancy loop formation in zirconium-based on atomic-scale modeling, *Acta Materialia* 179 (2019) 93–106.
- [43] C. Dai, C. Varvenne, P. Saidi, Z. Yao, M. R. Daymond, L. K. Béland, Stability of vacancy and interstitial dislocation loops in α -zirconium: atomistic calculations and continuum modelling, *Journal of Nuclear Materials* 554 (2021) 153059.
- [44] C. Varvenne, O. Mackain, E. Clouet, Vacancy clustering in zirconium: An atomic-scale study, *Acta materialia* 78 (2014) 65–77.
- [45] R. Holt, R. Gilbert, < c > component dislocations in annealed zircaloy irradiated at about 570 k, *Journal of Nuclear Materials* 137 (3) (1986) 185–189.
- [46] R. A. Holt, A. R. Causey, Irradiation creep and growth of Zr-2.5Nb pressure tubes, *Journal of Nuclear Materials* 209 (1993) 221–231.
- [47] R. Voskoboinikov, Y. N. Osetsky, D. Bacon, Identification and morphology of point defect clusters created in displacement cascades in α -zirconium, *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* 242 (1-2) (2006) 530–533.
- [48] R. E. Voskoboinikov, Y. N. Osetsky, D. J. Bacon, Identification of primary damage in displacement cascades in α -zirconium, *Journal of Nuclear Materials* 377 (2) (2008) 385–395. doi:10.1016/j.jnucmat.2008.04.002.
- [49] C. Björkäs, K. Nordlund, K. Arstila, J. Keinonen, V. N. Konobeev, C. S. Becquart, Influence of the picosecond defect distribution on damage accumulation in irradiated α -Zr, *Journal of Nuclear Materials* 385 (1) (2009) 154–158. doi:

- [10.1016/j.jnucmat.2008.09.025](https://doi.org/10.1016/j.jnucmat.2008.09.025).
- [50] N. De Diego, Y. N. Osetsky, D. Bacon, Mobility of interstitial clusters in alpha-zirconium, *Metallurgical and materials transactions A* 33 (2002) 783–789.
 - [51] C. Domain, A. Legris, Ab initio atomic-scale determination of point-defect structure in hcp zirconium, *Philosophical Magazine* 85 (4–7) (2005) 569–575. [doi:10.1080/14786430412331334625](https://doi.org/10.1080/14786430412331334625).
 - [52] Q. Peng, W. Ji, H. Huang, S. De, Stability of self-interstitial atoms in hcp-zr, *Journal of Nuclear Materials* 429 (1-3) (2012) 233–236.
 - [53] N. M. Ghoniem, A generalized graph-based cluster dynamics framework for irradiated materials, *Journal of Nuclear Materials* Submitted (2026).
 - [54] S.-M. Liu, I. J. Beyerlein, W.-Z. Han, Two-dimensional vacancy platelets as precursors for basal dislocation loops in hexagonal zirconium, *Nature Communications* 11 (1) (2020) 5766.
 - [55] J. F. March-Rico, S. Blondel, B. D. Wirth, Atomistically-informed cluster dynamics modeling of concurrent interstitial and vacancy a-loop evolution in irradiated alpha-zirconium, *Acta Materialia* 288 (2025) 120777. [doi:10.1016/j.actamat.2025.120777](https://doi.org/10.1016/j.actamat.2025.120777).
 - [56] J. F. March-Rico, B. D. Wirth, Stress states and point defect capture radii of dislocation a-loops and c-loops in alpha-zirconium, *Journal of Nuclear Materials* 587 (2023) 154752. [doi:10.1016/j.jnucmat.2023.154752](https://doi.org/10.1016/j.jnucmat.2023.154752).
 - [57] B. Singh, On the influence of grain boundaries on void growth, *Philosophical Magazine* 28 (6) (1973) 1409–1413.
 - [58] X.-M. Bai, A. F. Voter, R. G. Hoagland, M. Nastasi, B. P. Uberuaga, Efficient annealing of radiation damage near grain boundaries via interstitial emission, *Science* 327 (5973) (2010) 1631–1634.
 - [59] C. Sun, M. Song, K. Yu, Y. Chen, M. Kirk, M. Li, H. Wang, X. Zhang, In situ evidence of defect cluster absorption by grain boundaries in kr ion irradiated nanocrystalline ni, *Metallurgical and Materials Transactions A* 44 (2013) 1966–1974.
 - [60] G. Cheng, W. Xu, Y. Wang, A. Misra, Y. Zhu, Grain size effect on radiation tolerance of nanocrystalline mo, *Scripta Materialia* 123 (2016) 90–94.
 - [61] E. S. Fisher, C. J. Renken, Single-crystal elastic moduli and the hcp $\rightarrow$ bcc transformation in Ti, Zr, and Hf, *Physical Review* 135 (2A) (1964) A482–A494. [doi:10.1103/PhysRev.135.A482](https://doi.org/10.1103/PhysRev.135.A482).

Supplementary Material

The material collected here supports the main text and is not needed to follow it. [Appendix A](#) carries out in full the two reductions that take the general master equation [Eq. \(9\)](#) to the mobile and immobile blocks of [Section 3.6](#), and supplies along the way the definitions of the two flux sums Σ_k^{net} and Σ_k^{gro} and of the floor current Φ_{min}^k , which the main text uses without deriving; it closes with the three rules that keep the moment triple realizable, two of them consequences of the reduction and the third the one closure the immobile block needs beyond the log-normal itself. [Appendix B](#) sets out the finite-element discretization of the mobile block and the Newton iteration that solves it [Appendix C](#) gives the continuum-to-discrete conversion algorithm, from the restricted Voronoi tessellation to the loops the solver is handed. [Appendix D](#) gives the orientation factors that build the capture efficiencies and the co-growth criterion that follows from them; [Appendix E](#) the peripheral emission channel and the loop binding energy it is computed from; and [Appendix F](#) the layout of the integrated state vector.

Appendix A. Derivation of the Two-Block Equations

[Equation \(9\)](#) is one equation per species label α , per defect content n and per mesh node, and [Section 3.1](#) states the two reductions that bring it to [Eqs. \(12\)](#) and [\(26\)](#) to [\(28\)](#) without carrying either out. They are carried out here. Both start from one partition of the index set (α, n) : the mobile set $\mathcal{M}$ of [Eq. \(3\)](#), four species of fixed defect content and finite diffusivity, and its complement, the immobile clusters, grouped by habit and character into the nine families $\mathcal{F}$ of [Eq. \(4\)](#) and indexed within a family by the defect content m alone. That partition is what separates the two reductions, because the terms of [Eq. \(9\)](#) that couple different indices — the pairwise kernel $\mathcal{K}$, the coagulation kernel $\mathcal{C}$, the emission rates ε , the transfer θ and the absorption $\mathcal{D}$ at fixed sinks — each split according to how many of their partners lie in $\mathcal{M}$, and every piece lands in exactly one of the two blocks. [Table A.1](#) is that accounting, and the rest of this appendix is its justification.

One claim is worth making before the algebra, because it is what the accounting is for. Nothing below is added to [Eq. \(9\)](#). The mobile block is that equation restricted to $\mathcal{M}$; the immobile block is the same equation restricted to one family and integrated against m^j for $j = 0, 1, 2$. Every term of [Eqs. \(26\)](#) to [\(28\)](#) is therefore the moment of a term of [Eq. \(9\)](#), including the four — coalescence, the basal transfers, the grain-boundary removal and the dissolution channel — that an earlier presentation of this model appended to the moment equations after the fact. Appending them is what makes their weights look like conventions; deriving them is what fixes those weights, and in one case fixes a sign.

Take the mobile block first. Restricting [Eq. \(9\)](#) to $\alpha \in \mathcal{M}$ and writing c^x for the per-atom concentration of species x , the pairwise sum contributes to the x line every reaction whose two partners are both mobile. Collecting the rate constants [Eq. \(13\)](#) into one matrix $\mathbf{R}_2^x$ per line and stacking the four into $\mathbf{R}_2$, that contribution is the quadratic form whose x -th entry is

$$\left[\frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M \right]_x = \frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M, \quad (\text{A.1})$$

in which the $\frac{1}{2}$ removes the double count that the unrestricted double sum of [Eq. \(9\)](#) performs on each unordered pair $\{m, n\}$ with $m \neq n$. The like-species channels $i+i$ and $2i+2i$ are counted once by that same sum and so carry their combinatorial factors explicitly instead.

Written out, [Eq. \(A.1\)](#) is four symmetric 4×4 matrices of the pairwise constants [Eq. \(13\)](#), one per mobile line. Order the rows and columns as $\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$ and abbreviate $\beta^{x,y} = \beta^{y,x}$ for the constant of the channel $x+y$, so that $\beta^{i,v}$ carries the recombination factor β_r and every other $\beta^{x,y}$ is $\omega^x + \omega^y$ with $\omega^x = z_c \nu_x \exp(-E_x^m/k_B T)$ the jump rate of species x — the geometric prefactor $A^{x,y} a^2/z_c \Omega$ times the capture radius $(r^x + r^y)$ of [Eq. \(13\)](#) being absorbed into $A^{x,y}$, which is unity on every channel but $i+v$. Then

$$\mathbf{R}_2^v = - \begin{bmatrix} 0 & \beta^{i,v} & \beta^{2i,v} & \beta^{3i,v} \\ \beta^{i,v} & 0 & 0 & 0 \\ \beta^{2i,v} & 0 & 0 & 0 \\ \beta^{3i,v} & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{R}_2^i = \begin{bmatrix} 0 & -\beta^{i,v} & +\beta^{2i,v} & 0 \\ -\beta^{i,v} & -2\beta^{i,i} & -\beta^{i,2i} & -\beta^{i,3i} \\ +\beta^{2i,v} & -\beta^{i,2i} & 0 & 0 \\ 0 & -\beta^{i,3i} & 0 & 0 \end{bmatrix}, \quad (\text{A.2})$$

Table A.1: Where each term of Eq. (9) goes. The coupling terms split by how many of their partners are mobile, no piece appears twice and none is left over — which is the bookkeeping statement behind the identity between Eq. (16) and Eq. (18), and behind the claim that the immobile block is a moment of Eq. (9) rather than a moment plus four appended channels.

Term of Eq. (9)	Mobile block, Eq. (12)	Immobile block, Eqs. (26) to (28)
$-\nabla \cdot \mathbf{J}_n^\alpha$	$\nabla \cdot (\mathbf{D}_{\text{ten}} \nabla \mathbf{c}_M)$	absent: $\mathbf{D}_n^\alpha = \mathbf{0}$ for an immobile cluster
G_n^α	$\mathbf{P}_M = G \varepsilon \mathbf{f}$	the nucleation currents J_{sL}^k at m_k^{nuc}
$\mathcal{K}$, both partners mobile	$\frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M$	feeds J_{iL}^{ak} through the $2i+2i$, $i+3i$ and $2i+3i$ channels
$\mathcal{K}$, one partner mobile	$-\mathbf{D}_d \mathbf{K}_L \mathbf{c}_M$	the drift and diffusion of the size ladder, F_k and D_k
$\mathcal{C}$, both partners immobile	—	like-loop coalescence: $-\Delta N_{LL}$ on n , zero on c , $+\Delta_k \bar{m}^2 \Delta N_{LL}$ on q
ε , mobile emitter	$\mathbf{E} \mathbf{c}_M$	—
ε , loop emitter	$\mathbf{S}_{vL}$	the emission channel inside Σ_k^{net}
$\theta_n^{\beta \rightarrow \alpha}$	—	the basal transfers $\pm \Gamma_\theta^n, \pm \Gamma_\theta^c, \pm \Gamma_\theta^q$
$\mathcal{D}_N^\alpha(n)$	$-\text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} \mathbf{c}_M$	loop–network coalescence, ΔN_{LN} and ΔC of Eq. (23)
$\mathcal{D}_{gb}^\alpha(n)$	Eq. (10), the Dirichlet condition	the gated boundary removal $\nu_{gb} \Theta_j^k$
$\mathcal{D}_{\text{van}}^\alpha(n)$	the returned defects, in $\mathbf{S}_{vL}$	the complementary gate $\nu_{\text{van}} \Xi_j^k$

$$\mathbf{R}_2^{2i} = \begin{bmatrix} 0 & 0 & -\beta^{2i,v} & +\beta^{3i,v} \\ 0 & +\beta^{i,i} & -\beta^{i,2i} & 0 \\ -\beta^{2i,v} & -\beta^{i,2i} & -4\beta^{2i,2i} & -\beta^{2i,3i} \\ +\beta^{3i,v} & 0 & -\beta^{2i,3i} & 0 \end{bmatrix}, \quad \mathbf{R}_2^{3i} = \begin{bmatrix} 0 & 0 & 0 & -\beta^{3i,v} \\ 0 & 0 & +\beta^{i,2i} & -\beta^{i,3i} \\ 0 & +\beta^{i,2i} & 0 & -\beta^{2i,3i} \\ -\beta^{3i,v} & -\beta^{i,3i} & -\beta^{2i,3i} & 0 \end{bmatrix}. \quad (\text{A.3})$$

Each matrix is read one entry at a time, and the entry is a stoichiometric coefficient rather than a rate: the (x, y) position of $\mathbf{R}_2^\alpha$ is $\beta^{x,y}$ times the net number of α produced by one $x+y$ event, negative when α is one of the two reactants. The four off-diagonal patterns are the four channels that keep a mobile product. Recombination $i+v \rightarrow \emptyset$ appears as $-\beta^{i,v}$ in $\mathbf{R}_2^v$ and in $\mathbf{R}_2^i$ and nowhere else. Its two cluster analogues do keep a product: $2i+v \rightarrow i$ debits c^v and c^{2i} and credits c^i , which is the $+\beta^{2i,v}$ sitting in $\mathbf{R}_2^i$, and $3i+v \rightarrow 2i$ likewise puts $+\beta^{3i,v}$ into $\mathbf{R}_2^{2i}$. Clustering $i+2i \rightarrow 3i$ is the $-\beta^{i,2i}$, $-\beta^{i,2i}$, $+\beta^{i,2i}$ column of entries running down the three interstitial matrices.

The diagonals are where the combinatorial factors of the preceding paragraph sit, and they are the reason the four matrices cannot be read off the reaction list by inspection. For $i+i \rightarrow 2i$ the event rate is $\frac{1}{2}\beta^{i,i}(c^i)^2$, each event consumes *two* monomers and makes one dimer, so the (i, i) entry is $-2\beta^{i,i}$ in $\mathbf{R}_2^i$ and $+\beta^{i,i}$ in $\mathbf{R}_2^{2i}$: a factor of two apart, and of opposite sign, from one channel. The $(2i, 2i)$ entry is $-4\beta^{2i,2i}$ for the same reason once, and once again because $2i+2i$ leaves the mobile set entirely — it nucleates a loop, so there is no positive counterpart anywhere.

That last remark is the structural statement the matrices make. Five of the eight channels stay inside $\mathcal{M}$ and conserve defect content exactly: with $\mathbf{m}_M = [-1, +1, +2, +3]^T$ the signed content of Eq. (3), the identity $\sum_\alpha m^\alpha [\mathbf{R}_2^\alpha]_{xy} = 0$ holds at the (i, v) , $(2i, v)$, $(3i, v)$, $(i, 2i)$ and (i, i) positions — the $i+2i$ column, for instance, gives $1(-\beta) + 2(-\beta) + 3(+\beta) = 0$. It fails at the remaining three, $(i, 3i)$, $(2i, 2i)$ and $(2i, 3i)$, and that failure is not an error: those are the clustering channels, they have no mobile product, and the content they remove from $\mathbf{c}_M$ is exactly what Table A.1 sends to the nucleation current J_{iL}^{ak} of the immobile block — 4 defects per $i+3i$ event, 2 per $2i$ lost to $2i+2i$, and 5 per $2i+3i$. The mobile block does not conserve defects on its own; the two blocks together do, and the three non-vanishing entries of $\sum_\alpha m^\alpha \mathbf{R}_2^\alpha$ are the audit of that transfer.

Reactions with one mobile and one immobile partner are, at frozen immobile state, linear in $\mathbf{c}_M$, and so are the fixed-sink terms $\mathcal{D}_n^\alpha C_n^\alpha$. Together they give $-\mathcal{D} \mathbf{c}_M$ of Eq. (14): the network through the sink concentrations Eq. (15), the loops through the sink-strength block Eq. (16). This is the only place the

immobile state enters the mobile block, and Eq. (16) builds it from the same efficiencies $Z_{sk,m}$ that drive growth in Eq. (18). What a family absorbs is therefore what the mobile species lose, family by family and species by species, which is the operational content of the word self-consistent and the reason Table A.1 has no row appearing twice.

The emission terms divide by emitter. A mobile cluster emitting a point defect moves within $\mathcal{M}$ and gives the linear term $\mathbf{E}\mathbf{c}_M$, with $\alpha^{m,n} = \beta^{m,n} \exp(-E_{m,n}^b/k_B T)$; a loop emitting a vacancy across its own perimeter puts that vacancy into the pool and gives the source $\mathbf{S}_{vL}$ of Appendix E. What is left is the transport term and the time derivative. Since $\mathbf{D}_n^\alpha$ vanishes on every immobile index, $-\nabla \cdot \mathbf{J}_n^\alpha$ survives on $\mathcal{M}$ alone and becomes $\nabla \cdot (\mathbf{D}_{\text{ten}} \nabla \mathbf{c}_M)$; and the time derivative is dropped by the scale separation of Section 4, with $\tau_M \sim (k^2 \bar{D})^{-1}$ of order microseconds against immobile fields evolving over 10^7 s. Assembling the pieces gives Eq. (12), closed by Eq. (17). Note what that limit does and does not say: the mobile concentrations are not constant in time, since they track the immobile state through $\mathbf{D}$ at every instant, but they carry no memory of it, which is what permits the fast solve to be called at intervals rather than at every step.

The second reduction applies to the complement of $\mathcal{M}$, where transport is absent outright and the remaining question is what becomes of the size index. Fix a family $(s, k) \in \mathcal{F}$ and let $f_k(m, t) dm$ be the number of its loops per host atom holding between m and $m + dm$ defects, so that the fields the block carries are its first three moments Eq. (11). The pairwise terms with exactly one mobile partner move a loop along m and nowhere else: absorbing a mobile species x carries the loop by $\mu_x = |m^x| \in \{1, 2, 3\}$, the defect content of the arriving cluster, upward when the polarity of x matches the family's character and downward when it does not, and a thermal emission is the same move with $\mu = 1$. Restricted to one family, Eq. (9) is therefore a birth–death process on the integer m ,

$$\begin{aligned} \frac{\partial f_k}{\partial t} = \sum_x \Big[& W_x^+(m - \mu_x) f_k(m - \mu_x) + W_x^-(m + \mu_x) f_k(m + \mu_x) \\ & - (W_x^+(m) + W_x^-(m)) f_k(m) \Big] + J_{sL}^k \delta_{m, m_k^{\text{nuc}}}, \end{aligned} \quad (\text{A.4})$$

with W_x^+ and W_x^- the rates of the matching-polarity and the opposite-polarity step. Those rates are not free. A loop captures on its perimeter, so $W_x^\pm \propto R_{sL}^k = \lambda_k \sqrt{m}$ by Eq. (5), and the constant of proportionality is the same efficiency and the same mobility that Eq. (18) already carries,

$$W_x^\pm(m) = \pi_k \sqrt{m} \bar{D}^x Z_{sk,x} c^x, \quad W^{\text{emit}}(m) = \pi_k \sqrt{m} \bar{D}^v Z_{sk,v} c_k^{v,\text{eq}}(m), \quad \pi_k = \frac{\lambda_k \xi_k}{\ell}, \quad (\text{A.5})$$

so that the geometric prefactor $\Pi_{sL}^k = \pi_k \sqrt{n_{sL}^k c_{sL}^k}$ of Eq. (19) is $\pi_k n_{sL}^k \langle m^{1/2} \rangle_k$ evaluated on a monodisperse family. No parameter enters the size distribution that the growth law does not already contain.

W^{emit} is peripheral emission, Appendix E, and it belongs in Eq. (A.4) as a fourth process rather than as a correction applied afterwards. It has the same prefactor and the same efficiency as vacancy capture — it is the same event run backwards — and it steps the loop by $\mu_v = 1$ in the direction opposite to that capture, against the loop's own equilibrium concentration $c_k^{v,\text{eq}}$ of Eq. (E.1) rather than against the field. Carrying it here rather than bolting it on is what makes the opposite signs it acquires in the two flux sums below a consequence instead of a convention. Thermal *interstitial* emission has no counterpart term: it is $\sim 10^{-8}$ of this at 873 K and is dropped.

Treating m as continuous and expanding the shifted terms of Eq. (A.4) about m , the zeroth-order terms cancel the loss term identically — which is the statement that a size jump neither creates nor destroys a loop — and what survives is the Kramers–Moyal series

$$\frac{\partial f_k}{\partial t} = \sum_{p \geq 1} \frac{(-1)^p}{p!} \frac{\partial^p}{\partial m^p} [a_p(m) f_k] + J_{sL}^k \delta(m - m_k^{\text{nuc}}), \quad a_p = \sum_x \mu_x^p [W_x^+ + (-1)^p W_x^-]. \quad (\text{A.6})$$

Truncating Eq. (A.6) at $p = 2$ is the Fokker–Planck equation for the size density,

$$\frac{\partial f_k}{\partial t} = -\frac{\partial}{\partial m} [F_k f_k] + \frac{\partial^2}{\partial m^2} [D_k f_k] + J_{sL}^k \delta(m - m_k^{\text{nuc}}), \quad (\text{A.7})$$

whose drift is the difference of the two directions and whose diffusion is half their sum,

$$F_k = a_1 = \pi_k \sqrt{m} \Sigma_k^{\text{net}}, \quad D_k = \frac{1}{2} a_2 = \frac{1}{2} \pi_k \sqrt{m} \Sigma_k^{\text{gro}}, \quad (\text{A.8})$$

$$\Sigma_k^{\text{net}} = \eta_s \sum_{x \in \mathcal{M}} \varsigma_s(x) \mu_x \bar{D}^x Z_{sk,x} (c^x - c_k^{v,\text{eq}} \delta_{x,v}), \quad \Sigma_k^{\text{gro}} = \sum_{x \in \mathcal{M}} \mu_x^2 \bar{D}^x Z_{sk,x} (c^x + c_k^{v,\text{eq}} \delta_{x,v}). \quad (\text{A.9})$$

Equation (A.9) is where the two flux sums the main text uses are defined, and the two of them differ in *three* places, not one. The obvious one is the sign: Σ_k^{net} carries $\varsigma_s(x)$ and Σ_k^{gro} does not, because a defect of either polarity moves the loop along m and a move in either direction spreads the distribution. The second is the power of the step. Σ_k^{gro} is not the drift sum with its signs made positive: the second Kramers–Moyal coefficient weights each channel by μ_x^2 , one power more than the drift, because variance accumulates as the square of the step. For the monomer channels the two weightings coincide and the distinction is invisible, and for $2i$ and $3i$ it is not — those two carry 3.1% of the interstitial arrival in Σ_k^{net} and between 6% and 9% of Σ_k^{gro} , so using the drift weighting for both under-estimates the spreading by 3 to 6%.

The third is the one most easily missed, and it is the reason $c_k^{v,\text{eq}}$ of Eq. (E.1) appears in both with *opposite* signs, both of them inherited from W^{emit} through a_1 and a_2 of Eq. (A.6). The drift subtracts it, a_1 differencing the two directions, so the loop is driven not by c^v but by its departure from the loop’s own equilibrium and the channel vanishes identically at $c^v = c_k^{v,\text{eq}}$ — detailed balance by construction. That is the same statement as Eq. (E.4) entering Eq. (27) with the opposite sign, written as one substitution instead of two terms, and with it Σ_k^{net} is exactly the signed sum multiplying Π_{sL}^k in Eq. (18) together with that emission channel: the drift of Eq. (A.7) *is* the growth law and carries no independent content. The spreading takes the *sum* $c^v + c_k^{v,\text{eq}}$, because absorption and emission are independent events that each add variance; they do not cancel. A family sitting exactly at its own equilibrium therefore has $\Sigma_k^{\text{net}} = 0$ and $\Sigma_k^{\text{gro}} > 0$: it does not grow, and it keeps broadening. That is the same conclusion Eq. (41) reaches from the other side, and it is the strongest argument in the formulation for carrying q at all. Only the vacancy species carries the substitution, thermal interstitial emission being $\sim 10^{-8}$ of it (Appendix E); and both reduce to the plain c^x when the emission channel is switched off, which is what makes that switch a clean regression.

The truncation is second order in a parameter that is not the jump size but the jump size measured against the width of the distribution: successive terms of Eq. (A.6) carry one more power of $\mu_x \partial_m$, so the ratio of term $p+1$ to term p is of order μ_x / σ_m with $\sigma_m^2 = (\bar{m}_{sL}^k)^2 (\Delta_k - 1)$, which is where the dispersion enters this appendix and is worth restating. Δ_k is the third carried moment expressed as a shape: by Eq. (43) it is $\Delta_k = q_{sL}^k n_{sL}^k / (c_{sL}^k)^2 = \langle m^2 \rangle_k / \langle m \rangle_k^2$, the second moment of the family’s size distribution divided by the square of its first, so it is a pure number, it is 1 exactly when every loop in the family has the same size, and it exceeds 1 by the relative variance, $\Delta_k - 1 = \text{var}(m) / \bar{m}^2$. It is dimensionless because q , n and c enter it in the combination that cancels m ; it carries no information about how big the loops are, only about how unequal they are; and under the log-normal closure it is the shape parameter outright, $\Delta_k = e^{s_k^2}$, so $s_k = \sqrt{\ln \Delta_k}$ is the standard deviation of $\ln m$. Three things make that small here, and only the first is about the absorption step alone. The mobile set closes at $3i$, so $\mu_x \leq 3$ against family means of 10^2 to 10^5 defects, a ratio whose worst value anywhere on the reference march is 1.6×10^{-2} — reached by the prismatic interstitial family at 10^{-4} dpa, where it has barely left its nucleus of 300. The rates go as $\sqrt{m}$ by Eq. (A.5), so their relative change over one event is $\mu_x / 2m$, the same smallness again and half of it; a rate varying exponentially in m would not permit the truncation however small the step. And the distribution is broad in units of the step, which is the condition the expansion parameter actually names: a loop that has made N_{ex} exchange events has $\sigma_m = \mu \sqrt{N_{\text{ex}}}$ and hence $\mu / \sigma_m = N_{\text{ex}}^{-1/2}$, with $N_{\text{ex}} = 2\mu^{-2} (D_k / F_k) \Delta m$ — so the same near-cancellation that makes the second moment necessary, D_k / F_k between 7 and 44, also multiplies the number of exchanges a loop has made for a given net growth. The basal family reaches $N_{\text{ex}} \simeq 1.5 \times 10^7$ and the prismatic one 3.5×10^4 , giving μ / σ_m of 2.6×10^{-4} and 5×10^{-3} .

Two places where the expansion is worst are worth naming, because at both of them the model works outside Eq. (A.7) rather than relying on it. A loop that has just nucleated has made no exchanges at all, its distribution is a delta function, and no gradient expansion describes it; that defect is real, is confined to a region of width $\sim \mu_x$ about m_k^{nuc} , which is a percent of the nucleus, and is cured by the first few exchange

events. The floor $m_{\min}$ is the other, where f_k is cut off and its gradient is largest, and there the model carries an explicit current, derived next, instead of a Fokker–Planck flux.

Multiplying Eq. (A.7) by m^j and integrating over $[m_{\min}, \infty)$ turns it into a hierarchy in the moments. Integrating the drift term once by parts and the diffusion term twice, with f_k and its derivative vanishing as $m \rightarrow \infty$,

$$\frac{d}{dt} \int_{m_{\min}}^{\infty} m^j f_k dm = j \int m^{j-1} F_k f_k dm + j(j-1) \int m^{j-2} D_k f_k dm + J_{sL}^k (m_k^{\text{nuc}})^j - m_{\min}^j \Phi_{\min}^k, \quad (\text{A.10})$$

the last term being the boundary contribution at the floor, with

$$\Phi_{\min}^k = \left[D_k \partial_m f_k + (\partial_m D_k - F_k) f_k \right]_{m=m_{\min}}, \quad f_k(m_{\min}) = \frac{n_{sL}^k}{m_{\min} s_k \sqrt{2\pi}} \exp \left[-\frac{(\ln m_{\min} - \mu_k)^2}{2s_k^2} \right], \quad (\text{A.11})$$

the Fokker–Planck flux through the floor, evaluated on Eq. (43) and therefore analytic in the three carried moments. Equation (A.11) is the floor current that Eqs. (26) and (28) write as $\Phi_{\min}^k$. Two properties of it should be recorded. Because a delta function has no tail, $\Phi_{\min}^k \equiv 0$ at $\Delta_k = 1$: in a two-moment model the dissolution rate is not small but identically zero, which is why such a model under a pure anneal gives a falling mean size at exactly constant density, where a measurement reports the density falling and the surviving mean rising. And the exact integration by parts leaves one further boundary term, $j m_{\min}^{j-1} D_k f_k$ evaluated at $m = m_{\min}$, which Eq. (A.10) drops. Dropping it is the statement that loops leaving through the floor are booked at exactly $m_{\min}$ defects each, so that they debit $\Phi_{\min}^k$, $m_{\min} \Phi_{\min}^k$ and $m_{\min}^2 \Phi_{\min}^k$ from the three moments and release $m_{\min} \Phi_{\min}^k$ vacancies to the pool. That is exactly conservative, which retaining the term would not be.

The three cases are then taken in turn. At $j = 0$ both integrals vanish, j and $j(j-1)$ being zero together: drift and diffusion move loops along the size axis and can neither create nor destroy one, so the density changes only by nucleation and by the floor current. At $j = 1$ the diffusion integral still vanishes — spreading a distribution at fixed mean moves no defects — and the drift integral is, with F_k from Eq. (A.8),

$$\int_{m_{\min}}^{\infty} F_k f_k dm = \pi_k \Sigma_k^{\text{net}} n_{sL}^k \langle m^{1/2} \rangle_k = \Pi_{sL}^k \Sigma_k^{\text{net}} \Delta_k^{-1/8}, \quad (\text{A.12})$$

which is the growth law Eq. (18), reduced by the one factor $\Delta_k^{-1/8}$ that a spread family contributes because $\sqrt{\cdot}$ is concave and a spread family therefore presents less perimeter than a monodisperse one of the same density and content. At $j = 2$ both integrals survive,

$$2 \int m F_k f_k dm = 2 \Pi_{sL}^k \Sigma_k^{\text{net}} \bar{m}_{sL}^k \Delta_k^{3/8}, \quad 2 \int D_k f_k dm = \Pi_{sL}^k \Sigma_k^{\text{gro}} \Delta_k^{-1/8}, \quad (\text{A.13})$$

and their sum is Eq. (41) term for term.

Each of Eqs. (A.12) and (A.13) needs a half-integer moment of f_k , and supplying those is the whole work of the closure. With f_k log-normal in m and its two parameters fixed by the three carried moments through Eq. (43),

$$\langle m^j \rangle_k = (\bar{m}_{sL}^k)^j \Delta_k^{j(j-1)/2} \quad \text{for every real } j, \quad (\text{A.14})$$

which returns $\bar{m}^{1/2} \Delta_k^{-1/8}$ at $j = \frac{1}{2}$ and $\bar{m}^{3/2} \Delta_k^{3/8}$ at $j = \frac{3}{2}$ — the two exponents that appear in Eq. (41) and the only ones the model needs. A Gaussian closure gives neither in closed form and admits $m < 0$, and that, rather than positivity alone, is why the closure is log-normal. At $\Delta_k = 1$, Eq. (A.14) returns $\bar{m}^j$ for every j , and everything in this appendix collapses onto the two-moment model exactly, term by term and without any rate being set to zero, which is the regression test on the third moment.

Every power of Δ_k in Eqs. (A.12) and (A.13) arrives the same way, and none of them is an additional modelling assumption. A loop captures on its perimeter, so by Eqs. (5) and (A.5) both Fokker–Planck coefficients carry $\sqrt{m}$, and each of the three integrals is therefore a *fractional* moment of f_k : $\langle m^{1/2} \rangle_k$ from

the drift at $j = 1$, $\langle m^{3/2} \rangle_k$ from the drift at $j = 2$, and $\langle m^{1/2} \rangle_k$ again from the diffusion at $j = 2$, D_k carrying one power of m less than mF_k . The prefactor is the same integral evaluated as though every loop in the family had the mean size, $\Pi_{sL}^k = \pi_k \sqrt{n_{sL}^k c_{sL}^k} = \pi_k n_{sL}^k (\bar{m}_{sL}^k)^{1/2}$, so writing an integral as Π_{sL}^k times something is precisely dividing the true moment by its monodisperse value, and by Eq. (A.14) that ratio is a power of Δ_k and of nothing else:

$$\frac{\langle m^j \rangle_k}{(\bar{m}_{sL}^k)^j} = \Delta_k^{j(j-1)/2}, \quad j = \frac{1}{2} \Rightarrow \Delta_k^{-1/8}, \quad j = \frac{3}{2} \Rightarrow \Delta_k^{+3/8}. \quad (\text{A.15})$$

That is the whole origin of the two exponents. The $-1/8$ of Eq. (A.12) and of the second half of Eq. (A.13) is $j(j-1)/2$ at $j = \frac{1}{2}$; the $+3/8$ of the first half of Eq. (A.13) is the same expression at $j = \frac{3}{2}$.

The sign of the exponent is the sign of the curvature, which is what makes the two factors pull in opposite directions. $j(j-1)/2$ is negative for $0 < j < 1$ and positive for $j > 1$, so by Jensen's inequality the concave $m^{1/2}$ gives a family that is short of its monodisperse perimeter and the convex $m^{3/2}$ one that is over it: at fixed density and content, spreading a family reduces what it absorbs and increases how fast its second moment grows. The magnitudes are not equal — $3/8$ against $1/8$ — so broadening feeds back on q three times as strongly, in the exponent, as it withdraws from c . A family whose size distribution has a relative variance of one, $\Delta_k = 2$, therefore presents $\Delta_k^{-1/8} = 0.92$ of the perimeter a monodisperse family of the same n and c would, and broadens at $\Delta_k^{+3/8} = 1.30$ of the monodisperse rate. Both factors are 1 at $\Delta_k = 1$, which is the two-moment model.

Collecting the three cases gives the Fokker–Planck part of the block. Writing Eq. (A.10) at $j = 0, 1, 2$ and substituting Eqs. (A.12) to (A.14),

$$\dot{n}_{sL}^k|_{\text{FP}} = J_{sL}^k - \Phi_{\min}^k, \quad (\text{A.16})$$

$$\dot{c}_{sL}^k|_{\text{FP}} = \underbrace{\Pi_{sL}^k \Sigma_k^{\text{net}} \Delta_k^{-1/8}}_{= \dot{c}_{sL}^k|_{\text{growth}}} + \underbrace{J_{sL}^k m_k^{\text{nuc}}}_{= G_k} - m_{\min} \Phi_{\min}^k, \quad (\text{A.17})$$

$$\dot{q}_{sL}^k|_{\text{FP}} = \underbrace{2\Pi_{sL}^k \left[\Sigma_k^{\text{net}} \bar{m}_{sL}^k \Delta_k^{3/8} + \frac{1}{2} \Sigma_k^{\text{gro}} \Delta_k^{-1/8} \right]}_{\text{Eq. (41)}} + J_{sL}^k (m_k^{\text{nuc}})^2 - m_{\min}^2 \Phi_{\min}^k. \quad (\text{A.18})$$

Three identifications close the gap between these and Eqs. (26) to (28), and each is a definition rather than a step. The drift integral at $j = 1$ is the growth law: Eq. (A.12) returns $\Pi_{sL}^k \Sigma_k^{\text{net}}$ reduced by $\Delta_k^{-1/8}$, and that factor is carried on Π_{sL}^k itself, so the growth term of Eq. (27) and the drift of Eq. (A.7) are one quantity written twice. The cascade content rate is the nucleation current weighted by the size it deposits at, $G_k = J_{sL}^k m_k^{\text{nuc}}$, which is why Eq. (27) needs no separate nucleation symbol where Eqs. (26) and (28) carry J_{sL}^k and $J_{sL}^k (m_k^{\text{nuc}})^2$. And the $j = 2$ pair is Eq. (41) term for term, the first half the drift and the second the spreading.

Equations (A.16) to (A.18) are everything Eq. (A.7) can produce, and they are not yet the block. What is missing is not missing physics: it is the four terms of Eq. (9) that the restriction to one family leaves standing but that do not move a loop along m , and so never entered Eq. (A.4). They are the coagulation $\mathcal{C}$, the transfer θ , and the removals $\mathcal{D}_N$, $\mathcal{D}_{gb}$ and $\mathcal{D}_{\text{van}}$. Each is taken in turn by the same operation as before — multiply by m^j , integrate — and the weights that Eqs. (26) to (28) carry come out of the integral rather than being assigned to it.

Coalescence has both partners immobile, which is what makes it a coagulation rather than a transition on the size ladder. Restricting the $\mathcal{C}$ terms of Eq. (9) to one family and writing $M_j = \int m^j f_k dm$, the gain and loss combine under the substitution $m \rightarrow m + m'$ into a single symmetric double integral,

$$\left. \frac{dM_j}{dt} \right|_{\mathcal{C}} = \frac{1}{2} \iint \mathcal{C}_k(m, m') f_k(m) f_k(m') [(m + m')^j - m^j - m'^j] dm dm', \quad (\text{A.19})$$

whose bracket is -1 at $j = 0$, 0 at $j = 1$ and $2mm'$ at $j = 2$. The three statements the model makes about coalescence are therefore all properties of Eq. (9): one loop is lost per event; content is conserved *identically*,

not approximately; and the second moment *rises*, the integrand at $j = 2$ being non-negative. That last is the sign most easily inverted, and taking it from the integral rather than from a rule about a debit is what makes it a theorem about a coagulation operator; the consequences of getting it wrong are collected at the end of this appendix.

What Eq. (A.19) does not do on its own is close. The model carries no kernel $\mathcal{C}_k$; it carries the mean-field Avrami rate of Eq. (23), which is the $j = 0$ integral, $\Delta N_{LL} = \frac{1}{2} \iint \mathcal{C}_k f_k f_k$. The $j = 1$ result needs nothing further. The $j = 2$ result needs $\iint mm' \mathcal{C}_k f_k f_k$, which asks which *pairs* coalesce and is not determined by ΔN_{LL} alone. Two closures bracket it. A size-independent kernel gives $\iint mm' \mathcal{C}_k f_k f_k = \mathcal{C}_k (c_{sL}^k)^2$ against $\Delta N_{LL} = \frac{1}{2} \mathcal{C}_k (n_{sL}^k)^2$, hence $2\bar{m}^2 \Delta N_{LL}$; the shape-preserving closure stated below as rule 3, which holds Δ_k fixed through the merge and is the one implemented, gives $\Delta_k \bar{m}^2 \Delta N_{LL}$. Both are positive, and the choice between them moves a magnitude and never the sign.

What the choice does move is the *shape*, and the two closures differ there in kind rather than in degree. Under the implemented one the merge is exactly shape-neutral. With $dc_{sL}^k = 0$ on that channel,

$$\left. \frac{\dot{\Delta}_k}{\Delta_k} \right|_{LL} = \frac{\dot{q}_{sL}^k}{q_{sL}^k} + \frac{\dot{n}_{sL}^k}{n_{sL}^k} - \frac{2\dot{c}_{sL}^k}{c_{sL}^k} = \frac{\Delta_k \bar{m}^2 \Delta N_{LL}}{\Delta_k \bar{m}^2 n_{sL}^k} - \frac{\Delta N_{LL}}{n_{sL}^k} = 0, \quad (\text{A.20})$$

and the loop-network channel, being a removal in proportion, leaves Δ_k alone for the same reason. So coalescence as this model carries it is a pure translation of the family to larger sizes: the density falls, the content is untouched, the mean $\bar{n}_{sL}^k = c_{sL}^k / n_{sL}^k$ rises, and the width in units of the mean does not change. That is also why ΔQ is not an optional refinement. Delete it and Eq. (A.20) returns $-\Delta N_{LL} / n_{sL}^k$ instead of zero, so Δ_k falls at the rate at which coalescence removes loops — 98% of the loop-number loss — and reaches the unrealizable region in a few dpa.

A true coagulation is not shape-neutral, and the same combination evaluated on the size-independent kernel says where it goes instead:

$$\left. \frac{\dot{\Delta}_k}{\Delta_k} \right|_{LL} = \frac{\Delta N_{LL}}{n_{sL}^k} \left(\frac{2}{\Delta_k} - 1 \right), \quad (\text{A.21})$$

which vanishes only at $\Delta_k = 2$ and carries the sign of $2 - \Delta_k$ everywhere else. The three moment equations integrate in closed form on that kernel — $n(t) = n_0 / (1 + \frac{1}{2} \mathcal{C}_k n_0 t)$, c constant, $q(t) = q_0 + \mathcal{C}_k c^2 t$ — and give $\Delta_k(t) \rightarrow 2$ from either side whatever $\Delta_k(0)$ is. That number is not arithmetic: the self-similar solution of constant-kernel coagulation is exponential in m , and an exponential has $\langle m^2 \rangle = 2\langle m \rangle^2$. $\Delta_k = 2$ is therefore the coagulation's own attractor, and the two closures agree exactly there because that is the one shape a merge leaves alone.

Naming the approximation that way is more useful than a bound on it. The shape-preserving closure holds Δ_k where it finds it, where the kernel would relax it toward 2; on the two interior families of Section 7 it therefore under-broadens the basal one, at $\Delta_k = 1.59$, and under-narrows the prismatic interstitial one, at 2.71. Both errors are in the width alone, neither touches n or c directly, and both reach the rest of the model only through $\Delta_k^{-1/8}$ on Π_{sL}^k and through the gates. Taking the distance to the fixed point as the largest error the closure can make in Δ_k bounds the first of those at $(2/\Delta_k)^{-1/8}$, which is 0.972 on the basal family and 1.039 on the prismatic one — under 4% either way, because a one-eighth power is a poor lever. That is why this is the last approximation in the appendix worth worrying about rather than the first, and it is a bound on the *prefactor*: the gates carry Δ_k through $s_k = \sqrt{\ln \Delta_k}$ and are not bounded by the same argument.

The loop-network channel is not a coagulation at all, and separating it from the like-loop channel is what makes its weights exact. The network is a fixed sink, so it enters Eq. (9) as $\mathcal{D}_N^\alpha(n)$, and over one family the Avrami factor of Eq. (23) is built from the family mean and so carries no dependence on m . A first-order removal at an m -independent rate ν takes every moment down in proportion,

$$\left. \frac{dM_j}{dt} \right|_{\mathcal{D}_N} = -\nu M_j \implies (-\Delta N_{LN}, -\Delta C, -\Delta_k \bar{m}^2 \Delta N_{LN}) \quad (\text{A.22})$$

on (n, c, q) , using $\Delta C = \nu c_{sL}^k = \bar{m} \Delta N_{LN}$ and $q_{sL}^k = \Delta_k \bar{m}^2 n_{sL}^k$. No closure is involved: the q debit is $-\nu q$ and nothing else. The implementation carries the two channels as one shape-preserving expression, $\Delta_k(\bar{m}^2 \Delta N - 2\bar{m} \Delta C)$ with $\Delta N = \Delta N_{LL} + \Delta N_{LN}$, and splitting it recovers exactly the two results above — $+\Delta_k \bar{m}^2 \Delta N_{LL}$ from the merge and $-\Delta_k \bar{m}^2 \Delta N_{LN}$ from the removal. That the second is exact and only the first rests on a closure is the useful thing to know about that expression, and it is not visible until the two channels are derived apart.

The basal transfers Eq. (25) are the θ term of Eq. (9), which moves a cluster between families at fixed defect content. Its m^j integral over the donor's gated distribution is $\Gamma_\theta^n, \Gamma_\theta^c, \Gamma_\theta^q$, and because the same term appears in the donor's equation with a minus and the receiver's with a plus, the $j = 1$ pair cancels identically: content is conserved across the chain by the structure of the term, not by a constraint imposed on its rates. Only the number changes, the two states storing different content per unit area.

The last two are the removals $\mathcal{D}_{gb}$ and $\mathcal{D}_{van}$, and they are one integral read in opposite senses because they are the same kind of term gated on opposite sides. With $\mathcal{D}_{gb}^\alpha(m) = \nu_{gb} \mathbf{1}[m \geq m_{gb}]$ from Eq. (45),

$$\left. \frac{dM_j}{dt} \right|_{gb} = -\nu_{gb} \int_{m_{gb}}^\infty m^j f_k dm = -\nu_{gb} \Phi_k^{(j)}(m_{gb}) M_j = -\nu_{gb} \Theta_j^k, \quad (\text{A.23})$$

by the definition Eq. (46) of $\Phi_k^{(j)}$ as the fraction of the j -th moment above a threshold, and those defects leave the crystal. $\mathcal{D}_{van}^\alpha(m) = \nu_{van} \mathbf{1}[m \leq m_{van}]$ gives the complement, $-\nu_{van}[1 - \Phi_k^{(j)}(m_{van})]M_j = -\nu_{van} \Xi_j^k$ of Eq. (24), and those defects return to the mobile pool through $\mathbf{S}_{vL}$. Both are realizable without a rule: what remains after removing the mass of a distribution on one side of a threshold is a genuine sub-measure, so $\Delta_k \geq 1$ survives by construction. That is the same argument as Eq. (A.19) at $j = 2$ made on the other side — these two channels cannot break realizability, and the coagulation cannot either, so the only channels that need a rule are the ones that carry no distribution and no gate.

Collecting Eqs. (A.19), (A.22) and (A.23) and the transfer with Eqs. (A.16) to (A.18) gives

$$\begin{aligned} \dot{n}_{sL}^k &= J_{sL}^k - \Delta N - \Phi_{\min}^k \pm \Gamma_\theta^n - \nu_{gb} \Theta_0^k - \nu_{van} \Xi_0^k, \\ \dot{c}_{sL}^k &= \dot{c}_{sL}^k|_{\text{growth}} + G_k - \Delta C - m_{\min} \Phi_{\min}^k \pm \Gamma_\theta^c - \nu_{gb} \Theta_1^k - \nu_{van} \Xi_1^k, \\ \dot{q}_{sL}^k &= \dot{q}_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 - m_{\min}^2 \Phi_{\min}^k \pm \Gamma_\theta^q - \nu_{gb} \Theta_2^k - \nu_{van} \Xi_2^k, \end{aligned} \quad (\text{A.24})$$

which are Eqs. (26) to (28) term for term, and every one of those terms is now a moment of a term of Eq. (9). One closure was needed along the way and only one — the pair-weighted second moment of the coagulation, Eq. (A.19) at $j = 2$ — and it is stated below alongside the two channels that carry no distribution of their own and so must not be allowed to imply one.

What remains is to record why the triple (n, c, q) the reduction produces stays a triple of moments. $\Delta_k = q_{sL}^k n_{sL}^k / (c_{sL}^k)^2 \geq 1$ is the Cauchy–Schwarz inequality on a non-negative measure. A state with $\Delta_k < 1$ — or with $q < 0$ outright — is therefore not an inaccurate description of a distribution; it is not the moment triple of any distribution. Most of the channels keep the triple realizable on their own, and the reduction above is where that is shown: the gated removals leave a genuine sub-measure, Eq. (A.23); the loop–network removal is first order at a rate independent of m and takes every moment down in proportion, Eq. (A.22); the coagulation raises q by Eq. (A.19); and the Fokker–Planck part is the moment of a non-negative density. What is left needs stating, and three rules cover it — the first two exact, the third a closure.

1. **Nucleation deposits at its declared size:** $J_{sL}^k (m_k^{\text{nuc}})^2$. This is genuine broadening, $J(m^{\text{nuc}} - \bar{m})^2 \geq 0$, and it is correct because the channel really does deposit at m_k^{nuc} .
2. **The floor current debits at $m_{\min}$:** $-m_{\min}^2 \Phi_{\min}^k$. This is a genuine statement about the lower tail.
3. **Channels with no distribution of their own are shape-preserving:** $dq = \Delta_k(2\bar{m} dc - \bar{m}^2 dn)$. This is a closure and the only one the immobile block needs beyond the log-normal itself. It is what supplies the pair-weighted second moment of the coagulation, Eq. (A.19) at $j = 2$, which ΔN_{LL} alone does not determine; and it is what the legacy annealing surrogate would need and cannot have.

Rule 3 fixes a magnitude, not a sign, and the distinction matters because the sign is where this term was got wrong. That coalescence *raises* q is Eq. (A.19): a coalescing loop does not leave the family, it *merges*, its defects stay, and fewer loops holding the same content are *larger* loops, so the $j = 2$ integrand is $2mm' \geq 0$ whatever the kernel. Debiting it instead by the removed number times $\langle m^2 \rangle$, which is the natural-looking reading, inverts the sign of the term that carries 98% of the loop-number loss. The symptom in a calculation is not an error but a slow drift: on one nine-family march, q crossed zero at 1 dpa and was negative on 40% of the basal nodes at 10 dpa. What rule 3 then settles is how much, and Eqs. (A.20) and (A.21) say it in the form that carries meaning: the rule makes coalescence exactly shape-neutral, where a true coagulation relaxes Δ_k toward the fixed point $\Delta_k = 2$ at which a merge changes no shape. It is therefore not a bound but a substitution of one attractor for none, and its cost is $\Delta_k^{-1/8}$ on the sink prefactor and nothing else.

The legacy annealing surrogate cannot be made consistent, only avoided. It debits content at a *fixed* size, so the pair (dn, dc) it declares corresponds to a removed sub-population of mean $M \neq \bar{m}$. Debiting q proportionally over-removes; debiting at M — the smallest value Cauchy–Schwarz permits for the removed part — lowers Δ_k by $a(M/\bar{m} - 1)^2$ on a narrow family. No debit is realizable, because the surrogate is removing loops of a size that is not there. That is an independent argument for the peripheral-emission channel of Appendix E, which deletes the fitted lifetimes outright.

The two blocks differ, in the end, in exactly one structural respect, and it is the transport term. It survives on $\mathcal{M}$, where it couples every node to its neighbors and makes Eq. (12) a global nonlinear problem; and it is absent on $\mathcal{F}$, where $\mathbf{D}_n^\alpha = \mathbf{0}$ leaves a system of ordinary differential equations in which no neighbor's state appears. That the immobile block has no spatial coupling is not an approximation made for the convenience of the solver. It is a property of Eq. (9) restricted to clusters that do not diffuse, and it is what the operator split of Section 4 exploits.

Appendix B. Iterative FEM Solution Algorithm for Mobile Species

The mobile block, Eq. (12), is the only part of the model solved as a globally coupled problem, and this appendix takes it from the strong form to the linear system that is actually assembled and factorized. Nothing in it is specific to zirconium; what is specific is that the immobile state is a frozen coefficient throughout, which is what makes the block algebraic rather than transient.

What the code assembles is the Galerkin weak form of that statement. Multiplying by a test function $\mathbf{w}$ from the same finite-element space, integrating over the crystal and moving one derivative onto $\mathbf{w}$ gives: find $\mathbf{c}_M \in V_h$ with $\mathbf{c}_M = \mathbf{c}_M^{\text{eq}}$ on $\partial\Omega_D$ such that, for every $\mathbf{w} \in V_h$ vanishing there,

$$\int_{\Omega} \Omega_{at} \nabla \mathbf{w} : (\mathbf{D}_{\text{ten}} \nabla \mathbf{c}_M) dV = \int_{\Omega} \mathbf{w} \cdot \left[\mathbf{P}_M + \frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathcal{D}) \mathbf{c}_M + \mathbf{S}_{vL} \right] dV,$$

the surface term having dropped because $\mathbf{w}$ vanishes on the Dirichlet boundary and no flux is imposed anywhere else. The factor Ω_{at} is the atomic volume, present because the fields are atom fractions and the flux is per unit volume.

The discrete problem follows from restricting both fields to the same finite-element space. With $N_A(\mathbf{x})$ the scalar shape function of node A and c_A^x the atom fraction of species x there,

$$c^x(\mathbf{x}) = \sum_{A=1}^{N_{\text{node}}} N_A(\mathbf{x}) c_A^x, \quad w^x(\mathbf{x}) = \sum_{A=1}^{N_{\text{node}}} N_A(\mathbf{x}) w_A^x, \quad (\text{B.1})$$

with x running over $\{v, i, 2i, 3i\}$ and A over the nodes of the second-order tetrahedral mesh, so that the unknowns are the $4N_{\text{node}}$ nodal values collected in $\mathbf{c}$. Discrete arrays indexed by node are written in upright sans-serif here, to keep them apart from the species-space arrays of the continuous problem. Because the nodal coefficients w_A^x of the test function are arbitrary, each of them yields one equation, and the weak form becomes the algebraic system $\mathbf{r}(\mathbf{c}) = \mathbf{0}$ with

$$\mathbf{r}_A^x(\mathbf{c}) = \sum_B \mathbf{K}_{AB}^x c_B^x - \mathbf{f}_A^x(\mathbf{c}), \quad (\text{B.2})$$

in which the transport matrix and the reaction load are

$$\mathbf{K}_{AB}^x = \int_{\Omega} \Omega_{at} \frac{\partial N_A}{\partial x_i} D_{ij}^x \frac{\partial N_B}{\partial x_j} dV, \quad (\text{B.3})$$

$$\mathbf{f}_A^x(\mathbf{c}) = \int_{\Omega} N_A \left[P_M^x + \frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M + ((\mathbf{E} - \mathbf{D})\mathbf{c}_M)^x + S_{vL}^x \right] dV, \quad (\text{B.4})$$

repeated Cartesian indices i, j being summed. The transport matrix carries no species coupling, because $\mathbf{D}_{\text{ten}}$ is diagonal in species space and $\mathbf{K}$ is therefore one scalar Laplacian per species; all coupling between species sits in $\mathbf{f}$.

Equation (B.2) is nonlinear only through the quadratic term, and it is solved by Newton's method. Differentiating with respect to the nodal unknowns gives the tangent

$$\mathbf{T}_{AB}^{xy} = \frac{\partial \mathbf{r}_A^x}{\partial c_B^y} = \delta^{xy} \mathbf{K}_{AB}^x - \int_{\Omega} N_A \left[(\mathbf{R}_2^x \mathbf{c}_M)^y + (\mathbf{E} - \mathbf{D})^{xy} \right] N_B dV, \quad (\text{B.5})$$

the quadratic term contributing $\mathbf{R}_2^x \mathbf{c}_M$ rather than $\frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$ because $\mathbf{R}_2^x$ is symmetric. The linear block contributes itself, with no factor of $\mathbf{c}_M$ left over, which is what being linear in $\mathbf{c}_M$ amounts to; that it nonetheless carries two species indices is a separate property, and the two are worth holding apart. Linearity is a statement about the dependence on $\mathbf{c}_M$, whereas the indices xy report the shape of the operator in species space, and a linear operator there may perfectly well be off-diagonal. Of the pair, only $\mathbf{E}$ is: dissociation of a $2i$ returns an i , so the $2i$ column feeds the i row, and likewise for the $3i$. The absorption matrix $\mathbf{D}$ of Eq. (14) is diagonal by construction, being a sum of two diag blocks in Eqs. (14) and (16), so each species is absorbed at sinks in proportion to its own concentration alone and $(\mathbf{E} - \mathbf{D})^{xy}$ is off-diagonal through $\mathbf{E}$ only. Each pass then solves for an increment and updates,

$$\sum_{y,B} \mathbf{T}_{AB}^{xy}(\mathbf{c}^{(k)}) \Delta c_B^y = -\mathbf{r}_A^x(\mathbf{c}^{(k)}), \quad c_A^{x,(k+1)} = c_A^{x,(k)} + \Delta c_A^x, \quad (\text{B.6})$$

started from $\mathbf{c}^{(0)} = \mathbf{c}_M^{\text{eq}}$ and iterated until $\|\mathbf{r}\|$ falls below tolerance. Rows belonging to nodes on $\partial\Omega_D$ are replaced by $\Delta c_A^x = 0$, so the Dirichlet data are imposed once, at $k = 0$, and held thereafter.

Two properties of $\mathbf{T}$ decide what the solve costs. It is sparse in the node indices, $\mathbf{T}_{AB}^{xy}$ being non-zero only when A and B share an element, and dense in the species indices at every such pair, because recombination couples all four species; the tangent is therefore a sparse graph of 4×4 blocks rather than four decoupled scalar problems. And it must be rebuilt at every pass, because $\mathbf{c}_M$ enters the quadratic term — but $\mathbf{c}_M$ is the only thing in it that moves, the immobile state being held fixed for the whole of the fast solve.

Appendix C. Continuum-to-Discrete Conversion Algorithm

This appendix gives the construction that turns the three carried moments of a loop family into the individual loops a discrete-dislocation solver can be handed. Figure C.1 draws the tessellation the first step rests on and Fig. C.2 the four steps together; the requirements the result has to satisfy are those of Section 5, and they are checked on the transfer ledger rather than assumed here.

The conversion proceeds in four steps. The fields are known at unevenly spaced nodes, so each must first be given the volume it stands for, which a restricted Voronoi tessellation supplies — restricted in that sample points outside the crystal are rejected rather than charged to the nearest boundary node. On a hexagonal prism the bounding box is $4/3$ of the crystal, so using it would over-count by a third and scatter loops through six empty wedges.

The tessellation is never built explicitly. V_j is estimated by Monte Carlo: points are drawn uniformly in the bounding box of the node cloud, those failing the half-space test $\mathbf{N}\mathbf{x} \leq \mathbf{b}$ of the convex hull are discarded, and each survivor is assigned to its nearest node by a k -d tree search, so that $V_j = (M_j/M) V_{\Omega}$ with M_j the count at node j and M the number of accepted samples. The estimator is unbiased and converges as

$M^{-1/2}$; the production runs use $M = 4 \times 10^6$, which on the 9.1×10^4 nodes here is some 44 samples per node on average and a few percent on a typical cell. Nearest-node assignment is what makes the cells a genuine Voronoi partition, and the only condition the construction has to meet is that they tile the crystal exactly — which they do by construction, every accepted sample being charged to exactly one node, so that $\sum_j V_j = V_\Omega$ identically and no volume is created or lost however coarse the sampling is. Sampling error moves volume *between* neighbouring cells; it cannot change their sum.

Two properties of the partition are worth stating because they are easy to assume wrongly. First, the cells are *not* the finite elements. The tessellation is built on the cluster-dynamics nodes, which are the trial-function nodes of the concentration field and include the mid-edge nodes of every second-order element; the elements are tetrahedra, the Voronoi cells are convex polyhedra around points, and neither is a refinement of the other. What the two share is only the point set. Nodal volumes are not read from the mesh because the output the conversion consumes carries node positions and no connectivity, and a Voronoi volume is the natural quantity to reconstruct from positions alone. Second, a node can legitimately receive zero volume — on the runs here about 12% do, being nodes whose neighbours crowd them out at a refinement boundary. That is the correct answer rather than a failure: a node with no surrounding volume contributes nothing to a volume average, and it is given weight zero.

The expected number of loops of family (s, k) at node j is then

$$\lambda_{sj}^k = \frac{n_{sL}^k(\mathbf{x}_j)}{\Omega} V_j, \quad \Lambda_s^k = \sum_j \lambda_{sj}^k, \quad N_{\text{draw}} = \text{round}(\Lambda_s^k). \quad (\text{C.1})$$

Both factors are needed: sampling on n_{sL}^k alone would over-populate a refined region in exact proportion to how well it was resolved. Rounding rather than drawing a Poisson total is deliberate — $\text{Var} = \Lambda$ there, and a dose sweep whose count fluctuates by $\sqrt{\Lambda}$ between snapshots is dominated by counting noise.

Almost no cell expects a whole loop, and the construction does not require one. At 10 dpa the basal family converts to some 7×10^2 loops spread over 9×10^4 nodes, so the typical λ_{sj}^k is of order 10^{-2} and every cell is empty in expectation. That is why the draw is taken once for the family and not once per node: the rounded total N_{draw} fixes how many loops exist, and the nodes are then sampled with replacement from the multinomial with probabilities $\lambda_{sj}^k / \Lambda_s^k$, so a cell with $\lambda_{sj}^k = 10^{-2}$ simply wins a loop one time in a hundred rather than contributing a hundredth of one. The continuum density is recovered in the mean over that draw, exactly, and the alternative — rounding node by node — would not recover it at all: it would return zero everywhere and convert nothing, which is the failure mode this ordering exists to avoid.

Each loop's size is drawn from its home node's log-normal [Eq. \(43\)](#) rather than set to the node mean, two moments having left nothing to draw from and every loop at a node identical. The sample is rescaled so that $\sum_\ell m_\ell$ matches the continuum content exactly, which is requirement 1, leaving a residual at machine precision. Geometry closes the construction: a loop becomes a polygon on its habit plane of an area matching πR^2 rather than of that circumradius — an inscribed hexagon has only 82.7% of the disc, and it is the stored content that must match — elliptical with an axis ratio set by the anisotropy, and centered by rejection sampling inside the crystal.

What the domain refuses must be predicted rather than discovered, and that is the fourth step. A discrete-dislocation solver silently declines a loop whose nodes leave the grain; the conversion applies the same test in advance, so that the refused share stays in the continuum instead of falling between the two representations. It has to be counted in defects and not in loops, the two being very different measures of one refusal, as the figures quoted in [Section 5](#) show.

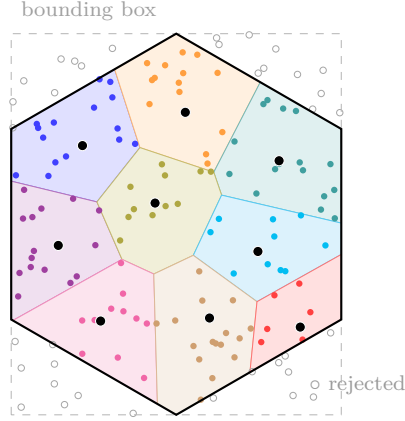
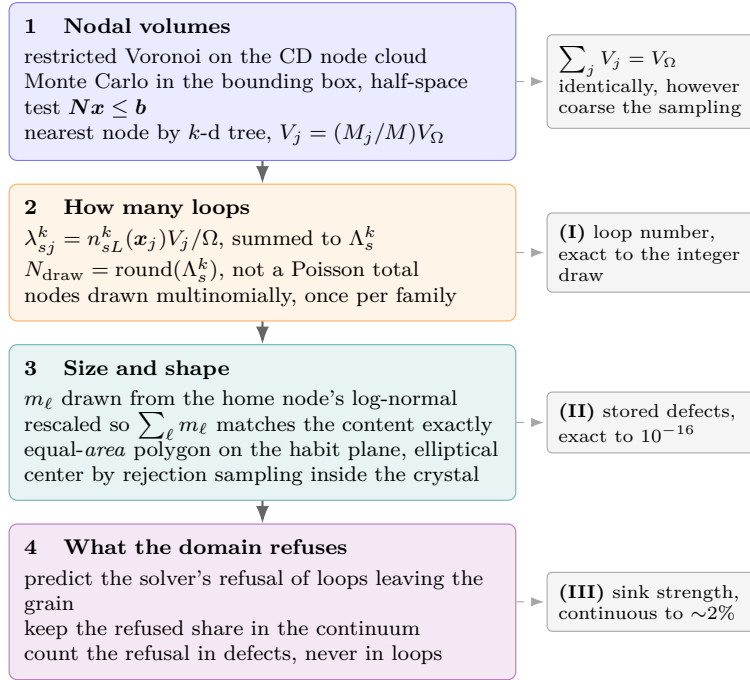


Figure C.1: The restricted Voronoi tessellation and the Monte-Carlo estimator of the nodal volumes, drawn in two dimensions. Black discs are cluster-dynamics nodes; the shaded polygons are their Voronoi cells, each the set of points nearer that node than any other, clipped to the crystal. Samples are drawn uniformly in the dashed bounding box: those failing the half-space test $N\mathbf{x} \leq \mathbf{b}$ are rejected (open circles) and the survivors are charged to their nearest node (filled, colored by cell), giving $V_j = (M_j/M)V_\Omega$. *Restricted* is the whole point — a sample in a corner of the box lies outside the crystal and belongs to no node, where an unrestricted tessellation would charge it to whichever boundary node happened to be nearest. For this hexagon the box carries 4/3 of the crystal area, and the same ratio holds for the prism in three dimensions, so a third of all samples are discarded by construction.



The right-hand column is what each step exists to protect: the three requirements of [Section 5](#), checked on the transfer ledger afterwards rather than assumed here.

Figure C.2: The continuum-to-discrete conversion, step by step, with the methods each step uses and the invariant it protects. The order is not arbitrary: the volumes must precede the counts, because sampling on the density alone would over-populate a refined region in exact proportion to how well it was resolved; and the count must precede the sizes, because it is the rounded family total, not a per-node rounding, that recovers the continuum density in the mean.

Appendix D. Capture efficiencies from the diffusion tensor

The orientation factors $A_h(p)$ of Eq. (32) follow from Woo’s treatment of a dislocation loop as an absorber in an anisotropically diffusing medium [28–30], and in the tensorial form given by Po [31] the correction to the sink strength of a loop of axis α is

$$\frac{\text{tr } \mathbf{D} + \alpha^T \mathbf{D} \alpha}{4\bar{D}}, \quad (\text{D.1})$$

which is universal in loop orientation and in the degree of anisotropy. For a diagonal tensor with basal components $D_\perp$ and axial component $D_\parallel$, and with $p = (D_\parallel/D_\perp)^{1/6}$ and $\bar{D} = (\det \mathbf{D})^{1/3} = D_\perp^{2/3} D_\parallel^{1/3}$, Eq. (D.1) evaluates on a basal loop ($\alpha \parallel [0001]$) to $A_c(p) = p$ and on a prismatic loop ($\alpha \perp [0001]$) to $A_a(p) = (p + p^{-2})/2$.

The two forms differ in a way that has quantitative consequences and is easy to overlook. A_c is monotone increasing, so raising the anisotropy of a species monotonically raises its capture by basal loops. A_a has a *minimum* at $p = 2^{1/3} = 1.2599$, so raising the anisotropy from $p = 1$ first *lowers* prismatic capture. A single change in p^v therefore moves the two habits in opposite directions. Setting $p^v = 1$ where a fit had placed it at 1.179 lowers $Z_{\text{basal}}(v)$ by a factor of 0.848 while *raising* $Z_{\text{prism}}(v)$ by 1.053 — starving basal growth and accelerating prismatic shrinkage at once. Any statement about the effect of anisotropy that does not distinguish the habits is incomplete.

The co-growth condition of Section 3.4 follows directly. Both growth inequalities in Eq. (35) reduce to bounds on the single arrival ratio x , and the window between them has width $g(p^v)/g(p^i)$ with $g(p) = 2/(1 + p^{-3})$, strictly increasing — so the window is open exactly when $p^i < p^v$.

Two limits on this result should be stated. It governs the *absorption* flux only: cascade nucleation G_k is added independently and no anisotropy affects it, so the criterion does not govern net population evolution. And an isotropic calculation is not a zero of the criterion but another parameter point, so a ratio taken against one cannot test a statement about signs.

Appendix E. Thermal Emission of Vacancies

The mean-field model represents loop annealing by two fitted lifetimes, one per character. They are replaced here by an emission channel with no fitted rate, in which a loop exchanges vacancies across its own perimeter against its own equilibrium concentration.

The binding energy of a vacancy to a loop of m defects is the difference of the loop self-energies, signed by the family’s character and shifted by the work the applied stress does on the defect it adds:

$$E_k^b(m) = E_v^f - \varsigma_s(v) [E_L^k(m) - E_L^k(m-1)] - \Sigma_k \Omega, \quad (\text{E.1})$$

with $\varsigma_s(v) = \pm 1$ the polarity product of Eq. (18) — adding a vacancy grows a vacancy family and shrinks an interstitial one, which is the whole of the sign — and $\Sigma_k = (\mathbf{b}_k \cdot \boldsymbol{\sigma} \cdot \mathbf{n}_k)/(\mathbf{b} \cdot \mathbf{n})_k$ the resolved stress the tensor supplies.

E_L^k is the *anisotropic* elastic self-energy of a circular loop of radius $R = \lambda_k \sqrt{m}$, plus the family’s own fault term:

$$E_L^k(m) = \frac{\bar{K}_k |\mathbf{b}_k|^2 R}{2} \left[\ln \frac{8R}{|\mathbf{b}_k|} - 2 \right] + \pi R^2 \gamma_k, \quad \lambda_k = \sqrt{\frac{\Omega}{\pi(\mathbf{b} \cdot \mathbf{n})_k}}. \quad (\text{E.2})$$

Three things in Eq. (E.2) are not the textbook isotropic loop and are the reason it is written out. The prefactor is $\bar{K}_k$, the perimeter-averaged *anisotropic* energy factor — the Stroh/Barnett–Lothe $K(\theta)$ for the family’s own $(\mathbf{b}_k, \mathbf{n}_k)$, integrated over the loop line and evaluated at Fisher and Renken’s 300 K constants for α -Zr [61] — and *not* $\mu/(1 - \nu)$. The logarithm is scaled by $|\mathbf{b}_k|$, the full Burgers magnitude, while the *area* the loop presents per stored defect is set by the edge component $(\mathbf{b} \cdot \mathbf{n})_k$; the two differ for the faulted basal state, whose $\frac{1}{6}\langle 20\bar{2}3 \rangle$ vector carries a basal partial. And γ_k is per family rather than one γ_{sf} : only c_f is faulted, but nothing in the expression assumes that.

Family	$\bar{K}_k$ [GPa]	$ \mathbf{b}_k $ [nm]	$(\mathbf{b} \cdot \mathbf{n})_k$ [nm]	λ_k [nm]	γ_k [J m ⁻²]	$\bar{K}_k \mathbf{b}_k ^2$ [nJ m ⁻¹]	$\bar{K}_k / \frac{\mu}{1-\nu}$
a_1 – a_3 , interstitial	53.1	0.3230	0.3230	0.1514	0	5.54	1.062
a_1 – a_3 , vacancy	53.1	0.3230	0.3230	0.1514	0	5.54	1.062
c_f , faulted basal	53.0	0.3179	0.2575	0.1696	0.124	5.36	1.060
c_p , perfect basal	57.8	0.5150	0.5150	0.1199	0	15.33	1.156

$|\mathbf{b}_{c_f}| = \sqrt{(c/2)^2 + a^2/3}$ and every length follows from the same $a = 3.23 \text{ \AA}$, $c = 5.15 \text{ \AA}$ and Ω the rest of the model uses; $\mu/(1-\nu) = 50.0 \text{ GPa}$ at $\mu = 33 \text{ GPa}$, $\nu = 0.34$. The anisotropy is not a uniform rescaling, which is why it cannot be absorbed into the fit: it is +6% on the prismatic families and +16% on the perfect basal one, so an isotropic prefactor would not merely shift the emission barrier, it would misorder the two habits by 9%. The fault energy is the one number here worth a sensitivity check — *ab initio* values for the I_1 intrinsic basal fault scatter over 0.10–0.15 J m⁻² and Eq. (E.2) is strictly linear in it.

What the model evaluates is not Eq. (E.2) but its derivative, the difference in Eq. (E.1) taken in the continuum limit:

$$\frac{dE_L^k}{dm} = \underbrace{\frac{\gamma_k \Omega}{(\mathbf{b} \cdot \mathbf{n})_k}}_{\text{fault, size-independent}} + \underbrace{\frac{\bar{K}_k |\mathbf{b}_k|^2 \lambda_k^2}{4R} \left[\ln \frac{8R}{|\mathbf{b}_k|} - 1 \right]}_{\text{capillary, } \sim \ln R/R}, \quad (\text{E.3})$$

which follows from $dR/dm = \lambda_k^2/2R$. The split is what separates the two basal states. At $R = 5 \text{ nm}$ and zero stress the capillary term is 0.184 eV for c_f against 0.231 eV for c_p — the faulted state has the *smaller* capillary term, its $\bar{K}_k |\mathbf{b}_k|^2$ being 2.9 times smaller — but c_f carries a further 0.070 eV of fault that does not fall with size at all, and that reverses the total: 0.254 eV against 0.231 eV, so E^b is 1.346 eV for c_f and 1.369 eV for c_p and the faulted state is the *less* strongly bound of the two. The gap widens monotonically with size, 0.011 eV at $R = 3 \text{ nm}$ to 0.061 eV at 50 nm, because only the capillary halves decay; that widening is the driving force behind the unfaulting transfer of Eq. (25), recovered here rather than imposed. The prismatic families sit at 0.151 eV with no fault term at all, and their E^b splits by character: 1.449 eV for a vacancy family and 1.751 eV for an interstitial one, the $\varsigma_s(v)$ of Eq. (E.1) carrying the same 0.151 eV in opposite directions.

The local equilibrium vacancy concentration at the loop periphery is then $c_k^{v,\text{eq}}(m) = \exp[-E_k^b(m)/k_B T]$, and the emission current is the same capture expression run backwards against it,

$$S_{vL}^k = \Pi_{sL}^k \bar{D}^v Z_{vk,v} c_k^{v,\text{eq}}(\bar{m}_{sL}^k), \quad (\text{E.4})$$

which enters Eq. (12) through $\mathbf{S}_{vL}$ and Eq. (27) with the opposite sign. It is not assembled as a separate term: the solver carries it as a signed driving concentration $c^v - c_k^{v,\text{eq}}(\bar{m}_{sL}^k)$ inside the same capture sum that Eq. (18) already forms, which is the same quantity and makes detailed balance exact by construction — the channel vanishes identically at $c^v = c_k^{v,\text{eq}}$ — rather than by the cancellation of an absorption and an emission term carrying different prefactors. Thermal *interstitial* emission is not carried at all: at 873 K it is $e^{-1.4/k_B T} \approx 10^{-8}$ of the vacancy channel.

Three properties recommend this over a fitted lifetime. *One expression serves both characters*: the same mechanism shrinks a vacancy loop, by emitting the vacancies it holds, and grows an interstitial one, by emitting vacancies that recombine with its interstitials, so no separate interstitial-loop annealing model is needed. *The climb threshold emerges rather than being imposed*: growth requires the arriving flux to exceed Eq. (E.4), which reproduces the classical critical radius without it appearing as a parameter. And *it removes content, not loops*, which is physically correct — emission shrinks a loop continuously — but has a consequence that must be recognized: after this replacement, coalescence becomes the *only* sink on vacancy loop number. That is stated in Section 8 as the leading open problem, and it is a consequence of doing the emission physics correctly rather than an artifact of it.

Appendix F. State-vector layout and the conservation accumulators

The integrated state at each node is a single flat vector whose width depends on the model controls. The layout is fixed and every index keeps its meaning as the controls add fields:

Indices	Contents	Count
0–3	mobile: c^v, c^i, c^{2i}, c^{3i}	4
4–7	number of families 0–3	4
8–11	content of families 0–3	4
12–17	accumulators: production, recombination, sink absorption, each per character	6
18	network dislocation density ρ_N	1
19–23	number of families 4–8	5
24–28	content of families 4–8	5
29–37	second content q of all nine families	9
38–39	grain-boundary loop ledger, per character	2

Widths are therefore 19 (four families, two moments), 29 (nine families, two moments), 38 (nine families, three moments) and 40 with the boundary ledger. Two properties of this layout are worth recording because both were established by their failure modes.

The appended families are appended, not interleaved. Every index below 19 means what it meant in the four-family model, so a state written by one configuration is readable by another up to its own width. The corollary is that a narrower state is always *accepted* by a wider solver, the appended slots defaulting to zero — which is exactly why a mismatch between the width a driver allocates and the width the model requires is silent rather than fatal, and why the checkpoint fingerprint records the actual width.

The block written to the spatial code is ordered the other way. It is moment-major — all nine numbers, then all nine contents, then all nine q — so the bridge between the two is a scatter and not a copy. Exactly one routine knows that mapping; performing it at a call site is what once put an 18-wide block into an 8-wide slice.

The accumulators are per-interval; the ledger is cumulative. The six conservation accumulators are reset at the start of each dose interval, so a stored snapshot holds one interval’s production and must be summed to give a running total. The two grain-boundary accumulators are never reset and are already cumulative. Differencing the wrong one against the wrong other gives a channel that appears to carry 6600 times total production, or none at all, depending on which way the error runs.